Structure-Preserving Modeling, Hamiltonian Neural Networks, and Local KAM Analysis for Magnetically Actuated Flexible Catheters

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Abstract: Magnetically actuated flexible catheters combine large-deformation rod mechanics with magnetic forcing, fluid loading, and contact. We develop a structure-preserving reduced Hamiltonian model and use it to examine how physical priors and complex representations affect Hamiltonian learning. The geometrically exact formulation permits a configuration-dependent cotangent pullback, whereas the learning experiments use a local fixed-basis reduction with constant reduced mass matrix and $p = M_4\dot{\mathbf{q}}$. We evaluate five models on 96 COMSOL trajectories, partitioned by complete trajectories into 64 training, 16 validation, and 16 independent-test cases. An approximately parameter-matched 2×2 design separates real and complex representations from pure and gray-box Hamiltonians, with five 300-epoch runs per architecture. Gray Real HNN and Gray CHNN achieve mean rollout errors of 12.6587 and 12.6640, compared with 1665.33 and 1665.28 for their pure counterparts. The gray-box prior therefore accounts for the dominant improvement. The paired gray-model difference is -0.00532, with a trajectory-bootstrap 95% confidence interval of $[-0.14161, 0.12748]$, so an additional benefit from complex representation is not statistically resolved. Auxiliary nested-basis data reduce generalized-force representation error from 0.20153 at $N = 2$ to 0.09037 at $N = 4$, relative to $N = 6$; predictive modal-order sensitivity was not evaluated. High-precision computation supports a local, conditional KAM result for the selected quartic conservative approximation.

Keywords: magnetically actuated flexible catheter; geometrically exact rod; Hamiltonian neural network; complex-valued neural network; symplectic algorithm; KAM theory

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Introduction

Magnetically actuated flexible catheters in microgravity combine magnetic actuation, background flow, vessel-wall contact, and boundary manipulation. Their motion couples large rotations, geometric nonlinearity, bending–torsion, and energy exchange with external inputs. A useful learning model must therefore connect the continuum mechanics, finite-dimensional reduction, data representation, and long-time analysis without changing the physical meaning of the state variables.

Geometrically exact beam and rod theories provide the kinematic and variational basis for large-deformation structures, while magnetic, fluid, and contact models describe the surrounding environment[1–18]. We retain translation and cross-sectional rotation during reduction, separate conservative loads from dissipation and external work, and use compatible variables in the weak form, Galerkin projection, constraint treatment, and Legendre transformation.

Hamiltonian neural networks encode conservative dynamics through the symplectic gradient of a real-valued energy. Dissipative and port-based extensions accommodate forcing and energy loss, while complex-valued networks offer a representation for amplitude–phase coupling[19–24]. A complex internal representation does not make the physical Hamiltonian complex, and one scalar cannot represent dissipation and forcing without additional channels. Physical components in a multi-component model must be identified using the known mechanics, port powers, paired operating conditions, and ablations.

KAM theory and symplectic numerical methods provide tools for studying quasiperiodic dynamics in nearly integrable Hamiltonian systems and maps[25–33]. Continuous-time KAM, symplectic-map KAM, two-dimensional twist-map results, and infinite-dimensional KAM address different objects. A training loss or trajectory residual cannot replace the smallness conditions expressed in higher-order or analytic norms. We therefore treat learning, physical, coordinate, truncation, and time-discretization errors in compatible spaces.

This study makes four contributions.

(1) We formulate a geometrically exact multiphysics catheter model and a joint translational–rotational Galerkin reduction through consistent energy and power identities. The full nonlinear material-reference cotangent pullback is distinguished from the fixed-basis finite-dimensional canonical approximation used for the primary trajectories.

(2) Real and complex Hamiltonian networks share a real-valued physical Hamiltonian. An approximately parameter-matched 2×2 factorial design then separates representation from the gray-box prior.

(3) We evaluate five architectures on 96 COMSOL trajectories split by complete trajectories into 64 training, 16 validation, and 16 independent-test cases. Five seeds are retained per architecture, and uncertainty is evaluated at the trajectory level. The results support the gray-box prior, whereas an additional complex-representation effect remains statistically unresolved.

(4) We combine a local quartic Birkhoff-normal-form and KAM analysis with high-precision conditioning checks. The conclusion is restricted to the selected quartic conservative approximation and is not transferred to the full material-reference system without a uniform higher-order bound.

The workflow in Figure 6.1 follows this modeling chain: geometrically exact rod mechanics, Galerkin reduction, a fixed-basis canonical representation, factorial Hamiltonian learning, auxiliary modal-order sensitivity, and local Birkhoff/KAM analysis. The full nonlinear cotangent pullback remains a theoretical generalization rather than a numerical data path in the primary study. We compare Residual MLP, Real HNN, Gray Real HNN, Standard CHNN, and Gray CHNN under one prespecified protocol. The analysis reports per-trajectory five-seed medians, paired trajectory-bootstrap intervals, computation cost, and the boundary imposed by the unavailable predictive $N = 2, 4, 6$ comparison.

The remainder of the paper is organized as follows. Sections 1 and 2 develop the continuous multiphysics model and its joint Galerkin discretization. Section 3 derives the finite-dimensional Hamiltonian structure and the canonical state mapping. Section 4 presents the complex-representation-enhanced structured HNN and its error analysis. Section 5 discusses

symplectic discretization, Birkhoff normal forms, and invariant-torus analysis. Numerical results are reported in Section 6; limitations and scope are discussed in Section 7; and Section 8 summarizes the main findings.

1. Geometrically Exact Bending–Torsion Dynamics of a Flexible Catheter under Multiphysics Loading

1.1. **Modeling framework.** We consider a magnetically actuated flexible catheter undergoing three-dimensional large deformation in a space-station microgravity environment. Magnetic fields, fluid flow, and vessel-wall contact act simultaneously. The centerline position and cross-sectional orientation are retained as independent fields, together with axial extension, transverse shear, two bending directions, independent torsion, rotary inertia, and distributed couples.

We first formulate the continuum model with a geometrically exact Cosserat rod. For a slender catheter, a Kirchhoff shear-free constraint can be imposed later. It removes transverse shear strain but retains cross-sectional orientation, torsion, angular velocity, and rotary inertia. The translational–rotational model is therefore not replaced by a centerline-only equation.[1, 2, 17]

Let the reference material coordinate be

$$S \in [0, L], \qquad t \in [0, T], \tag{1.1}$$

where L is the catheter length in the reference configuration. Unless otherwise stated, all cross-sectional parameters may vary with S to represent nonuniform geometry, spatially varying material properties, or localized magnetic components.

Magnetic, fluid, and contact effects enter the rod equations as distributed forces and couples per unit reference length. A strictly coupled multiphysics PDE system would require simultaneous solution of the electromagnetic, fluid, and structural field equations. The formulation developed in this section is instead a geometrically exact rod-dynamics model driven by multiphysics load inputs.

1.1.1. *Model hierarchy and computational branches.* The analysis proceeds through four linked levels. We combine geometrically exact rod mechanics with Maxwell magnetic actuation, Morison-type fluid loading, and catheter–vessel-wall contact; Galerkin projection then produces finite-dimensional dynamics for the Hamiltonian, learning, and KAM analyses.

1. **Geometrically exact physical parent model.** The coupled translational–cross-sectional-orientation Cosserat rod, independent torsion, rotary inertia, distributed forces, and distributed couples are retained. This level preserves the complete mechanical information and provides the common energy and virtual-work basis for all reduced branches[1, 2].
2. **Strict Kirchhoff theoretical branch.** For slender, low-shear catheters, the shear-free constraint is imposed without removing cross-sectional orientation, torsion, angular velocity, or rotary inertia. Before CHNN or KAM analysis, the constrained finite-dimensional system must be expressed in independent local coordinates on the constraint manifold or approximated by a regularized model whose convergence to the constrained system has been verified. A differential–algebraic system with Lagrange multipliers must not be treated directly as a canonical Hamiltonian system on the full coordinate space.
3. **COMSOL Kirchhoff-consistent training branch.** The primary training data preferentially use the three-dimensional Euler–Bernoulli Beam interface with geometric nonlinearity, dynamic inertia, and torsional degrees of freedom, which is the closest built-in COMSOL realization of the Kirchhoff shear-free assumptions. Because this interface permits large displacements and rotations while retaining a small-strain, element-local linearization, it is referred to as a "Kirchhoff-consistent engineering model" rather than being identified unconditionally with the geometrically exact parent model[4–6]. When term-by-term reproduction of the coupled weak form is required, custom Weak Form PDE or Weak Contribution equations are used[7, 8].

4. **High-fidelity validation and open-port branch.** Timoshenko beams, three-dimensional Solid Mechanics, or a custom Cosserat weak form are used to assess shear effects, local cross-sectional effects, and model discrepancy. Magnetic control, background-flow input, passive drag, contact dissipation, friction, and boundary actuation enter the finite-dimensional model through mutually exclusive power ports. High-fidelity data are used for external validation and model-discrepancy learning and are not mixed with conservative Kirchhoff samples without explicit labels. The complete open system is used for structured CHNN identification and safe-domain computation; only the autonomous conservative backbone obtained by closing the ports, or an independently proven exact symplectic extension, is a direct object of classical KAM theory.

Together, these levels connect the continuous equations and Galerkin system to the conservative energies, constraint-reduced states, port decompositions, and representation labels used in Sections 3 to 5. Subsequent reductions are interpreted within this shared physical–mathematical framework.

1.2. Geometry and kinematics.

1.2.1. *Centerline and cross-sectional frame.* Let the catheter centerline be described by the position vector

$$\mathbf{r} = \mathbf{r}(S,t) \in \mathbb{R}^3. \tag{1.2}$$

Introduce a right-handed orthonormal frame on each cross section,

$$\{\mathbf{d}_1(S,t), \mathbf{d}_2(S,t), \mathbf{d}_3(S,t)\}, \tag{1.3}$$

and define the rotation matrix

$$\mathbf{R}(S,t) = [\mathbf{d}_1\ \mathbf{d}_2\ \mathbf{d}_3] \in SO(3), \tag{1.4}$$

which satisfies

$$\mathbf{R}^T\mathbf{R} = \mathbf{I}, \qquad \det \mathbf{R} = 1. \tag{1.5}$$

The full model does not impose $\mathbf{d}_3$ to be parallel to the centerline tangent; transverse shear is therefore admissible. The centerline stretch and unit tangent are defined, respectively, by

$$\lambda(S,t) = \| \mathbf{r}_{,S} \|, \qquad \mathbf{t}(S,t) = \frac{\mathbf{r}_{,S}}{\lambda}. \tag{1.6}$$

1.2.2. *Skew-symmetric operator.* For any $\mathbf{a} \in \mathbb{R}^3$, define the skew-symmetric matrix $\widehat{\mathbf{a}} \in \mathfrak{so}(3)$ such that[9, 17]

$$\widehat{\mathbf{a}}\,\mathbf{b} = \mathbf{a} \times \mathbf{b}, \qquad \forall \mathbf{b} \in \mathbb{R}^3. \tag{1.7}$$

Its inverse mapping is denoted by $(\cdot)^\vee$. Hence, both spatial and temporal derivatives of the rotation matrix can be represented by three-dimensional vectors.

1.2.3. *Shear–extension strain.* Define the shear–extension strain vector in the material frame as

$$\boldsymbol{\nu}(S,t) = \mathbf{R}^T\mathbf{r}_{,S} = \begin{bmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{bmatrix}. \tag{1.8}$$

where ν_1 and ν_2 are the two transverse shear-strain components and ν_3 is the axial extension component along the third director. The reference strain is denoted by

$$\boldsymbol{\nu}_0(S) = \begin{bmatrix} \nu_{01} \\ \nu_{02} \\ \nu_{03} \end{bmatrix}. \tag{1.9}$$

For a straight rod with no initial shear and no reference-length extension, one may take

$$\boldsymbol{\nu}_0 = (0, 0, 1)^T. \tag{1.10}$$

In the fully shearable Cosserat parent model, the axial material-strain component along the third director must be distinguished from the centerline stretch strain. The former is defined by

$$\varepsilon_3 = \nu_3 - \nu_{03}. \tag{1.11}$$

On the other hand, $\mathbf{r}_{,S} = \mathbf{R}\boldsymbol{\nu}$ together with $\mathbf{R}^T\mathbf{R} = \mathbf{I}$ gives $\lambda = \| \mathbf{r}_{,S} \| = \| \boldsymbol{\nu} \|$. The centerline extension strain is therefore defined as

$$\varepsilon_\lambda = \lambda - \lambda_0 = \| \boldsymbol{\nu} \| - \| \boldsymbol{\nu}_0 \| . \tag{1.12}$$

Under the Kirchhoff shear-free condition $\nu_1 = \nu_2 = 0$, if the reference configuration also satisfies $\nu_{03} = \lambda_0$, then $\varepsilon_3 = \varepsilon_\lambda$. The constitutive matrices used below remain formulated in terms of the material strain $\boldsymbol{\nu} - \boldsymbol{\nu}_0$.

1.2.4. *Curvature–torsion kinematics.* Define the curvature–torsion vector in the material frame by

$$\widehat{\boldsymbol{\kappa}} = \mathbf{R}^T\mathbf{R}_{,S}, \qquad \boldsymbol{\kappa} = \begin{bmatrix} \kappa_1 \\ \kappa_2 \\ \kappa_3 \end{bmatrix}. \tag{1.13}$$

where κ_1 and κ_2 are the bending curvatures about the two principal directions and κ_3 is the independent twist rate. The variation of the local frame along the reference coordinate satisfies

$$\mathbf{d}_{i,S} = (\mathbf{R}\boldsymbol{\kappa}) \times \mathbf{d}_i, \qquad i = 1, 2, 3. \tag{1.14}$$

The reference curvature–torsion vector is denoted by

$$\boldsymbol{\kappa}_0(S) = (\kappa_{01}, \kappa_{02}, \kappa_{03})^T. \tag{1.15}$$

1.2.5. *Cross-sectional angular velocity and compatibility.* Define the cross-sectional angular velocity in the material frame by

$$\widehat{\boldsymbol{\omega}} = \mathbf{R}^T\mathbf{R}_{,t}. \tag{1.16}$$

The corresponding angular velocity in the spatial frame is

$$\boldsymbol{\omega}_s = \mathbf{R}\boldsymbol{\omega}. \tag{1.17}$$

The identities $\mathbf{r}_{,St} = \mathbf{r}_{,tS}$ and $\mathbf{R}_{,St} = \mathbf{R}_{,tS}$ yield the kinematic compatibility relations[1, 2, 9]

$$\dot{\boldsymbol{\nu}} = \mathbf{R}^T\dot{\mathbf{r}}_{,S} - \boldsymbol{\omega} \times \boldsymbol{\nu}, \tag{1.18}$$

$$\dot{\boldsymbol{\kappa}} = \boldsymbol{\omega}_{,S} + \boldsymbol{\kappa} \times \boldsymbol{\omega}. \tag{1.19}$$

The second relation enforces compatibility between the spatial and temporal evolution of the cross-sectional orientation and is indispensable to the complete bending–torsion dynamics.

1.2.6. *Motion of material points within a cross section.* Let the in-plane material coordinate be

$$\boldsymbol{\xi} = \xi_1 \mathbf{e}_1 + \xi_2 \mathbf{e}_2, \tag{1.20}$$

The spatial position of a material point in the rod is then

$$\mathbf{x}\left(S, \boldsymbol{\xi}, t\right) = \mathbf{r}(S,t) + \mathbf{R}(S,t)\boldsymbol{\xi}. \tag{1.21}$$

and its velocity is

$$\dot{\mathbf{x}} = \dot{\mathbf{r}} + \mathbf{R}\left(\boldsymbol{\omega} \times \boldsymbol{\xi}\right) = \dot{\mathbf{r}} - \mathbf{R}\widehat{\boldsymbol{\xi}}\boldsymbol{\omega}. \tag{1.22}$$

This expression contains both centerline translation and cross-sectional rotational velocity and is the starting point for the complete kinetic energy and the configuration-dependent mass matrix.

1.3. **Complete kinetic energy, elastic potential, and constitutive relations.**

1.3.1. *Cross-sectional mass parameters and the mass-centerline convention.* Let the material density be $\rho\left(S, \boldsymbol{\xi}\right)$. Define the mass per unit reference length as

$$\mu(S) = \int_{A(S)} \rho \, dA, \tag{1.23}$$

Define the first mass moment of the cross section by

$$\mathbf{c}_\rho(S) = \int_{A(S)} \rho \boldsymbol{\xi} \, dA, \tag{1.24}$$

and the mass-inertia tensor per unit reference length in the material frame by

$$\mathbf{J}_c(S) = \int_{A(S)} \rho \left(\| \boldsymbol{\xi} \|^2 \mathbf{I} - \boldsymbol{\xi}\boldsymbol{\xi}^T \right) dA. \tag{1.25}$$

The preceding definitions retain a general cross-sectional mass distribution. To ensure that the subsequent momentum balance, weak form, and discrete mass matrix use a common inertial reference, the centerline $\mathbf{r}(S,t)$ is taken, from Section 1.4 onward, to pass through the mass center of each cross section. The distributed rod therefore satisfies

$$\boxed{\mathbf{c}_\rho(S) = \mathbf{0}.} \tag{1.26}$$

This convention removes neither cross-sectional rotational degrees of freedom nor rotary inertia. A magnetic tip, sensor, or other localized component whose center of mass is offset from the centerline is not absorbed into $\mathbf{c}_\rho$; it is introduced as an attached rigid body with its own coupled translational–rotational kinetic energy. This treatment preserves both closure of the base-rod equations and the physical inertial coupling of eccentric attachments.

If the principal cross-sectional axes coincide with $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, then

$$\mathbf{J}_c = diag\left(\rho I_1, \rho I_2, \rho I_3\right), \tag{1.27}$$

where I_1, I_2 are the area moments of inertia about the two principal directions and $I_3 = I_1 + I_2$ is the polar moment. For a nonuniform density distribution, the area-integral definitions must be used directly; the density cannot in general be factored out of the integrals.

1.3.2. *Complete structural kinetic energy.* The three-dimensional kinetic energy of the base rod follows from the material-point velocity as

$$T_{str} = \frac{1}{2}\int_0^L \int_{A(S)} \rho \left|\dot{\mathbf{r}} + \mathbf{R}\left(\boldsymbol{\omega}\times\boldsymbol{\xi}\right)\right|^2 dA\, dS. \tag{1.28}$$

After integration over the cross section, the general form becomes

$$T_{str} = \frac{1}{2}\int_0^L \left[\mu\left|\dot{\mathbf{r}}\right|^2 + 2\dot{\mathbf{r}}^T\mathbf{R}\left(\boldsymbol{\omega}\times\mathbf{c}_\rho\right) + \boldsymbol{\omega}^T\mathbf{J}_c\boldsymbol{\omega}\right] dS. \tag{1.29}$$

Under the mass-centerline convention of Section 1.3.1, $\mathbf{c}_\rho = \mathbf{0}$, and the kinetic energy of the base rod reduces to

$$\boxed{T_{str} = \frac{1}{2}\int_0^L \left[\mu\left|\dot{\mathbf{r}}\right|^2 + \boldsymbol{\omega}^T\mathbf{J}_c\boldsymbol{\omega}\right] dS.} \tag{1.30}$$

If eccentric attached components are present, their kinetic energy T_e is added separately and generally reintroduces translational–rotational cross terms. Regardless of whether the base rod uses a mass-centerline convention, the kinetic energy retains rotary inertia about both transverse axes and torsional inertia about the tangent direction. The model must not subsequently be reduced to a centerline-only field by deleting the rotational terms.

1.3.3. *Elastic strain-energy density.* Let the shear–extension and bending–torsion stiffness matrices be, respectively,

$$\mathbf{K}_\nu(S) = diag\left(k_1GA, k_2GA, EA\right), \tag{1.31}$$

$$\mathbf{K}_\kappa(S) = diag\left(EI_1, EI_2, GJ\right), \tag{1.32}$$

where E is Young's modulus, G is the shear modulus, k_1, k_2 are shear correction factors, and J is the cross-sectional torsional constant. For asymmetric materials or cross sections, $\mathbf{K}_\nu$ and $\mathbf{K}_\kappa$ may be general symmetric positive-definite matrices so that directional coupling is retained.

The elastic strain-energy density per unit reference length is defined by

$$W\left(\boldsymbol{\nu},\boldsymbol{\kappa};S\right) = \frac{1}{2}\left(\boldsymbol{\nu}-\boldsymbol{\nu}_0\right)^T\mathbf{K}_\nu\left(\boldsymbol{\nu}-\boldsymbol{\nu}_0\right) + \frac{1}{2}\left(\boldsymbol{\kappa}-\boldsymbol{\kappa}_0\right)^T\mathbf{K}_\kappa\left(\boldsymbol{\kappa}-\boldsymbol{\kappa}_0\right). \tag{1.33}$$

The internal elastic potential energy is

$$V_{int} = \int_0^L W\, dS. \tag{1.34}$$

1.3.4. *Constitutive laws for internal force and couple.* The resultant internal force and couple in the material frame are obtained from the gradient of the strain-energy density:

$$\mathbf{n}_b = \frac{\partial W}{\partial \boldsymbol{\nu}} = \mathbf{K}_\nu\left(\boldsymbol{\nu}-\boldsymbol{\nu}_0\right), \tag{1.35}$$

$$\mathbf{m}_b = \frac{\partial W}{\partial \boldsymbol{\kappa}} = \mathbf{K}_\kappa\left(\boldsymbol{\kappa}-\boldsymbol{\kappa}_0\right). \tag{1.36}$$

Their counterparts in the spatial frame are

$$\mathbf{n} = \mathbf{R}\mathbf{n}_b, \qquad \mathbf{m} = \mathbf{R}\mathbf{m}_b. \tag{1.37}$$

These constitutive equations retain extension, shear in both transverse directions, bending about two axes, and independent torsion. For a general nonlinear hyperelastic material, the quadratic form W may be replaced by the corresponding objective strain-energy density, with the constitutive resultants still obtained from its derivatives.

1.4. **Momentum balance and multiphysics loading.**

1.4.1. *Translational balance.* The continuous balance laws below use the mass centerline specified in Section 1.3.1 as the centerline of the base catheter. The linear-momentum balance per unit reference length is

$$\mu\ddot{\mathbf{r}} = \mathbf{n}_{,S} + \mathbf{f}_{ext}, \tag{1.38}$$

where

$$\mathbf{f}_{ext} = \mathbf{f}_{mag} + \mathbf{f}_{hydro} + \mathbf{f}_{cont} + \mathbf{f}_{g}. \tag{1.39}$$

All terms represent spatially distributed forces per unit reference length. The inertia of eccentric attachments is not contained implicitly in $\mu\ddot{\mathbf{r}}$; it is introduced consistently at the discrete level through their independent kinetic energy. Under the energy-consistent fluid formulation of Section 1.4.4, the added-inertia operator is placed on the left-hand side, while the fluid-acceleration input on the right-hand side contains only the pure Froude–Krylov term and drag. The complete engineering Morison branch and this theoretical branch must not be activated simultaneously.

1.4.2. *Rotational balance.* The angular-momentum balance about the centerline is written as

$$\mathbf{m}_{,S} + \mathbf{r}_{,S} \times \mathbf{n} + \mathbf{l}_{ext} = \frac{\partial}{\partial t}\left(\mathbf{R}\mathbf{J}_{c}\boldsymbol{\omega}\right), \tag{1.40}$$

where

$$\mathbf{l}_{ext} = \mathbf{l}_{mag} + \mathbf{l}_{hydro} + \mathbf{l}_{cont}. \tag{1.41}$$

Using $\dot{\mathbf{R}} = \mathbf{R}\widehat{\boldsymbol{\omega}}$, the expanded form is

$$\mathbf{m}_{,S} + \mathbf{r}_{,S} \times \mathbf{n} + \mathbf{l}_{ext} = \mathbf{R}\left[\mathbf{J}_{c}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times \left(\mathbf{J}_{c}\boldsymbol{\omega}\right)\right]. \tag{1.42}$$

where $\boldsymbol{\omega} \times \left(\mathbf{J}_{c}\boldsymbol{\omega}\right)$ is the gyroscopic inertia term. It follows from the exact time derivative of angular momentum and is not an empirical correction that may be added or removed arbitrarily.

1.4.3. *Magnetic force and couple from the Maxwell stress tensor.* In the air or vacuum region surrounding an infinitesimal magnetic catheter segment, the Maxwell stress tensor is[10, 11]

$$\mathbf{T}_{M} = \frac{1}{\mu_{0}}\left(\mathbf{B} \otimes \mathbf{B} - \frac{1}{2}\left|\mathbf{B}\right|^{2}\mathbf{I}\right), \tag{1.43}$$

where $\mathbf{B}$ is the magnetic flux density and μ_0 is the permeability of free space. For a segment of reference length ΔS, the magnetic force per unit reference length and the magnetic couple about the centerline are defined by

$$\mathbf{f}_{mag} = \lim_{\Delta S \to 0} \frac{1}{\Delta S} \int_{\partial\Omega_{\Delta S}} \mathbf{T}_{M}\mathbf{n}_{A}\, dA, \tag{1.44}$$

$$\mathbf{l}_{mag} = \lim_{\Delta S \to 0} \frac{1}{\Delta S} \int_{\partial\Omega_{\Delta S}} \left(\mathbf{x} - \mathbf{r}\right) \times \left(\mathbf{T}_{M}\mathbf{n}_{A}\right)\, dA, \tag{1.45}$$

where $\mathbf{n}_A$ is the outward unit normal to the closed integration surface.

When a magnetic segment can be represented by a dipole approximation, let $\boldsymbol{\mu}_{0}^{b}(S)$ be the magnetic moment per unit reference length in the material frame. The corresponding spatial magnetic moment is[10, 11]

$$\boldsymbol{\mu}(S,t) = \mathbf{R}(S,t)\boldsymbol{\mu}_{0}^{b}(S). \tag{1.46}$$

For a prescribed external magnetic field $\mathbf{B}\left(\mathbf{r},t\right)$, the magnetic potential energy is

$$V_{mag} = -\int_{0}^{L} \boldsymbol{\mu} \cdot \mathbf{B}\left(\mathbf{r},t\right)\, dS. \tag{1.47}$$

The associated magnetic force and couple are

$$\mathbf{f}_{mag} = \nabla_{\mathbf{r}} \left(\boldsymbol{\mu} \cdot \mathbf{B} \right), \qquad \mathbf{l}_{mag} = \boldsymbol{\mu} \times \mathbf{B}. \tag{1.48}$$

The Maxwell-stress integral and the magnetic-dipole potential are two scale-dependent representations of the same magnetic action. One should be selected as the primary load model according to the required fidelity; they must not be superposed and counted twice.

1.4.4. *Modified Morison loading and energy-consistent added mass.* Let the fluid velocity be $\mathbf{v}_f(\mathbf{r}, t)$, with material acceleration

$$\mathbf{a}_f = \frac{\partial \mathbf{v}_f}{\partial t} + (\mathbf{v}_f \cdot \nabla)\, \mathbf{v}_f. \tag{1.49}$$

Define the fluid velocity relative to the catheter by

$$\mathbf{u}_r = \mathbf{v}_f - \dot{\mathbf{r}}. \tag{1.50}$$

If only the component normal to the current centerline is retained, define

$$\mathbf{P}_\perp = \mathbf{I} - \mathbf{t} \otimes \mathbf{t}, \qquad \mathbf{u}_{r\perp} = \mathbf{P}_\perp \mathbf{u}_r. \tag{1.51}$$

An engineering Morison validation branch may be written as[3, 12, 13]

$$\mathbf{f}^M_{hydro} = \mathbf{f}^M_{in} + \mathbf{f}^M_{add} + \mathbf{f}_{drag}, \tag{1.52}$$

where

$$\mathbf{f}^M_{in} = \rho_f C_M A_d \mathbf{P}_\perp \mathbf{a}_f, \tag{1.53}$$

$$\mathbf{f}^M_{add} = -\rho_f C_A A_d \mathbf{P}_\perp \ddot{\mathbf{r}}, \tag{1.54}$$

$$\mathbf{f}_{drag} = \frac{1}{2} \rho_f C_D D \parallel \mathbf{u}_{r\perp} \parallel \mathbf{u}_{r\perp}. \tag{1.55}$$

Here, ρ_f is the fluid density, A_d is the characteristic displaced-fluid area per unit length, D is the catheter outer diameter, and C_A, C_M, C_D are the added-mass, inertia, and drag coefficients, respectively; under a common convention, $C_M = 1 + C_A$. This empirical branch must be used as a complete formulation and must not be combined with the functional T_A introduced below.

To incorporate configuration-dependent added mass into the full inertial structure, the theoretical branch is instead defined through a velocity-dependent added-fluid inertia functional. Let[12, 13]

$$m_A(S) = \rho_f C_A A_d, \qquad \mathbf{w}_\perp = \mathbf{P}_\perp \left(\dot{\mathbf{r}} - \mathbf{v}_f \right), \tag{1.56}$$

and take

$$T_A \left[\mathbf{r}, \dot{\mathbf{r}}; t \right] = \frac{1}{2} \int_0^L m_A \, \mathbf{w}_\perp^T \mathbf{w}_\perp \, dS. \tag{1.57}$$

The corresponding inertial action is generated consistently by the functional Euler–Lagrange derivative

$$\mathbf{I}_A = \frac{d}{dt} \frac{\delta T_A}{\delta \dot{\mathbf{r}}} - \frac{\delta T_A}{\delta \mathbf{r}} \tag{1.58}$$

When $\mathbf{P}_\perp$ is held fixed, the leading term is

$$\mathbf{I}_A = \rho_f C_A A_d \mathbf{P}_\perp \left(\ddot{\mathbf{r}} - \mathbf{a}_f \right), \tag{1.59}$$

For the full pointwise variation, define $\mathbf{A}_A = m_A \mathbf{P}_\perp$, $\mathbf{u} = \dot{\mathbf{r}} - \mathbf{v}_f$, and $\mathbf{g}_{A,i} = \mathbf{u}^T [(\partial_{r_i} m_A) \mathbf{P}_\perp] \mathbf{u}$. With $\dot{\mathbf{A}}_A = \partial_t \mathbf{A}_A + (\nabla \mathbf{A}_A) \dot{\mathbf{r}}$, the Euler–Lagrange contribution is

$$\mathbf{I}_A = \mathbf{A}_A (\ddot{\mathbf{r}} - \mathbf{a}_f) + \left[\dot{\mathbf{A}}_A + (\nabla \mathbf{v}_f)^T \mathbf{A}_A - \mathbf{A}_A \nabla \mathbf{v}_f \right] \mathbf{u} - \frac{1}{2} \mathbf{g}_A + \mathbf{G}_P. \tag{1.60}$$

Here $\mathbf{G}_P$ denotes the local contribution before the configuration-dependent projector variation is fully expanded. It includes the quadratic-form and derivative terms collected at this stage, but not the complete projector variation. Equation (1.59) is recovered only when the projector and added-mass tensor are held fixed, or when the displayed gradient terms vanish. Spatial velocity gradients and vorticity remain in the general prescribed-flow branch.

Variations of $\mathbf{P}_\perp$ with configuration and spatial variations of the prescribed flow generate additional geometric terms. When the theoretical branch based on T_A is used, the explicit fluid-acceleration input on the right-hand side retains only the pure Froude–Krylov term

$$\boxed{\mathbf{f}_{FK} = \rho_f A_d \mathbf{P}_\perp \mathbf{a}_f}, \tag{1.61}$$

with

$$\mathbf{f}_{hydro,rem} = \mathbf{f}_{FK} + \mathbf{f}_{drag}. \tag{1.62}$$

In this formulation, T_A already supplies the complete added-mass action associated with $C_A(\mathbf{a}_f - \ddot{\mathbf{r}})$. Together with the Froude–Krylov term having coefficient 1, it recovers $C_M = 1 + C_A$; consequently, $\mathbf{f}_{in}^M$ must no longer retain its $C_A\mathbf{a}_f$ term, and no additional $\mathbf{f}_{add}^M$ may be included. When $\mathbf{v}_f = \mathbf{0}$ and m_A is time independent, T_A reduces to an added kinetic energy that can be included in the conservative inertial backbone. For a general prescribed flow, its cross terms and explicit time dependence are treated collectively as a flow port.

For quadratic drag, define the nonnegative instantaneous state-dependent coefficient

$$d_f = \frac{1}{2}\rho_f C_D D \parallel \mathbf{u}_{r\perp} \parallel \geq 0. \tag{1.63}$$

The drag force must then be decomposed exclusively into a passive dissipative kernel and a background-flow input:

$$\boxed{\mathbf{f}_{drag} = d_f \mathbf{P}_\perp (\mathbf{v}_f - \dot{\mathbf{r}}) = \underbrace{-d_f \mathbf{P}_\perp \dot{\mathbf{r}}}_{\mathbf{f}_{drag,diss}} + \underbrace{d_f \mathbf{P}_\perp \mathbf{v}_f}_{\mathbf{f}_{drag,flow}}.} \tag{1.64}$$

Only the passive kernel has a definite dissipative sign with respect to the catheter mechanical energy:

$$\dot{\mathbf{r}}^T \mathbf{f}_{drag,diss} = -d_f \parallel \mathbf{P}_\perp \dot{\mathbf{r}} \parallel^2 \leq 0, \tag{1.65}$$

The term $\dot{\mathbf{r}}^T \mathbf{f}_{drag,flow}$ is generally sign-indefinite and is therefore assigned to the flow-input port. The complete drag force also satisfies the relative-motion passivity identity

$$(\dot{\mathbf{r}} - \mathbf{v}_f)^T \mathbf{f}_{drag} = -d_f \parallel \mathbf{P}_\perp (\dot{\mathbf{r}} - \mathbf{v}_f) \parallel^2 \leq 0, \tag{1.66}$$

This identity, however, does not replace the port accounting in the structural mechanical-energy balance. Thus, when $\mathbf{v}_f \neq \mathbf{0}$, the complete quadratic drag must not be assigned entirely to $\mathbf{Q}_{diss}$. Because d_f depends on the relative velocity, the passive kernel is a statewise positive-semidefinite nonlinear operator rather than a constant damping matrix.

If damping of cross-sectional spin by the fluid is relevant, an additional fluid couple per unit reference length may be introduced, for example,

$$\mathbf{l}_{hydro} = -c_\omega(S) \left|\omega_3 - \omega_{f3}\right| (\omega_3 - \omega_{f3}) \mathbf{d}_3, \tag{1.67}$$

where ω_{f3} is the effective fluid angular velocity about the catheter tangent. If this effect is negligible compared with the magnetic and structural torsional moments, one may set $\mathbf{l}_{hydro} = \mathbf{0}$ for the operating condition under consideration, while retaining the cross-sectional rotational degree of freedom.

The baseline computational route assumes a stationary vessel wall and Morison loading associated with normal cross-flow, neglecting axial tangential skin friction; the effective fluid spin about the catheter tangent is set to $\omega_{f3} = 0$. These are medium-fidelity reductions of the environmental loads and do not remove the structural variables describing cross-sectional

orientation, independent torsion, angular velocity, or rotary inertia. Moving walls, tangential friction, or fluid spin can be incorporated later through additional ports following the same input–dissipation accounting.

1.4.5. *Catheter–vessel-wall contact force and couple.* Let $g(S,t)$ be the gap function between the catheter surface and the vessel wall, and let g_0 be the contact-detection threshold. Define the penetration measure

$$\delta_c = \max\left(g_0 - g, 0\right). \tag{1.68}$$

To represent Hertzian elasticity, normal-impact dissipation, and regularized friction simultaneously, let $\dot{\delta}_c$ denote the material time derivative of the penetration at the contact point. When contact is active, $\delta_c > 0$, the normal force is taken as[14, 16]

$$f_n = k_c \delta_c^{\alpha} + c_n \dot{\delta}_c, \qquad \delta_c > 0. \tag{1.69}$$

This is nonlinear Hertz-type penalty contact with linear viscous normal damping. It is not the Hunt–Crossley penetration-dependent damping law $k\delta^{\alpha} + \chi\delta^{\alpha}\dot{\delta}$; Hunt–Crossley is cited only for contrast ([15]).

If the preceding expression is negative or if $\delta_c = 0$, set $f_n = 0$. This nonnegative truncation prevents nonphysical tensile contact forces during separation or noncontact. The tangential friction force and total contact force are, respectively,

$$\mathbf{f}_t = -\mu_c f_n \frac{\mathbf{v}_t}{\sqrt{\mathbf{v}_t \cdot \mathbf{v}_t + v_0^2}}, \tag{1.70}$$

$$\mathbf{f}_{cont} = f_n \mathbf{n}_w + \mathbf{f}_t. \tag{1.71}$$

Here, k_c is the contact stiffness, c_n is the normal contact-damping coefficient, α is the contact exponent, and Hertzian contact commonly uses $\alpha = 3/2$. The vector $\mathbf{n}_w$ is the unit normal directed away from the wall, $\mathbf{v}_t$ is the tangential velocity of the contact point relative to the wall, μ_c is the friction coefficient, and v_0 is a regularization parameter. For automatic differentiation or a consistent Newton iteration, the contact activation and nonnegative truncation may be replaced by smooth regularizations. For conservative contact, one may set $c_n = \mu_c = 0$, in which case the elastic contact term is derivable from a potential.

If the contact force acts at an offset vector $\boldsymbol{\rho}_c$ from the centerline, the contact couple is

$$\mathbf{l}_{cont} = \boldsymbol{\rho}_c \times \mathbf{f}_{cont} + \mathbf{l}_{spin}, \tag{1.72}$$

where $\mathbf{l}_{spin}$ may represent additional damping or frictional torque associated with cross-sectional spin in the contact region. Normal damping, tangential friction, and $\mathbf{l}_{spin}$ are all included in the nonconservative power.

1.4.6. *Microgravity term.* If a residual acceleration $\mathbf{g}_{res}(t)$ is present in the space station, the residual gravitational force per unit reference length is

$$\mathbf{f}_g = \mu \mathbf{g}_{res}. \tag{1.73}$$

Under the ideal microgravity approximation, set

$$\mathbf{g}_{res} = \mathbf{0}. \tag{1.74}$$

1.4.7. *Complete strong-form system.* Collecting the kinematics, balance laws, and constitutive equations gives the geometrically exact system for the base rod:

$$\begin{cases} \mathbf{r}_{,S} = \mathbf{R}\boldsymbol{\nu}, \\ \mathbf{R}_{,S} = \mathbf{R}\widehat{\boldsymbol{\kappa}}, \\ \mathbf{R}_{,t} = \mathbf{R}\widehat{\boldsymbol{\omega}}, \\ \mu\ddot{\mathbf{r}} = \mathbf{n}_{,S} + \mathbf{f}_{ext}, \\ \mathbf{m}_{,S} + \mathbf{r}_{,S} \times \mathbf{n} + \mathbf{l}_{ext} = \mathbf{R}\left[\mathbf{J}_c\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{J}_c\boldsymbol{\omega})\right], \\ \mathbf{n} = \mathbf{R}\mathbf{K}_\nu\left(\boldsymbol{\nu} - \boldsymbol{\nu}_0\right), \\ \mathbf{m} = \mathbf{R}\mathbf{K}_\kappa\left(\boldsymbol{\kappa} - \boldsymbol{\kappa}_0\right). \end{cases} \tag{1.75}$$

with the compatibility relations

$$\dot{\boldsymbol{\nu}} = \mathbf{R}^T\dot{\mathbf{r}}_{,S} - \boldsymbol{\omega} \times \boldsymbol{\nu}, \qquad \dot{\boldsymbol{\kappa}} = \boldsymbol{\omega}_{,S} + \boldsymbol{\kappa} \times \boldsymbol{\omega}. \tag{1.76}$$

Under the energy-consistent added-mass formulation, $\mathbf{I}_A$ is placed on the left-hand side together with the structural inertia, and only $\mathbf{f}_{FK}$, drag, and the remaining fluid ports are retained on the right-hand side. Under the engineering Morison validation branch, the complete combination $\mathbf{f}_{in}^M + \mathbf{f}_{add}^M$ is retained and T_A is omitted. Exactly one of these two representations must be used.

The primary unknowns are $\mathbf{r}, \mathbf{R}, \boldsymbol{\nu}, \boldsymbol{\kappa}, \boldsymbol{\omega}, \mathbf{n}, \mathbf{m}$. The system is a geometrically nonlinear mixed partial differential–algebraic system with Lie-group-valued orientation variables.

1.5. **Variational formulation and complete weak form.**

1.5.1. *Virtual displacement and virtual rotation.* Let $\delta\mathbf{r}$ be the virtual centerline displacement and represent the orientation variation by the material virtual rotation $\boldsymbol{\eta}$:

$$\delta\mathbf{R} = \mathbf{R}\widehat{\boldsymbol{\eta}}. \tag{1.77}$$

The corresponding spatial virtual rotation is

$$\boldsymbol{\eta}_s = \mathbf{R}\boldsymbol{\eta}. \tag{1.78}$$

The strain definitions give

$$\delta\boldsymbol{\nu} = \mathbf{R}^T\delta\mathbf{r}_{,S} - \boldsymbol{\eta} \times \boldsymbol{\nu}, \tag{1.79}$$

$$\delta\boldsymbol{\kappa} = \boldsymbol{\eta}_{,S} + \boldsymbol{\kappa} \times \boldsymbol{\eta}. \tag{1.80}$$

1.5.2. *Internal virtual work.* The internal virtual work is

$$\delta W_{int} = \int_0^L \left(\mathbf{n}_b \cdot \delta\boldsymbol{\nu} + \mathbf{m}_b \cdot \delta\boldsymbol{\kappa}\right) dS. \tag{1.81}$$

Substituting the strain variations and integrating the spatial derivatives by parts gives

$$\delta W_{int} = \left[\mathbf{n} \cdot \delta\mathbf{r} + \mathbf{m}_b \cdot \boldsymbol{\eta}\right]_0^L - I_r - I_R, \tag{1.82}$$

where

$$I_r = \int_0^L \mathbf{n}_{,S} \cdot \delta\mathbf{r}\, dS, \tag{1.83}$$

$$I_R = \int_0^L \left(\mathbf{m}_{b,S} + \boldsymbol{\kappa} \times \mathbf{m}_b + \boldsymbol{\nu} \times \mathbf{n}_b\right) \cdot \boldsymbol{\eta}\, dS. \tag{1.84}$$

The material-form rotational balance term satisfies

$$\mathbf{m}_{b,S} + \boldsymbol{\kappa} \times \mathbf{m}_b + \boldsymbol{\nu} \times \mathbf{n}_b + \mathbf{l}_b = \mathbf{J}_c\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times \left(\mathbf{J}_c\boldsymbol{\omega}\right), \tag{1.85}$$

and $\mathbf{l}_b = \mathbf{R}^T\mathbf{l}_{ext}$.

1.5.3. *Inertial and external virtual work.* The inertial virtual work of the base rod is

$$\delta W_{in,str} = \int_0^L \left[\mu \ddot{\mathbf{r}} \cdot \delta \mathbf{r} + (\mathbf{J}_c \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{J}_c \boldsymbol{\omega})) \cdot \boldsymbol{\eta}\right] dS. \tag{1.86}$$

If the energy-consistent added-mass formulation of Section 1.4.4 is used, the added-inertia virtual work is

$$\delta W_{in,A} = \left\langle \frac{d}{dt}\frac{\delta T_A}{\delta \dot{\mathbf{r}}} - \frac{\delta T_A}{\delta \mathbf{r}},\, \delta \mathbf{r} \right\rangle, \tag{1.87}$$

where $\langle \cdot, \cdot \rangle$ denotes the duality pairing along the rod. If eccentric attachments are present, their inertial virtual work is generated by the Euler–Lagrange derivative of the corresponding rigid-body kinetic energy T_e. The total inertial virtual work is therefore

$$\delta W_{in} = \delta W_{in,str} + \delta W_{in,A} + \delta W_{in,e}. \tag{1.88}$$

The virtual work of distributed external loads is

$$\delta W_{ext} = \int_0^L \left(\mathbf{f}_{ext,rem} \cdot \delta \mathbf{r} + \mathbf{l}_{ext} \cdot \boldsymbol{\eta}_s\right) dS + \delta W_{\partial}, \tag{1.89}$$

where $\mathbf{f}_{ext,rem}$ no longer contains the added inertia already represented by T_A. In the energy-consistent fluid branch, the fluid contribution contains only $\mathbf{f}_{FK}$, drag, and other inputs not represented by T_A. Because $\boldsymbol{\eta}_s = \mathbf{R}\boldsymbol{\eta}$, it may equivalently be written as

$$\mathbf{l}_{ext} \cdot \boldsymbol{\eta}_s = \mathbf{l}_b \cdot \boldsymbol{\eta}. \tag{1.90}$$

1.5.4. *Coupled weak form.* By D'Alembert's principle, every admissible pair $\delta\mathbf{r}$ and $\boldsymbol{\eta}$ satisfying the essential boundary conditions obeys

$$\delta W_{in} + \delta W_{int} - \delta W_{ext} = 0. \tag{1.91}$$

Expansion yields the complete coupled translational–rotational weak form:

$$\begin{aligned} &\textstyle\int_0^L \mu \ddot{\mathbf{r}} \cdot \delta \mathbf{r}\, dS + \int_0^L \left[\mathbf{J}_c \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{J}_c \boldsymbol{\omega})\right] \cdot \boldsymbol{\eta}\, dS \\ &+ \delta W_{in,A} + \delta W_{in,e} + \textstyle\int_0^L \left(\mathbf{n}_b \cdot \delta \boldsymbol{\nu} + \mathbf{m}_b \cdot \delta \boldsymbol{\kappa}\right) dS \\ &- \textstyle\int_0^L \left(\mathbf{f}_{ext,rem} \cdot \delta \mathbf{r} + \mathbf{l}_b \cdot \boldsymbol{\eta}\right) dS - \delta W_{\partial} = 0. \end{aligned} \tag{1.92}$$

If neither the added-fluid inertia functional nor eccentric attachments are included, setting $\delta W_{in,A} = \delta W_{in,e} = 0$ recovers the base-rod weak form. Because the weak form contains only first spatial derivatives of the centerline and orientation fields, C^0-continuous approximation spaces for both fields are sufficient; no fourth-order centerline equation need be constructed by eliminating the rotational equation.

In functional terms, the centerline and orientation may be viewed as mappings in $\mathbf{r} \in H^1\left(0, L; \mathbb{R}^3\right)$ and $\mathbf{R} \in H^1\left(0, L; SO(3)\right)$, respectively. The virtual displacement $\delta\mathbf{r}$ and material virtual rotation $\boldsymbol{\eta}$ belong to the corresponding H^1 test spaces satisfying homogeneous essential boundary conditions. Thus, the coupled weak form requires only first-order spatial regularity.

1.5.5. *Boundary terms and boundary conditions.* Integration by parts produces the natural boundary term

$$\left[\mathbf{n} \cdot \delta \mathbf{r} + \mathbf{m} \cdot \boldsymbol{\eta}_s\right]_0^L. \tag{1.93}$$

At an endpoint $S = S_e$, one of the following two classes of boundary conditions may be prescribed:

$$\mathbf{r}\left(S_e, t\right) = \overline{\mathbf{r}}(t) \quad \text{or} \quad \mathbf{n}\left(S_e, t\right) = \overline{\mathbf{n}}(t), \tag{1.94}$$

$$\mathbf{R}\,(S_e,t)=\overline{\mathbf{R}}(t)\quad\text{or}\quad\mathbf{m}\,(S_e,t)=\overline{\mathbf{m}}(t). \tag{1.95}$$

At a clamped end, $\delta\mathbf{r}=\mathbf{0}$ and $\boldsymbol{\eta}=\mathbf{0}$; at a free end without applied terminal loads, the natural boundary conditions are $\mathbf{n}=\mathbf{0}$ and $\mathbf{m}=\mathbf{0}$. Boundary forces and moments must remain in the weak form unless the boundary conditions have been specified explicitly.

The complete initial–boundary-value problem also requires the initial conditions

$$\mathbf{r}(S,0)=\mathbf{r}^0(S),\qquad\dot{\mathbf{r}}(S,0)=\mathbf{v}^0(S), \tag{1.96}$$

$$\mathbf{R}(S,0)=\mathbf{R}^0(S)\in SO(3),\qquad\boldsymbol{\omega}(S,0)=\boldsymbol{\omega}^0(S). \tag{1.97}$$

The initial fields must satisfy the geometric compatibility relations $\boldsymbol{\nu}^0=\left(\mathbf{R}^0\right)^T\mathbf{r}^0_{,S}$ and $\widehat{\boldsymbol{\kappa}}^0=\left(\mathbf{R}^0\right)^T\mathbf{R}^0_{,S}$, and they must be consistent with the essential boundary conditions. The strict Kirchhoff branch additionally requires the initial position constraint and its velocity-level compatibility condition.

1.6. **Kirchhoff shear-free constraint.** For a slender catheter, impose the extensible Kirchhoff shear-free constraint

$$\nu_1=0,\qquad\nu_2=0,\qquad\nu_3=\lambda. \tag{1.98}$$

which gives

$$\mathbf{r}_{,S}=\lambda\mathbf{d}_3,\qquad\mathbf{d}_3=\mathbf{t}. \tag{1.99}$$

This condition aligns the third director with the centerline tangent but does not determine the rotation of $\mathbf{d}_1,\mathbf{d}_2$ about $\mathbf{d}_3$. Consequently, κ_3, ω_3, GJ, the polar rotary inertia, and the magnetic couple must all be retained.

In the strong form, the transverse shear force is the reaction enforcing the shear-free constraint and does not vanish merely because $\nu_1=\nu_2=0$. The constraint may be imposed exactly with Lagrange multipliers or regularized by retaining a large but finite shear stiffness. In either case, the cross-sectional orientation field must remain in the model.[17, 18]

1.7. **Continuous energy balance.**

1.7.1. *Structural energy and extended inertial energy.* The mechanical energy of the base rod is defined as

$$E_{str}=T_{str}+V_{int}. \tag{1.100}$$

If eccentric attachments are present, add their kinetic energy T_e. In the energy-consistent added-mass branch with a stationary flow $\mathbf{v}_f=\mathbf{0}$ and time-independent m_A, define the total kinetic energy as

$$T_{tot}=T_{str}+T_A+T_e. \tag{1.101}$$

If magnetic loading, frictionless contact, or residual gravity is derivable from potentials, define

$$V=V_{int}+V_{mag}+V_{cont}+V_g, \tag{1.102}$$

and the extended mechanical energy

$$\boxed{E=T_{tot}+V.} \tag{1.103}$$

For a general prescribed flow, T_A may depend explicitly on time and contain cross terms involving relative velocity. It is then a velocity-dependent added-fluid inertia functional in the Lagrangian rather than the ordinary mechanical kinetic energy of a closed system. Its contribution must remain explicit as an independent flow port in the power balance; it does not imply conservation of the mechanical energy of the combined structure–prescribed-flow system.

1.7.2. *Power balance.* Taking the inner product of the translational equation with $\dot{\mathbf{r}}$, the rotational equation with the spatial angular velocity $\boldsymbol{\omega}_s$, and integrating along the rod gives

$$\frac{d}{dt}\left(T_{tot}+V_{int}\right)=P_{\partial}+P_{mag}+P_{flow}+P_{cont}+P_{g}, \tag{1.104}$$

where the boundary power is

$$P_{\partial}=\left[\mathbf{n}\cdot\dot{\mathbf{r}}+\mathbf{m}\cdot\boldsymbol{\omega}_s\right]_0^L. \tag{1.105}$$

After all conservative external loads have been incorporated into V, one obtains

$$\boxed{\frac{dE}{dt}=P_{\partial}+P_{diss}+P_{in}+P_{flow}+\frac{\partial V_{ext}}{\partial t}.} \tag{1.106}$$

Here, P_{diss} contains only the passive components obtained after the input–dissipation split, including $P_{drag,diss}$, normal contact damping and friction for a fixed wall, and any retained spin damping, and it satisfies $P_{diss}\leq 0$. The term P_{flow} collects the background-flow drag power $P_{drag,flow}$, the Froude–Krylov input, and reactive-flow power assigned to the port under the selected representation; it is generally sign-indefinite. The term P_{in} contains only nonfluid nonconservative body inputs and control inputs. Terminal actuation contributes only through P_{∂}. If a nonautonomous magnetic action has already been represented by $V_{mag}\left(\mathbf{r},\mathbf{R},t\right)$, its explicit-time power appears only through $\partial V_{ext}/\partial t$ and must not be counted again in P_{in}. The extended mechanical energy is conserved only when $P_{diss}=P_{in}=P_{flow}=P_{\partial}=0$ and the potential is time independent.

1.8. **Section summary.** With the centerline $\mathbf{r}$ and cross-sectional orientation $\mathbf{R}$ as independent primary fields, this section established a geometrically exact rod-dynamics model retaining extension, shear, bending about two axes, independent torsion, cross-sectional rotary inertia, magnetic couples, fluid loading, and contact couples. The base catheter uses a mass-centerline convention, whereas eccentric magnetic tips and sensors retain translational–rotational coupling through independent rigid-body kinetic energies. In the energy-consistent fluid representation, T_A generates the added inertia and the pure Froude–Krylov term supplies the remaining fluid-acceleration input; this branch is mutually exclusive with the engineering Morison representation, thereby avoiding double counting of $C_A\mathbf{a}_f$.

The complete weak form retains translational and rotational test functions as well as terminal force and moment terms. The passive kernel of quadratic drag, normal contact damping, and friction are assigned to dissipative power, while background-flow drag input and prescribed-flow inertia are assigned to an independent flow port. The Kirchhoff shear-free condition is enforced as a constraint rather than by deleting cross-sectional orientation, independent torsion, or rotary inertia. This coupled weak form provides the theoretical basis for the joint translational–rotational Galerkin discretization in Section 2.

2. Joint Translational–Rotational Galerkin Discretization

2.1. **Discretization principles.** Section 1 established the coupled weak form with $\mathbf{r}(S,t)$ and $\mathbf{R}(S,t)\in SO(3)$ as primary fields. We now reduce this model without discarding the rotational dynamics. The Galerkin projection acts jointly in the translational space and in a local coordinate chart of the rotation group.

The discretization follows five principles. First, the centerline position and cross-sectional orientation retain independent generalized coordinates. Second, rotation matrices cannot be superposed linearly as ordinary vectors and must be updated through the exponential map or an equivalent multiplicative rule. Third, the mass–inertia matrix must be derived from the complete three-dimensional material-point kinetic energy rather than assumed a priori to be constant or configuration dependent. Fourth, geometric inertial terms must follow rigorously from the Euler–Lagrange derivative of the discrete kinetic energy. Fifth, internal forces are obtained from the gradient of the discrete potential, while boundary forces and moments enter the generalized forces naturally through the weak form.[9, 17, 18]

2.2. **Joint finite-dimensional approximation.**

2.2.1. *Translational approximation.* Let $\mathbf{N}(S)$ be matrix-valued basis functions satisfying the essential translational boundary conditions. Approximate the centerline by

$$\mathbf{r}_h(S,t) = \mathbf{r}_0(S) + \mathbf{N}(S)\mathbf{a}(t), \tag{2.1}$$

where $\mathbf{a}(t) \in \mathbb{R}^{n_a}$ contains the translational generalized coordinates. Hence,

$$\dot{\mathbf{r}}_h = \mathbf{N}\dot{\mathbf{a}}, \qquad \mathbf{r}_{h,S} = \mathbf{r}_{0,S} + \mathbf{N}_{,S}\mathbf{a}. \tag{2.2}$$

If the essential boundary conditions are time dependent, a known lifting function may be added to the approximation; its first and second time derivatives are treated as prescribed generalized loads. To emphasize the structural formulation, the boundary conditions are assumed homogeneous below.

2.2.2. *Approximation of cross-sectional orientation.* Let $\mathbf{R}_0(S)$ be the reference orientation. In a neighborhood of the reference state, use exponential coordinates for the rotation vector:

$$\mathbf{R}_h(S,t) = \mathbf{R}_0(S) \exp\left(\widehat{\boldsymbol{\theta}_h(S,t)}\right), \tag{2.3}$$

$$\boldsymbol{\theta}_h(S,t) = \boldsymbol{\Psi}(S)\mathbf{b}(t), \tag{2.4}$$

where $\mathbf{b}(t) \in \mathbb{R}^{n_b}$ contains the rotational generalized coordinates and $\boldsymbol{\Psi}(S)$ is a matrix-valued basis satisfying the essential rotational boundary conditions. Define

$$\mathbf{E}_h = \exp\left(\widehat{\boldsymbol{\theta}_h}\right). \tag{2.5}$$

so that

$$\mathbf{R}_h = \mathbf{R}_0\mathbf{E}_h. \tag{2.6}$$

Exponential coordinates form a local chart on $SO(3)$. Provided that the rotational neighborhood does not cross the singular set of the exponential map, this parameterization is smooth and one-to-one and is suitable for finite-dimensional modal truncation and local nonlinear analysis. Global time integration through large rotations may instead employ chart switching or unit quaternions without changing the underlying mechanical derivation.

2.2.3. *Right Jacobian and angular velocity.* Let $\theta = \| \boldsymbol{\theta}_h \|$. The right Jacobian of the rotation vector is defined by

$$\mathbf{J}_r(\boldsymbol{\theta}_h) = \mathbf{I} - \frac{1-\cos\theta}{\theta^2}\widehat{\boldsymbol{\theta}_h} + \frac{\theta - \sin\theta}{\theta^3}\widehat{\boldsymbol{\theta}_h}^2. \tag{2.7}$$

As $\theta \to 0$, continuous extension gives

$$\mathbf{J}_r(\boldsymbol{\theta}_h) = \mathbf{I} - \frac{1}{2}\widehat{\boldsymbol{\theta}_h} + \frac{1}{6}\widehat{\boldsymbol{\theta}_h}^2 + O\left(\theta^3\right). \tag{2.8}$$

The exact differential relation of the exponential map gives the material angular velocity

$$\boldsymbol{\omega}_h = \mathbf{J}_r(\boldsymbol{\theta}_h)\,\dot{\boldsymbol{\theta}}_h = \mathbf{J}_r(\boldsymbol{\theta}_h)\,\boldsymbol{\Psi}\dot{\mathbf{b}}. \tag{2.9}$$

Define the angular-velocity matrix

$$\mathbf{B}_\omega(S,\mathbf{b}) = \mathbf{J}_r(\boldsymbol{\theta}_h)\,\boldsymbol{\Psi}(S), \tag{2.10}$$

Then

$$\boldsymbol{\omega}_h = \mathbf{B}_\omega(S,\mathbf{b})\,\dot{\mathbf{b}}. \tag{2.11}$$

Thus, $\boldsymbol{\omega}_h = \boldsymbol{\Psi}\dot{\mathbf{b}}$ cannot in general be assumed; that relation is only a zeroth-order small-rotation approximation.

2.2.4. *Joint generalized coordinates and virtual variables.* Define the complete generalized coordinate vector

$$\mathbf{q} = \begin{bmatrix} \mathbf{a} \\ \mathbf{b} \end{bmatrix} \in \mathbb{R}^n, \qquad n = n_a + n_b. \tag{2.12}$$

The translational virtual displacement is

$$\delta \mathbf{r}_h = \mathbf{N}\, \delta \mathbf{a}. \tag{2.13}$$

From $\delta \mathbf{R}_h = \mathbf{R}_h \widehat{\boldsymbol{\eta}_h}$, the material virtual rotation satisfies

$$\boldsymbol{\eta}_h = \mathbf{J}_r\left(\boldsymbol{\theta}_h\right) \delta \boldsymbol{\theta}_h = \mathbf{B}_\omega\left(S, \mathbf{b}\right) \delta \mathbf{b}. \tag{2.14}$$

Accordingly, the translational and rotational test functions are $\mathbf{N}$ and $\mathbf{B}_\omega$, respectively, and both enter the coupled Galerkin condition.

For $\mathbf{q} = \mathbf{0}$ to represent a stress-free reference configuration, the reference centerline, reference orientation, and reference strains must satisfy the compatibility relations

$$\boldsymbol{\nu}_0(S) = \mathbf{R}_0^T(S)\mathbf{r}_{0,S}(S), \tag{2.15}$$

$$\widehat{\boldsymbol{\kappa}}_0(S) = \mathbf{R}_0^T(S)\mathbf{R}_{0,S}(S). \tag{2.16}$$

If these relations are not satisfied, $\mathbf{q} = \mathbf{0}$ describes a pre-extended, pre-sheared, pre-bent, or pre-twisted configuration, with the corresponding initial force and moment generated automatically by $V_{int,h}$. The equilibrium state, tangent stiffness, and center of the KAM expansion must use the same reference convention.

2.3. **Discrete strains and potential energy.**

2.3.1. *Shear–extension strain.* The discrete shear–extension strain is

$$\boldsymbol{\nu}_h\left(\mathbf{q}\right) = \mathbf{R}_h^T \mathbf{r}_{h,S} = \mathbf{E}_h^T \mathbf{R}_0^T \left(\mathbf{r}_{0,S} + \mathbf{N}_{,S}\mathbf{a}\right). \tag{2.17}$$

Its virtual variation can be written as

$$\delta \boldsymbol{\nu}_h = \mathbf{R}_h^T \mathbf{N}_{,S} \delta \mathbf{a} - \boldsymbol{\eta}_h \times \boldsymbol{\nu}_h. \tag{2.18}$$

Thus, there exists a strain–generalized-coordinate matrix $\mathbf{B}_\nu\left(S, \mathbf{q}\right)$ such that

$$\delta \boldsymbol{\nu}_h = \mathbf{B}_\nu\left(S, \mathbf{q}\right) \delta \mathbf{q}. \tag{2.19}$$

Explicitly,

$$\mathbf{B}_\nu = \begin{bmatrix} \mathbf{R}_h^T \mathbf{N}_{,S} & \widehat{\boldsymbol{\nu}_h} \mathbf{B}_\omega \end{bmatrix}, \tag{2.20}$$

because $-\boldsymbol{\eta}_h \times \boldsymbol{\nu}_h = \widehat{\boldsymbol{\nu}_h}\boldsymbol{\eta}_h$.

2.3.2. *Curvature–torsion strain.* Define the curvature–torsion vector of the reference orientation by

$$\widehat{\boldsymbol{\kappa}_R} = \mathbf{R}_0^T \mathbf{R}_{0,S}. \tag{2.21}$$

From $\mathbf{R}_h = \mathbf{R}_0 \mathbf{E}_h$, one obtains

$$\mathbf{R}_h^T \mathbf{R}_{h,S} = \mathbf{E}_h^T \widehat{\boldsymbol{\kappa}_R} \mathbf{E}_h + \mathbf{E}_h^T \mathbf{E}_{h,S}. \tag{2.22}$$

Hence, the discrete curvature–torsion vector is

$$\boxed{\boldsymbol{\kappa}_h\left(\mathbf{q}\right) = \mathbf{E}_h^T \boldsymbol{\kappa}_R + \mathbf{J}_r\left(\boldsymbol{\theta}_h\right) \boldsymbol{\theta}_{h,S}} \tag{2.23}$$

where

$$\boldsymbol{\theta}_{h,S} = \boldsymbol{\Psi}_{,S} \mathbf{b}. \tag{2.24}$$

The corresponding virtual variation satisfies

$$\delta\boldsymbol{\kappa}_h = \boldsymbol{\eta}_{h,S} + \boldsymbol{\kappa}_h \times \boldsymbol{\eta}_h. \tag{2.25}$$

Therefore, there exists $\mathbf{B}_\kappa(S, \mathbf{q})$ such that

$$\delta\boldsymbol{\kappa}_h = \mathbf{B}_\kappa(S, \mathbf{q})\,\delta\mathbf{q}. \tag{2.26}$$

In assembly, $\mathbf{B}_\kappa$ may be obtained by analytic differentiation of the preceding expression or by evaluating the Jacobian of $\boldsymbol{\kappa}_h(\mathbf{q})$ through automatic differentiation. In either implementation, the independent torsional contribution κ_{h3} must be retained.

2.3.3. *Discrete internal potential energy.* Substituting the discrete strains into the potential energy of Section 1, the discrete internal potential energy can be decomposed as

$$V_{int,h}(\mathbf{q}) = V_{\nu,h}(\mathbf{q}) + V_{\kappa,h}(\mathbf{q}), \tag{2.27}$$

where

$$V_{\nu,h}(\mathbf{q}) = \frac{1}{2}\int_0^L (\boldsymbol{\nu}_h - \boldsymbol{\nu}_0)^T \mathbf{K}_\nu (\boldsymbol{\nu}_h - \boldsymbol{\nu}_0)\, dS, \tag{2.28}$$

$$V_{\kappa,h}(\mathbf{q}) = \frac{1}{2}\int_0^L (\boldsymbol{\kappa}_h - \boldsymbol{\kappa}_0)^T \mathbf{K}_\kappa (\boldsymbol{\kappa}_h - \boldsymbol{\kappa}_0)\, dS. \tag{2.29}$$

The discrete internal generalized force is

$$\mathbf{f}_{int}(\mathbf{q}) = \nabla_{\mathbf{q}} V_{int,h}(\mathbf{q}). \tag{2.30}$$

Using the virtual-work expression, it can be written as

$$\mathbf{f}_{int} = \int_0^L \left[\mathbf{B}_\nu^T \mathbf{n}_b + \mathbf{B}_\kappa^T \mathbf{m}_b\right] dS. \tag{2.31}$$

This expression contains the restoring forces associated with extension, shear, bending about two axes, and torsion. In general, $\mathbf{f}_{int}$ is a nonlinear vector-valued function of $\mathbf{q}$ and cannot be written a priori as a constant stiffness matrix multiplied by $\mathbf{q}$.

2.4. **Complete discrete kinetic energy and mass–inertia matrix.**

2.4.1. *Material-point velocity matrix.* The discrete material-point position is

$$\mathbf{x}_h(S, \boldsymbol{\xi}, t) = \mathbf{r}_h(S, t) + \mathbf{R}_h(S, t)\boldsymbol{\xi}. \tag{2.32}$$

and its velocity is

$$\dot{\mathbf{x}}_h = \mathbf{N}\dot{\mathbf{a}} - \mathbf{R}_h\widehat{\boldsymbol{\xi}}\mathbf{B}_\omega\dot{\mathbf{b}}. \tag{2.33}$$

Define the material-point velocity matrix

$$\mathbf{B}_x(S, \boldsymbol{\xi}, \mathbf{q}) = \begin{bmatrix}\mathbf{N} & -\mathbf{R}_h\widehat{\boldsymbol{\xi}}\mathbf{B}_\omega\end{bmatrix}, \tag{2.34}$$

Then

$$\dot{\mathbf{x}}_h = \mathbf{B}_x(S, \boldsymbol{\xi}, \mathbf{q})\,\dot{\mathbf{q}}. \tag{2.35}$$

2.4.2. *Rigorous definition of the mass–inertia matrix* $\mathbf{M}(\mathbf{q})$. The discrete kinetic energy is

$$T_h(\mathbf{q},\dot{\mathbf{q}}) = \frac{1}{2}\int_0^L \int_{A(S)} \rho \dot{\mathbf{x}}_h^T \dot{\mathbf{x}}_h \, dA\, dS. \tag{2.36}$$

Substitution of the velocity matrix gives

$$\boxed{T_h(\mathbf{q},\dot{\mathbf{q}}) = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q})\,\dot{\mathbf{q}}} \tag{2.37}$$

where

$$\boxed{\mathbf{M}(\mathbf{q}) = \int_0^L \int_{A(S)} \rho \mathbf{B}_x^T \mathbf{B}_x \, dA\, dS} \tag{2.38}$$

is the mass–inertia matrix derived rigorously from the complete three-dimensional translational–rotational kinetic energy.

2.4.3. *Block form of the mass–inertia matrix.* With $\mathbf{q} = \left(\mathbf{a}^T, \mathbf{b}^T\right)^T$, write

$$\mathbf{M}(\mathbf{q}) = \begin{bmatrix} \mathbf{M}_{aa} & \mathbf{M}_{ab}(\mathbf{q}) \\ \mathbf{M}_{ba}(\mathbf{q}) & \mathbf{M}_{bb}(\mathbf{q}) \end{bmatrix}. \tag{2.39}$$

Using $\mu, \mathbf{c}_\rho, \mathbf{J}_c$ defined in Section 1, the blocks are

$$\mathbf{M}_{aa} = \int_0^L \mu \mathbf{N}^T \mathbf{N}\, dS, \tag{2.40}$$

$$\mathbf{M}_{ab}(\mathbf{q}) = -\int_0^L \mathbf{N}^T \mathbf{R}_h \widehat{\mathbf{c}}_\rho \mathbf{B}_\omega \, dS, \tag{2.41}$$

$$\mathbf{M}_{ba}(\mathbf{q}) = \mathbf{M}_{ab}^T(\mathbf{q}), \tag{2.42}$$

$$\boxed{\mathbf{M}_{bb}(\mathbf{q}) = \int_0^L \mathbf{B}_\omega^T \mathbf{J}_c \mathbf{B}_\omega \, dS} \tag{2.43}$$

Thus,

$$\mathbf{M}_{bb}(\mathbf{q}) = \int_0^L \mathbf{\Psi}^T \mathbf{J}_r^T(\boldsymbol{\theta}_h)\, \mathbf{J}_c \mathbf{J}_r(\boldsymbol{\theta}_h)\, \mathbf{\Psi}\, dS. \tag{2.44}$$

Whether the mass matrix depends on the configuration is therefore not an a priori assumption; it is determined jointly by the centerline definition, orientation parameterization, cross-sectional mass distribution, and attached components.

2.4.4. *Mass centerline and eccentric attachments.* Consistently with Section 1, the centerline of the base catheter is taken to pass through the mass center of each cross section, so the distributed rod satisfies

$$\mathbf{c}_\rho = \mathbf{0}, \tag{2.45}$$

and the translational–rotational cross block of the base rod vanishes:

$$\mathbf{M}_{ab} = \mathbf{M}_{ba} = \mathbf{0}. \tag{2.46}$$

In this case,

$$\mathbf{M}_{str}(\mathbf{q}) = \begin{bmatrix} \mathbf{M}_{aa} & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_{bb}(\mathbf{b}) \end{bmatrix}. \tag{2.47}$$

However, $\mathbf{M}_{bb}$ still depends on the rotational generalized coordinates through $\mathbf{J}_r(\boldsymbol{\theta}_h)$, and the complete structural mass–inertia matrix therefore generally remains $\mathbf{M}_{str}(\mathbf{q})$.

If the center of mass of a magnetic tip, sensor, or other attached component is offset from the centerline, its kinetic energy must be added separately. Let the center-of-mass position of the attachment be

$$\mathbf{x}_e = \mathbf{r}_h\left(S_e\right) + \mathbf{R}_h\left(S_e\right)\boldsymbol{\rho}_e, \tag{2.48}$$

If its mass is m_e and its material-frame rotary inertia is $\mathbf{J}_e$, then

$$T_e = \frac{1}{2} m_e \left|\dot{\mathbf{x}}_e\right|^2 + \frac{1}{2}\boldsymbol{\omega}_e^T \mathbf{J}_e \boldsymbol{\omega}_e. \tag{2.49}$$

Let $\mathbf{B}_e\left(\mathbf{q}\right)$ map $\dot{\mathbf{q}}$ to the attachment velocity. Then

$$T_e = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}_e\left(\mathbf{q}\right)\dot{\mathbf{q}}, \qquad \mathbf{M}_e = \mathbf{B}_e^T \mathbf{M}_e^{rig} \mathbf{B}_e. \tag{2.50}$$

This contribution generally produces a nonzero translational–rotational coupling block. Thus, use of the mass centerline for the base rod does not imply that the cross-mass block of the complete system must vanish.

2.4.5. *Scope of the small-rotation constant-mass approximation.* When $\| \boldsymbol{\theta}_h \| \ll 1$, if only

$$\mathbf{J}_r\left(\boldsymbol{\theta}_h\right) \approx \mathbf{I}, \tag{2.51}$$

is retained, then

$$\mathbf{M}_{bb} \approx \int_0^L \boldsymbol{\Psi}^T \mathbf{J}_c \boldsymbol{\Psi}\, dS, \tag{2.52}$$

becomes constant. This is a zeroth-order small-rotation inertial approximation rather than a general property of the complete model. The principal model retains the exact $\mathbf{J}_r$ and $\mathbf{M}\left(\mathbf{q}\right)$.

2.4.6. *Energy-consistent discretization of fluid added mass.* Define the joint translational velocity matrix

$$\mathbf{N}_q(S) = \begin{bmatrix}\mathbf{N}(S) & \mathbf{0}\end{bmatrix}, \tag{2.53}$$

such that $\dot{\mathbf{r}}_h = \mathbf{N}_q\dot{\mathbf{q}}$. Let

$$\mathbf{w}_{\perp,h} = \mathbf{P}_\perp\left(\mathbf{q}\right)\left[\mathbf{N}_q\dot{\mathbf{q}} - \mathbf{v}_f\left(\mathbf{r}_h, t\right)\right], \qquad m_A = \rho_f C_A A_d. \tag{2.54}$$

The added-fluid inertia functional of Section 1 is discretized as follows (and becomes an added kinetic energy for a stationary flow):

$$\boxed{T_{A,h} = \frac{1}{2}\int_0^L m_A\, \mathbf{w}_{\perp,h}^T \mathbf{w}_{\perp,h}\, dS.} \tag{2.55}$$

Expansion gives

$$T_{A,h} = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}_A\left(\mathbf{q}\right)\dot{\mathbf{q}} - \dot{\mathbf{q}}^T \mathbf{b}_A\left(\mathbf{q}, t\right) + c_A\left(\mathbf{q}, t\right), \tag{2.56}$$

where

$$\boxed{\mathbf{M}_A\left(\mathbf{q}\right) = \int_0^L m_A\, \mathbf{N}_q^T \mathbf{P}_\perp\left(\mathbf{q}\right)\mathbf{N}_q\, dS,} \tag{2.57}$$

$$\mathbf{b}_A\left(\mathbf{q}, t\right) = \int_0^L m_A\, \mathbf{N}_q^T \mathbf{P}_\perp\left(\mathbf{q}\right)\mathbf{v}_f\, dS, \tag{2.58}$$

$$c_A\left(\mathbf{q}, t\right) = \frac{1}{2}\int_0^L m_A\, \mathbf{v}_f^T \mathbf{P}_\perp\left(\mathbf{q}\right)\mathbf{v}_f\, dS. \tag{2.59}$$

The structural kinetic energy, eccentric-attachment kinetic energy, and added-fluid contribution are therefore collected into the velocity-dependent inertial functional

$$T_h = T_{str,h} + \sum_e T_e + T_{A,h}. \tag{2.60}$$

Its Hessian with respect to the generalized velocity gives the total effective mass–inertia matrix

$$\boxed{\mathbf{M}(\mathbf{q}) = \mathbf{M}_{str}(\mathbf{q}) + \sum_e \mathbf{M}_e(\mathbf{q}) + \mathbf{M}_A(\mathbf{q}).} \tag{2.61}$$

Only when $\mathbf{v}_f = \mathbf{0}$ is it physically valid to set $\mathbf{b}_A = \mathbf{0}$ and $c_A = 0$. For a general prescribed flow, moving the cross terms to an external port to construct an autonomous conservative backbone merely removes $\mathbf{b}_A$ and c_A from the selected Hamiltonian backbone and assigns their complete Euler–Lagrange action to $\mathbf{Q}_{flow}$; it does not assume that these quantities vanish in the original system. For a general flow, one cannot retain only $\mathbf{M}_A\dot{\mathbf{q}}$. The constant added mass obtained by freezing $\mathbf{P}_\perp$ is used only as an engineering validation branch.

2.4.7. *Symmetry and positive definiteness.* By definition of the kinetic-energy Hessian,

$$\mathbf{M}^T(\mathbf{q}) = \mathbf{M}(\mathbf{q}). \tag{2.62}$$

If the material and added-mass parameters are positive, the orientation coordinates remain in a local chart where $\mathbf{J}_r$ is nonsingular, and the joint velocity map does not degenerate almost everywhere for any nonzero generalized velocity, namely,

$$\dot{\mathbf{q}} \neq \mathbf{0} \quad \Rightarrow \quad T_h^{(2)}(\mathbf{q}, \dot{\mathbf{q}}) > 0, \tag{2.63}$$

then

$$\mathbf{y}^T \mathbf{M}(\mathbf{q})\, \mathbf{y} > 0, \qquad \forall \mathbf{y} \neq \mathbf{0}. \tag{2.64}$$

Therefore, $\mathbf{M}(\mathbf{q})$ is symmetric positive definite. What must be excluded is a redundant generalized coordinate or a degenerate velocity map, not a rigid-body mode under free boundary conditions. A rigid-body mode may give a zero eigenvalue of the stiffness matrix, while a valid rigid-body motion still has positive kinetic energy.

2.5. **Joint Galerkin projection.**

2.5.1. *Inertial term.* Substituting the finite-dimensional approximation into the total inertial virtual work of Section 1 gives

$$\delta W_{in,h} = \delta \mathbf{q}^T \left[\frac{d}{dt}\left(\frac{\partial T_h}{\partial \dot{\mathbf{q}}} \right) - \frac{\partial T_h}{\partial \mathbf{q}} \right]. \tag{2.65}$$

For a stationary flow, or for an autonomous conservative inertial backbone selected after assigning the complete Euler–Lagrange action of the prescribed-flow cross terms to $\mathbf{Q}_{flow}$,

$$T_h = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q})\, \dot{\mathbf{q}}, \tag{2.66}$$

the inertial generalized force can be arranged as

$$\mathbf{M}(\mathbf{q})\, \ddot{\mathbf{q}} + \mathbf{g}_M(\mathbf{q}, \dot{\mathbf{q}}). \tag{2.67}$$

For a general prescribed flow, T_h also contains $-\dot{\mathbf{q}}^T \mathbf{b}_A + c_A$, whose additional contribution is denoted by $\mathbf{g}_f(\mathbf{q}, \dot{\mathbf{q}}, t)$. The unified form is therefore

$$\boxed{\mathbf{M}(\mathbf{q})\, \ddot{\mathbf{q}} + \mathbf{g}_T(\mathbf{q}, \dot{\mathbf{q}}, t), \qquad \mathbf{g}_T = \mathbf{g}_M + \mathbf{g}_f.} \tag{2.68}$$

The terms are derived rigorously in Section 2.6.

2.5.2. *Internal-force term.* The discrete internal virtual work is

$$\delta W_{int,h} = \delta \mathbf{q}^T \int_0^L \left[\mathbf{B}_\nu^T \mathbf{n}_b + \mathbf{B}_\kappa^T \mathbf{m}_b \right] dS. \tag{2.69}$$

Therefore,

$$\mathbf{f}_{int}(\mathbf{q}) = \int_0^L \left[\mathbf{B}_\nu^T \mathbf{n}_b + \mathbf{B}_\kappa^T \mathbf{m}_b \right] dS = \nabla_{\mathbf{q}} V_{int,h}(\mathbf{q}). \tag{2.70}$$

2.5.3. *Joint projection of distributed forces and couples.* First, decompose the distributed loads that remain in the external-force ports through mutually exclusive accounting:

$$\mathbf{f}_{ext,rem} = \mathbf{f}_{cons} + \mathbf{f}_{nc}, \qquad \mathbf{l}_{ext} = \mathbf{l}_{cons} + \mathbf{l}_{nc}. \tag{2.71}$$

The conservative part is represented through an external potential in Section 2.5.5; only the nonconservative part is projected directly onto the right-hand side. A spatially distributed couple is first transformed to the material frame, $\mathbf{l}_{nc}^b = \mathbf{R}_h^T \mathbf{l}_{nc}$, and the nonconservative distributed generalized force is

$$\boxed{\mathbf{Q}_{nc,dist}(\mathbf{q}, \dot{\mathbf{q}}, t) = \int_0^L \begin{bmatrix} \mathbf{N}^T \mathbf{f}_{nc} \\ \mathbf{B}_\omega^T \mathbf{R}_h^T \mathbf{l}_{nc} \end{bmatrix} dS.} \tag{2.72}$$

This expression allows magnetic, contact, and fluid couples to enter the rotational generalized coordinates directly when they belong to nonconservative ports. If a load has already been represented by a potential, it must not be included again in $\mathbf{Q}_{nc,dist}$.

For normal quadratic drag acting only on the centerline translational velocity, define the joint velocity matrix

$$\boxed{\mathbf{N}_q = [\, \mathbf{N} \;\; \mathbf{0} \,] \in \mathbb{R}^{3 \times n}, \qquad \dot{\mathbf{r}}_h = \mathbf{N}_q \dot{\mathbf{q}}.} \tag{2.73}$$

Its consistent generalized-force projection is

$$\mathbf{Q}_{drag} = \int_0^L d_f \mathbf{N}_q^T \mathbf{P}_\perp (\mathbf{v}_f - \mathbf{N}_q \dot{\mathbf{q}}) \, dS = -\mathbf{D}_{drag}(\mathbf{q}, \dot{\mathbf{q}}, t) \, \dot{\mathbf{q}} + \mathbf{Q}_{drag,flow}(\mathbf{q}, \dot{\mathbf{q}}, t), \tag{2.74}$$

where

$$\boxed{\mathbf{D}_{drag} = \int_0^L d_f \mathbf{N}_q^T \mathbf{P}_\perp \mathbf{N}_q \, dS, \qquad \mathbf{Q}_{drag,flow} = \int_0^L d_f \mathbf{N}_q^T \mathbf{P}_\perp \mathbf{v}_f \, dS.} \tag{2.75}$$

For every $\mathbf{y} \in \mathbb{R}^n$,

$$\mathbf{y}^T \mathbf{D}_{drag} \mathbf{y} = \int_0^L d_f \parallel \mathbf{P}_\perp \mathbf{N}_q \mathbf{y} \parallel^2 \, dS \geq 0, \tag{2.76}$$

Hence, $\mathbf{D}_{drag} = \mathbf{D}_{drag}^T \succcurlyeq 0$. The matrix $\mathbf{N}_q^T$, rather than $\mathbf{N}^T$, must be used so that $\mathbf{Q}_{drag} \in \mathbb{R}^n$ and $\mathbf{D}_{drag} \in \mathbb{R}^{n \times n}$. Because $\mathbf{N}_q = [\mathbf{N} \; \mathbf{0}]$, the ab, ba, and bb blocks associated with centerline drag vanish. Eccentric fluid couples and cross-sectional spin damping still enter the orientation channels through independent distributed-couple projections. Since d_f depends on the relative velocity, $\mathbf{D}_{drag}$ is a statewise positive-semidefinite nonlinear matrix.

2.5.4. *Boundary generalized force.* The boundary virtual work in Section 1 is

$$[\mathbf{n} \cdot \delta\mathbf{r} + \mathbf{m}_b \cdot \boldsymbol{\eta}]_0^L . \tag{2.77}$$

Substitution of the finite-dimensional virtual variables gives

$$\boxed{\mathbf{Q}_\partial (\mathbf{q}, t) = \begin{bmatrix} \mathbf{N}^T \overline{\mathbf{n}} \\ \mathbf{B}_\omega^T (\mathbf{q}) \, \overline{\mathbf{m}}_b \end{bmatrix}_0^L .} \tag{2.78}$$

The boundary generalized force generally depends on the configuration because the rotational test matrix $\mathbf{B}_\omega$ and the transformation from a spatial moment to a material moment may both vary with orientation. At an essential boundary, the corresponding test function vanishes and the boundary term disappears automatically. At a natural boundary, the prescribed terminal force and moment must be retained.

2.5.5. *Conservative and nonconservative external loads.* If the conservative distributed and boundary loads admit a discrete external potential $V_{ext,h} (\mathbf{q}, t)$, their total conservative generalized force is

$$\mathbf{Q}_{cons} = -\nabla_{\mathbf{q}} V_{ext,h}, \tag{2.79}$$

and define

$$V_h (\mathbf{q}, t) = V_{int,h} (\mathbf{q}) + V_{ext,h} (\mathbf{q}, t) . \tag{2.80}$$

The nonconservative generalized force is denoted collectively by

$$\mathbf{Q}_{nc} = \mathbf{Q}_{nc,dist} + \mathbf{Q}_{control} + \mathbf{Q}_{disturbance} + \mathbf{Q}_{other,nc}. \tag{2.81}$$

The final equations use only $\nabla_{\mathbf{q}} V_h$ on the left-hand side and $\mathbf{Q}_{nc}$ on the right-hand side. The original load projection containing conservative components must not be added again in its entirety to the right-hand side. If a time-dependent magnetic action has already been included in $V_{ext,h} (\mathbf{q}, t)$, its explicit-time power is represented only by $\partial V_{ext,h} / \partial t$.

2.6. **Rigorous derivation of generalized momentum and geometric inertial terms.**

2.6.1. *Generalized momentum.* For the velocity-dependent inertial functional in Section 2.4.6,

$$T_h = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M} (\mathbf{q}) \, \dot{\mathbf{q}} - \dot{\mathbf{q}}^T \mathbf{b}_A (\mathbf{q}, t) + c_A (\mathbf{q}, t) , \tag{2.82}$$

define the generalized momentum

$$\boxed{\mathbf{p} = \frac{\partial T_h}{\partial \dot{\mathbf{q}}} = \mathbf{M} (\mathbf{q}) \, \dot{\mathbf{q}} - \mathbf{b}_A (\mathbf{q}, t) .} \tag{2.83}$$

For a general prescribed flow, one therefore cannot unconditionally write $\mathbf{p} = \mathbf{M}\dot{\mathbf{q}}$. The backbone momentum is defined only for a stationary flow or after the complete action of $\mathbf{b}_A$ has been moved to $\mathbf{Q}_{flow}$ in a deliberately selected autonomous conservative backbone:

$$\mathbf{p}_0 = \mathbf{M} (\mathbf{q}) \, \dot{\mathbf{q}}. \tag{2.84}$$

2.6.2. *Geometric inertia generated by the mass metric.* Taking the Euler–Lagrange derivative of the purely quadratic velocity term gives

$$\mathbf{M}(\mathbf{q})\,\ddot{\mathbf{q}} + \mathbf{g}_M(\mathbf{q},\dot{\mathbf{q}}), \tag{2.85}$$

where

$$\boxed{\mathbf{g}_M = \dot{\mathbf{M}}\dot{\mathbf{q}} - \frac{1}{2}\nabla_{\mathbf{q}}\left(\dot{\mathbf{q}}^T\mathbf{M}\dot{\mathbf{q}}\right).} \tag{2.86}$$

Define

$$\Gamma_{ijk} = \frac{1}{2}\left(\frac{\partial M_{ij}}{\partial q_k} + \frac{\partial M_{ik}}{\partial q_j} - \frac{\partial M_{jk}}{\partial q_i}\right), \tag{2.87}$$

Then

$$g_{M,i} = \sum_{j,k=1}^{n} \Gamma_{ijk}\dot{q}_j\dot{q}_k. \tag{2.88}$$

Let $C_{ij} = \sum_k \Gamma_{ijk}\dot{q}_k$. Then $\mathbf{g}_M = \mathbf{C}\dot{\mathbf{q}}$, and $\dot{\mathbf{M}} - 2\mathbf{C}$ is skew symmetric, which guarantees the kinetic-energy power identity.

2.6.3. *Additional action generated by prescribed-flow cross terms.* The term $-\dot{\mathbf{q}}^T\mathbf{b}_A + c_A$ gives

$$\boxed{\mathbf{g}_f = -\dot{\mathbf{b}}_A + (\nabla_{\mathbf{q}}\mathbf{b}_A)^T\,\dot{\mathbf{q}} - \nabla_{\mathbf{q}}c_A.} \tag{2.89}$$

Here, $\dot{\mathbf{b}}_A$ denotes the total derivative along the trajectory. The total inertial action is therefore

$$\boxed{\frac{d}{dt}\frac{\partial T_h}{\partial \dot{\mathbf{q}}} - \frac{\partial T_h}{\partial \mathbf{q}} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}_M + \mathbf{g}_f.} \tag{2.90}$$

Equivalently, $\mathbf{g}_f$ may be moved to the right-hand side and denoted as a flow input $\mathbf{Q}_{flow}$, but the entire term must be moved; retaining only selected components is inconsistent.

2.6.4. *Constant-mass special case.* If, under a specific small-rotation, zero-eccentricity, fixed-normal-projection, and fixed-added-mass approximation,

$$\frac{\partial \mathbf{M}}{\partial q_i} = \mathbf{0}, \qquad i = 1,\ldots,n, \tag{2.91}$$

then $\mathbf{g}_M = \mathbf{0}$. If $\mathbf{b}_A = \mathbf{0}$ as well, all geometric inertial terms vanish. This conclusion applies only to the corresponding local approximation and must not be used in reverse to deny configuration-dependent inertia in the complete model.

2.7. **Discrete treatment of the Kirchhoff shear-free constraint.**

2.7.1. *Discrete constraints.* The strict Kirchhoff model requires[17, 18]

$$\nu_{h1}(\mathbf{q}) = 0, \qquad \nu_{h2}(\mathbf{q}) = 0. \tag{2.92}$$

Collect all discrete constraints in

$$\mathbf{c}(\mathbf{q}) = \mathbf{0}. \tag{2.93}$$

Define the constraint Jacobian

$$\mathbf{G}(\mathbf{q}) = \frac{\partial \mathbf{c}}{\partial \mathbf{q}}. \tag{2.94}$$

2.7.2. *Lagrange-multiplier form.* Introduce the Lagrange multiplier $\boldsymbol{\lambda}_c$. The generalized constraint reaction is

$$\mathbf{Q}_c = -\mathbf{G}^T(\mathbf{q})\,\boldsymbol{\lambda}_c. \tag{2.95}$$

Equivalently, one may define $\mathbf{Q}_c = +\mathbf{G}^T\boldsymbol{\lambda}_c$ and move the multiplier term to the opposite side; the present manuscript uses the former convention consistently.

The complete constrained discrete system is

$$\begin{cases} \mathbf{M}(\mathbf{q})\,\ddot{\mathbf{q}} + \mathbf{g}(\mathbf{q},\dot{\mathbf{q}}) + \nabla_{\mathbf{q}} V_h(\mathbf{q},t) + \mathbf{G}^T(\mathbf{q})\,\boldsymbol{\lambda}_c = \mathbf{Q}_{nc}(\mathbf{q},\dot{\mathbf{q}},t), \\ \mathbf{c}(\mathbf{q}) = \mathbf{0}. \end{cases} \tag{2.96}$$

This is a nonlinear differential–algebraic system. The constraint eliminates shear deformation only; it does not remove rotational generalized coordinates or torsional modes.

2.7.3. *Finite-shear-stiffness form.* If the fully shearable Cosserat rod is used, or if the Kirchhoff constraint is approximated by a large but finite shear stiffness, no Lagrange multiplier is required because the shear energy is already included in $V_{int,h}$. The resulting system is an ordinary second-order nonlinear ODE. The two treatments have different numerical properties but share the same complete translational–rotational discretization.

2.8. **Global finite-dimensional equations of motion.**

2.8.1. *Form without explicit constraints.* Assembling the inertial term, internal conservative force, conservative external potential, nonconservative distributed loads, and boundary loads gives

$$\boxed{\mathbf{M}(\mathbf{q})\,\ddot{\mathbf{q}} + \mathbf{g}_T(\mathbf{q},\dot{\mathbf{q}},t) + \nabla_{\mathbf{q}} V_h(\mathbf{q},t) = \mathbf{Q}_{nc}(\mathbf{q},\dot{\mathbf{q}},t) + \mathbf{Q}_{\partial}(\mathbf{q},t).} \tag{2.97}$$

Here, $\mathbf{q}$ contains both translational and rotational generalized coordinates; $\mathbf{M}$ is generated by the velocity-quadratic terms of the base rod, eccentric attachments, and the energy-consistent added mass; $\mathbf{g}_T = \mathbf{g}_M + \mathbf{g}_f$; and V_h contains extension, shear, bending, torsion, and external potentials satisfying the potential condition.

If the prescribed-flow cross terms are moved collectively to the input port and dissipation is separated from control inputs, the system becomes

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}_M + \nabla_{\mathbf{q}} V_h = \mathbf{Q}_{diss} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_{\partial}. \tag{2.98}$$

The passively dissipative drag kernel obtained by exclusive splitting, normal contact damping and friction for a fixed wall, and any retained spin damping constitute $\mathbf{Q}_{diss}$, with $\dot{\mathbf{q}}^T\mathbf{Q}_{diss} \le 0$. The background-flow drag term $\mathbf{Q}_{drag,flow}$ must be assigned to $\mathbf{Q}_{flow}$, and its power is generally sign-indefinite; therefore, no global dissipativity claim may be made for the complete relative drag. Nor should $\mathbf{Q}_{diss}$ be forced into the form of a constant damping matrix times velocity, because the quadratic-drag matrix $\mathbf{D}_{drag}(\mathbf{q},\dot{\mathbf{q}},t)$ itself depends on state and relative velocity. The same fluid action may be counted only once in any selected representation.

2.8.2. *Constrained form.* The strict Kirchhoff shear-free model is finally written as

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}_T + \nabla_{\mathbf{q}} V_h + \mathbf{G}^T(\mathbf{q})\,\boldsymbol{\lambda}_c = \mathbf{Q}_{nc} + \mathbf{Q}_{\partial}, \tag{2.99}$$

$$\mathbf{c}(\mathbf{q}) = \mathbf{0}. \tag{2.100}$$

The system retains complete bending–torsion dynamics, cross-sectional orientation, and rotary inertia while enforcing the shear-free condition exactly.

2.9. **Discrete energy and power identities.**

2.9.1. *Discrete mechanical energy of the conservative inertial backbone.* For a stationary flow with $\mathbf{b}_A = \mathbf{0}$ and $c_A = 0$, or for an autonomous conservative backbone selected after the prescribed-flow cross terms have been moved completely to $\mathbf{Q}_{flow}$, define

$$E_h(\mathbf{q}, \dot{\mathbf{q}}, t) = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q})\,\dot{\mathbf{q}} + V_h(\mathbf{q}, t). \tag{2.101}$$

Because $\mathbf{g}_M$ is generated by the Euler–Lagrange derivative of the complete quadratic kinetic energy,

$$\dot{\mathbf{q}}^T [\mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}_M] = \frac{d}{dt}\left(\frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}\dot{\mathbf{q}}\right). \tag{2.102}$$

2.9.2. *General port-power balance.* Left-multiplication of the global equation by $\dot{\mathbf{q}}^T$ gives

$$\frac{dE_h}{dt} = \dot{\mathbf{q}}^T (\mathbf{Q}_{nc} + \mathbf{Q}_\partial - \mathbf{g}_f) + \frac{\partial V_h}{\partial t}. \tag{2.103}$$

If the complete Euler–Lagrange contribution of the prescribed-flow cross action is treated as part of the flow input, then

$$\boxed{\frac{dE_h}{dt} = \dot{\mathbf{q}}^T (\mathbf{Q}_{diss} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_\partial) + \frac{\partial V_h}{\partial t}.} \tag{2.104}$$

Here, $\mathbf{Q}_{diss}$ contains only the passive dissipative kernel after splitting and satisfies $\dot{\mathbf{q}}^T \mathbf{Q}_{diss} \leq 0$. The term $\mathbf{Q}_{flow}$ collects $\mathbf{Q}_{drag,flow}$, the Froude–Krylov input, and the reactive-flow contribution determined by the selected representation; its power is not assigned a fixed sign. The relative-motion identity verifies physical passivity of the drag law, whereas the dissipative structural constraint of the CHNN must be imposed on the preceding $\mathbf{Q}_{diss}$; the two must not be conflated. For an ideal constraint,

$$\mathbf{G}(\mathbf{q})\,\dot{\mathbf{q}} = \mathbf{0}, \tag{2.105}$$

Therefore,

$$\dot{\mathbf{q}}^T \mathbf{G}^T \boldsymbol{\lambda}_c = 0. \tag{2.106}$$

Only in the absence of nonconservative loads, flow inputs, boundary-power inputs, and explicit time dependence of the potential does $dE_h/dt = 0$.

2.9.3. *Consistent tangent matrix.* For nonlinear solution or local linearization, the consistent internal tangent matrix is defined by

$$\mathbf{K}_T(\mathbf{q}) = \frac{\partial \mathbf{f}_{int}}{\partial \mathbf{q}} = \nabla^2_{\mathbf{q}} V_{int,h}(\mathbf{q}). \tag{2.107}$$

This matrix contains both material stiffness and the geometric stiffness generated by variations in orientation, extension, and curvature. It is not the secant matrix obtained by multiplying a state-dependent scalar stiffness by an integral of fixed basis functions.

2.10. **Local linearization and engineering reductions.**

2.10.1. *Local linearization about an equilibrium configuration.* Use the dissipation–input decomposition

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}_M + \nabla V_h + \mathbf{G}^T \boldsymbol{\lambda}_c = \mathbf{Q}_{diss} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_\partial. \tag{2.108}$$

Let $\mathbf{q}_e$ be a static equilibrium configuration with $\dot{\mathbf{q}}_e = \mathbf{0}$. Then

Here, a "static equilibrium configuration" requires the explicit time-dependent input, control parameters, and prescribed flow to be held fixed at $t = t_e$, or that the system itself is autonomous. If the magnetic field or flow is periodically driven, a Floquet linearization about a periodic orbit must be used instead of a static equilibrium matrix.

$$\nabla V_h(\mathbf{q}_e) + \mathbf{G}^T(\mathbf{q}_e)\boldsymbol{\lambda}_{ce} = \mathbf{Q}_{diss}(\mathbf{q}_e, \mathbf{0}) + \mathbf{Q}_{in,e} + \mathbf{Q}_{flow,e} + \mathbf{Q}_{\partial,e}. \tag{2.109}$$

Usually, $\mathbf{Q}_{diss}(\mathbf{q}, \mathbf{0}) = \mathbf{0}$. Let $\mathbf{q} = \mathbf{q}_e + \boldsymbol{\eta}$ and neglect second- and higher-order perturbation terms. For the unconstrained system,

$$\mathbf{M}_e\ddot{\boldsymbol{\eta}} + \mathbf{C}_e\dot{\boldsymbol{\eta}} + \mathbf{K}_e\boldsymbol{\eta} = \delta\mathbf{Q}(t), \tag{2.110}$$

where

$$\mathbf{M}_e = \mathbf{M}(\mathbf{q}_e), \tag{2.111}$$

$$\mathbf{C}_e = -\left.\frac{\partial \mathbf{Q}_{diss}}{\partial \dot{\mathbf{q}}}\right|_e, \tag{2.112}$$

$$\boxed{\mathbf{K}_e = \left.\left[\nabla^2_{\mathbf{q}} V_h - \frac{\partial \mathbf{Q}_{diss}}{\partial \mathbf{q}} - \frac{\partial(\mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_{\partial})}{\partial \mathbf{q}}\right]\right|_e.} \tag{2.113}$$

Here, $\delta\mathbf{Q}(t)$ collects explicit time-dependent input perturbations not absorbed by the state Jacobian. Under the actual dissipative-force convention above, passive dissipation gives $\mathbf{C}_e \succcurlyeq \mathbf{0}$ near the equilibrium. If the load Jacobian contains nonsymmetric follower terms, $\mathbf{K}_e$ need not be symmetric.

If contact is active near the equilibrium, the preceding Jacobian and consistent tangent assume that the contact active set remains fixed. If the equilibrium lies on a contact opening/closing switching surface, a smooth contact regularization, semismooth Newton method, or piecewise linearization must be used; the max function cannot be treated as differentiable everywhere.

For a purely conservative system without boundary follower loads, the expression reduces directly to

$$\boxed{\mathbf{K}_e = \nabla^2_{\mathbf{q}} V_h(\mathbf{q}_e),} \tag{2.114}$$

No derivative of a conservative generalized force already included in V_h is subtracted again. If strict constraints are present, the system must be projected onto the constraint tangent space or first expressed in independent local coordinates satisfying $\mathbf{c}(\mathbf{q}) = 0$. The tangent stiffness at a constrained equilibrium must also include the second-order contribution of the constraint Lagrangian. A constant-matrix linear system is only a local approximation of the complete coupled nonlinear model near the specified equilibrium configuration.[17, 18]

2.10.2. *Scope of engineering reductions.* Small rotations, constant mass, a fixed normal projection, and low-order modal truncation are used only for numerical checks, parameter identification, or ablation experiments. The Euler–Bernoulli branch has two roles: theoretically, it is the Kirchhoff shear-free reduction of the geometrically exact parent model; within COMSOL, it is the Kirchhoff-consistent engineering realization used for the primary training data. It may be used as the primary training branch only after the slenderness ratio, the fraction of shear energy, and the comparison error relative to Timoshenko/solid models satisfy prescribed tolerances. The corotational Beam geometric-nonlinearity formulation in COMSOL permits large rotations but assumes locally small element strains; the mesh must therefore keep the rotation increment between adjacent nodes sufficiently small[4, 5].

2.10.3. *Modal truncation and selection of the principal computational order.* The CHNN and KAM analyses must use a finite-dimensional system of fixed dimension and cannot remain indefinitely at an abstract modal order n. For candidate truncation orders $n = 1, 2, \ldots, n_{max}$, compute at least the leading natural frequencies, terminal displacement, structural energy, torsional response, and principal port powers. Define the adjacent-order convergence indicators

$$e_n^{freq} = \max_{1\le j\le m} \frac{\left|\omega_j^{(n+1)} - \omega_j^{(n)}\right|}{\left|\omega_j^{(n+1)}\right|}, \qquad e_n^{traj} = \frac{\| \mathbf{q}^{(n+1)} - \Pi_{n+1\to n}\mathbf{q}^{(n+1)} \|}{\| \mathbf{q}^{(n+1)} \|}, \tag{2.115}$$

and define analogous indicators for energy and torsion. The principal computational order n_* is the first order at which all prescribed indicators simultaneously fall below their tolerances. Until this computation has been completed, the manuscript may retain n_* only as an undetermined parameter and must not describe an empirically selected order as proven sufficient.

The canonical state mapping is examined using a fixed $N = 2$ basis. The basis is extracted from the mesh-80 near-frequency cluster, orthonormalized with respect to the consistent finite-element mass metric, and fixed in sign, normalization, and degree-of-freedom ordering. It defines the paired canonical coordinates used in the P1 series and in the subsequent consistency analysis. The full COMSOL–Galerkin mapping still retains the first 12 production modes and the mesh-40/80 comparison, so $N = 2$ is a diagnostic choice rather than a replacement for the convergence criterion for n_* or a claim of high-fidelity modal sufficiency.

2.11. **Numerical implementation and finite-dimensional model outputs.**

2.11.1. *COMSOL model hierarchy, projection, and validation.* The COMSOL data use a three-level hierarchy: "primary training–strict reproduction–high-fidelity comparison.

1. **Kirchhoff-consistent primary training model.** Use the Euler–Bernoulli formulation of the three-dimensional Beam interface with Time Dependent dynamics and geometric nonlinearity, retaining axial deformation, bending about two axes, three-dimensional cross-sectional rotations, and torsion. The three-dimensional Euler–Bernoulli section stiffness in COMSOL includes axial force, torque, and two bending moments; shear-free therefore does not mean torsion-free[4, 6]. This model generates data under assumptions consistent with the strict Kirchhoff branch.
2. **Strict weak-form reproduction model.** When a term-by-term check of the coupled translational–orientation weak form, nonstandard constraints, or custom distributed couples is required, Weak Form PDE or Weak Contribution is used to specify test functions, boundary terms, and auxiliary variables[7, 8]. This branch checks discrepancies between the built-in Beam interface and the present equations; it is not the default model for every operating condition.
3. **High-fidelity comparison model.** A Timoshenko Beam model quantifies corrections from transverse shear and rotary inertia, while three-dimensional Solid Mechanics checks local cross-sectional stresses, contact regions, and attachment effects. These models output X_{HF}. Their discrepancy from the Kirchhoff-consistent model X_K is used for validation and model-discrepancy learning rather than directly replacing the primary KAM branch.

Every model should output the centerline position $\mathbf{r}_C(S,t)$, cross-sectional orientation $\mathbf{R}_C(S,t)$ or an equivalent rotation representation, centerline velocity, cross-sectional angular velocity, resultant section forces and moments, structural kinetic energy, and strain energy. The primary training branch should additionally output the separate powers associated with magnetic, fluid, contact, and boundary actions.

Define the translational Gram matrix

$$\mathbf{G}_a = \int_0^L \mathbf{N}^T \mathbf{W}_a \mathbf{N}\, dS, \tag{2.116}$$

Then the projection of the COMSOL centerline field is

$$\mathbf{a}(t) = \mathbf{G}_a^{-1} \int_0^L \mathbf{N}^T \mathbf{W}_a \left[\mathbf{r}_C(S,t) - \mathbf{r}_0(S)\right] dS. \tag{2.117}$$

Orientation must not be projected linearly by its Euler-angle components. First compute the logarithmic coordinates of the relative rotation

$$\boldsymbol{\theta}_C(S,t) = \log\left(\mathbf{R}_0^T(S)\mathbf{R}_C(S,t)\right)^\vee, \tag{2.118}$$

and then set

$$\mathbf{G}_b = \int_0^L \boldsymbol{\Psi}^T \mathbf{W}_b \boldsymbol{\Psi}\, dS, \qquad \mathbf{b}(t) = \mathbf{G}_b^{-1} \int_0^L \boldsymbol{\Psi}^T \mathbf{W}_b \boldsymbol{\theta}_C(S,t)\, dS. \tag{2.119}$$

This gives $\mathbf{q} = \left(\mathbf{a}^T, \mathbf{b}^T\right)^T$. The angular velocity should be checked through $\boldsymbol{\omega}_h = \mathbf{B}_\omega(\mathbf{q})\,\dot{\mathbf{b}}$, rather than by setting $\boldsymbol{\omega} = \boldsymbol{\Psi}\dot{\mathbf{b}}$.

The primary learning branch uses the fixed four-mode basis and constant reduced mass matrix described in Section 6.1. This finite-dimensional representation is distinct from the configuration-dependent full nonlinear cotangent pullback: it defines the canonical momentum directly by $p = M_4\dot{q}$ and does not require unobserved full-order transport fields. Modal-order sensitivity is assessed separately on nested auxiliary data in Section 6.5.

The fluid-loading implementation must match the selected theoretical branch. The theoretical route uses T_A (or its discrete equivalent mass and geometric terms), the pure Froude–Krylov term, and drag. The engineering Morison comparison uses the complete $C_M\mathbf{a}_f - C_A\ddot{\mathbf{r}}$ form but must not activate T_A simultaneously. The same physical action may be counted only once in any data sample.

2.11.2. *Finite-dimensional state, energy, port, and structure-learning outputs.* After the joint Galerkin discretization and the consistent Legendre postprocessing of Section 3, the numerical implementation should output unified finite-dimensional states, energies, ports, and representation labels:

$$\mathcal{O}_h = \left(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}, \mathbf{p}_{rep}, \dot{\mathbf{p}}_{rep}, \mathbf{M}(\mathbf{q}), V_h, E_h, \mathbf{Q}_{diss}, \mathbf{Q}_{in}, \mathbf{Q}_{flow}, \mathbf{Q}_\partial, \mathbf{c}, \mathbf{G}, representation\right). \tag{2.120}$$

Here, $\mathbf{p}_{rep}$ and $\dot{\mathbf{p}}_{rep}$ must be consistent with the selected Lagrangian representation. The *representation* label must distinguish at least the autonomous-backbone, exact-nonautonomous, and engineering-reduction representations. The strict Kirchhoff branch also stores the discrete constraint $\mathbf{c}(\mathbf{q})$ and its Jacobian $\mathbf{G}(\mathbf{q})$; these two components may be marked as empty in an unconstrained branch.

On this basis, Section 3 constructs two data objects that must not be mixed. The autonomous-backbone data are

$$\mathcal{D}_{sk} = \left(\mathbf{z}_0, \dot{\mathbf{z}}_0, H_{cons}, \nabla_{\mathbf{z}_0} H_{cons}, \mathbf{Q}_{diss}, \mathbf{Q}_{in}, \mathbf{Q}_{flow}^{sk}, \mathbf{Q}_\partial\right), \qquad \mathbf{z}_0 = (\mathbf{q}, \mathbf{p}_0), \tag{2.121}$$

and the exact nonautonomous data are

$$\mathcal{D}_{na} = \left(\mathbf{z}, \dot{\mathbf{z}}, t, H, \nabla_{\mathbf{z}} H, \partial_t H, \mathbf{Q}_{diss}, \mathbf{Q}_{in}, \mathbf{Q}_{flow}^{H}, \mathbf{Q}_\partial\right), \qquad \mathbf{z} = (\mathbf{q}, \mathbf{p}). \tag{2.122}$$

If $\dot{\mathbf{p}}_{rep}$ or $\dot{\mathbf{z}}_{rep}$ is obtained by numerical differentiation rather than from a solver residual, analytic equations, or automatic differentiation, the differentiation order, time step, and derivative-error estimate must also be recorded. CHNN derivative supervision, energy-balance supervision, and the subsequent KAM error budget may use only representation-consistent data with traceable errors.

2.11.3. *Training-data hierarchy and model labels.* Each sample must store

$$model_level \in \{K, WF, T, S\}, \qquad representation \in \{sk, na, red\}, \tag{2.123}$$

Here, K denotes the Kirchhoff-consistent Beam primary training model, WF the strict weak-form model, T the Timoshenko comparison, and S the three-dimensional solid comparison; sk denotes the autonomous-backbone representation, na the exact nonautonomous representation, and red an engineering reduction.

The data set used for the conservative CHNN and the statistical-learning–KAM interface is restricted to

$$\mathcal{D}_K^{cons} = \{(\mathbf{z}_0, \dot{\mathbf{z}}_0, H_{cons}, \nabla H_{cons}) : model_level = K,\ representation = sk,\ \mathbf{Q}_{port} = 0\}, \tag{2.124}$$

The open-port training set is

$$\mathcal{D}_K^{port} = \{(\mathbf{z}_{rep}, \dot{\mathbf{z}}_{rep}, \mathbf{Q}_{diss}, \mathbf{Q}_{in}, \mathbf{Q}_{flow}, \mathbf{Q}_{\partial}) : model_level = K\}, \tag{2.125}$$

The high-fidelity comparison set is

$$\mathcal{D}_{HF} = \{X_{HF} - X_K : model_level \in \{WF, T, S\}\}. \tag{2.126}$$

Only after closure, power-sign, and regularity tests may $\mathcal{D}_{HF}$ enter a conservative correction, dissipative correction, or general residual, respectively; the entire high-fidelity discrepancy must not be interpreted as a Hamiltonian perturbation. Samples with different *model_level* and *representation* labels must not be mixed directly without labels and pairing.

2.12. **Section summary.** Starting directly from the complete translational–rotational weak form, this section constructed a joint Galerkin approximation using a centerline expansion and an exponential-map representation of cross-sectional orientation. The compatibility conditions for the reference configuration and strains, the initial conditions, the contact active set, and the mutually exclusive accounting of conservative and nonconservative loads were specified explicitly. The base rod, eccentric attachments, and energy-consistent fluid added mass were collected in a velocity-dependent inertial functional, whose Hessian with respect to generalized velocity gives

$$\mathbf{M}(\mathbf{q}) = \mathbf{M}_{str}(\mathbf{q}) + \sum_e \mathbf{M}_e(\mathbf{q}) + \mathbf{M}_A(\mathbf{q}). \tag{2.127}$$

For a general prescribed flow, the generalized momentum is rigorously

$$\mathbf{p} = \mathbf{M}(\mathbf{q})\,\dot{\mathbf{q}} - \mathbf{b}_A(\mathbf{q}, t), \tag{2.128}$$

Both the geometric inertia and the prescribed-flow cross terms are generated by the complete Euler–Lagrange derivative rather than added empirically. The internal restoring force is obtained from the gradient of the full extension–shear–bending–torsion potential, while magnetic and other distributed couples are projected directly through the rotational test functions. The local linearization eliminates double counting of conservative external loads and explicitly includes boundary follower-load and constraint-tangent contributions.

The resulting finite-dimensional nonlinear dynamical systems are provided in unconstrained and Kirchhoff-constrained forms, together with a projection from COMSOL full-order fields to the joint generalized coordinates and a unified output specification. The energy-consistent fluid branch uses $T_A + \mathbf{f}_{FK} + \mathbf{f}_{drag}$, with the drag further split, under a general background flow, into $\mathbf{f}_{drag,diss} + \mathbf{f}_{drag,flow}$; the engineering Morison formulation is used only as a mutually exclusive validation branch. Small rotations, constant mass, and fixed added mass are retained only as reduced approximations. Euler–Bernoulli serves as the COMSOL Kirchhoff-consistent primary training branch after the slenderness and shear-error tolerances are satisfied. These discrete objects support not only the canonical momentum and Hamiltonian construction of Section 3, but also provide the sole upstream interface to the CHNN training in Section 4 and the structure-preservation and KAM validation in Section 5 through $\mathcal{D}_K^{cons}$, $\mathcal{D}_K^{port}$, $\mathcal{D}_{HF}$, port labels, model-level labels, and constraint labels.

3. Finite-Dimensional Hamiltonian Structure and Canonicalization

The Galerkin system now provides a finite-dimensional configuration, but learning and symplectic analysis require a compatible phase-space representation. This section derives the momentum, Hamiltonian, and port structure for both the nonlinear material-reference formulation and the fixed-basis approximation used in the experiments.

3.1. **Finite-Dimensional Phase-Space Variables and Hamiltonian Formulation.**

3.1.1. *Joint Generalized Coordinates.* The joint Galerkin projection of centerline position and cross-sectional orientation developed in Section 2 yields

$$\boxed{\mathbf{q} = \begin{bmatrix} \mathbf{a} \\ \mathbf{b} \end{bmatrix} \in \mathbb{R}^n, \qquad n = n_a + n_b,} \tag{3.1}$$

where $\mathbf{a}$ denotes the generalized coordinates of centerline translation and $\mathbf{b}$ contains the finite-dimensional coefficients of the local rotational coordinates for cross-sectional orientation. The vector $\mathbf{b}$ is an independent configuration variable describing the bending orientation of the cross section, torsion about the tangent, angular velocity, and rotary inertia; it is not an auxiliary variable derived from the centerline displacement. Thus, $\mathbf{q}$ is the complete joint generalized configuration coordinate for translation and cross-sectional orientation.

The discrete dynamics of the finite-shear-stiffness geometrically exact Cosserat parent model can be written as

$$\boxed{\mathbf{M}(\mathbf{q})\,\ddot{\mathbf{q}} + \mathbf{g}_M(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{g}_f(\mathbf{q}, \dot{\mathbf{q}}, t) + \nabla_{\mathbf{q}} V_h(\mathbf{q}, t) = \mathbf{Q}_{port}.} \tag{3.2}$$

The mass–inertia matrix

$$\mathbf{M}(\mathbf{q}) = \mathbf{M}_{str}(\mathbf{q}) + \sum_e \mathbf{M}_e(\mathbf{q}) + \mathbf{M}_A(\mathbf{q}) \tag{3.3}$$

contains the translational kinetic energy of the rod, cross-sectional rotational kinetic energy, translational–rotational coupling of eccentric attachments, and energy-consistent added mass. The Hamiltonian structure developed below is therefore always based on the complete bending–torsion model; it does not use a constant-mass, small-rotation, centerline-only, or torsion-deleted surrogate.

3.1.2. *Basic Framework of the Hamiltonian Formulation.* The finite-dimensional Hamiltonian formulation follows the chain[17, 29]

$$\mathbf{q} \to \mathbf{p} \to H(\mathbf{q}, \mathbf{p}) \to \begin{cases} \dot{\mathbf{q}} = \nabla_{\mathbf{p}} H, \\ \dot{\mathbf{p}} = -\nabla_{\mathbf{q}} H + \mathbf{Q}, \end{cases} \to \frac{dH}{dt}. \tag{3.4}$$

This derivation applies to the autonomous conservative backbone and its forced–dissipative extension. A general prescribed flow generates a velocity-linear term in the Lagrangian and consequently shifts the canonical momentum; Section 3.5 gives the complete nonautonomous Hamiltonian formulation.

3.2. **Lagrangian, Momentum, and Hamiltonian of the Autonomous Conservative Backbone.**

3.2.1. *Selection of the Autonomous Potential.* Let the external actuation be controlled by parameters $\boldsymbol{\lambda}(t)$, and choose a reference value $\bar{\boldsymbol{\lambda}}$. Write the discrete potential as

$$V_h(\mathbf{q}, t; \boldsymbol{\lambda}(t)) = V_{cons}\left(\mathbf{q}; \bar{\boldsymbol{\lambda}}\right) + V_{drv}\left(\mathbf{q}, t; \boldsymbol{\lambda}(t), \bar{\boldsymbol{\lambda}}\right), \tag{3.5}$$

and impose

$$V_{cons}\left(\mathbf{q}; \bar{\boldsymbol{\lambda}}\right) := V_h(\mathbf{q}, t; \boldsymbol{\lambda}(t))\,|_{\boldsymbol{\lambda}(t) \equiv \bar{\boldsymbol{\lambda}}}, \qquad V_{drv} := V_h - V_{cons}. \tag{3.6}$$

When the reference parameters are held fixed, no explicit time-dependent term remains; hence

$$\boxed{V_{drv}\left(\mathbf{q}, t; \bar{\boldsymbol{\lambda}}, \bar{\boldsymbol{\lambda}}\right) = 0, \qquad \partial_t V_{cons} = 0.} \tag{3.7}$$

The potential V_{cons} contains the discrete Cosserat strain energy and the conservative magnetic potential, elastic contact potential, and other conservative external potentials selected at the

reference operating condition. This definition holds the actuation parameters at their reference values rather than selecting an arbitrary time slice.

3.2.2. *Autonomous-Backbone Lagrangian and Generalized Momentum.* For the autonomous port-closed backbone, the Lagrangian is

$$\boxed{L_{cons}\left(\mathbf{q},\dot{\mathbf{q}}\right)=\frac{1}{2}\dot{\mathbf{q}}^{T}\mathbf{M}\left(\mathbf{q}\right)\dot{\mathbf{q}}-V_{cons}\left(\mathbf{q}\right).} \tag{3.8}$$

The canonical momentum conjugate to $\mathbf{q}$ is defined by[17, 29]

$$\mathbf{p}_0=\frac{\partial L_{cons}}{\partial\dot{\mathbf{q}}}, \tag{3.9}$$

which gives the relation between canonical momentum and generalized velocity

$$\boxed{\mathbf{p}_0=\mathbf{M}\left(\mathbf{q}\right)\dot{\mathbf{q}},\qquad\dot{\mathbf{q}}=\mathbf{M}^{-1}\left(\mathbf{q}\right)\mathbf{p}_0.} \tag{3.10}$$

The subscript 0 distinguishes this quantity from the shifted canonical momentum $\mathbf{p}$ arising under a general prescribed flow in Section 3.5. The $\mathbf{p}_0$ orientation component $\mathbf{p}_{0b}$ is the discrete momentum conjugate to the local coordinate $\mathbf{b}$; it is not, in general, equal to a component of the material angular momentum of any individual cross section.

3.2.3. *Legendre Regularity.* The velocity Hessian is[17, 29]

$$\frac{\partial^2 L_{cons}}{\partial\dot{\mathbf{q}}\,\partial\dot{\mathbf{q}}}=\mathbf{M}\left(\mathbf{q}\right). \tag{3.11}$$

If the joint basis contains no velocity redundancy, the material density, rotary inertia, and added-mass parameters satisfy the required physical positivity conditions, and throughout the configuration domain of interest

$$\boxed{\mathbf{y}^{T}\mathbf{M}\left(\mathbf{q}\right)\mathbf{y}>0\quad\left(\forall\mathbf{y}\neq\mathbf{0}\right),} \tag{3.12}$$

then $\mathbf{M}\left(\mathbf{q}\right)=\mathbf{M}^{T}\left(\mathbf{q}\right)\succ 0$, and the Legendre map is locally invertible. Geometrically, $\mathbf{M}$ is the mass metric on the tangent space of configuration space, while $\mathbf{M}^{-1}$ is the corresponding dual metric on the cotangent space.

For the fixed N-dimensional reduced manifold $\mathcal{Q}_N$, the configuration coordinate satisfies $q_N\in\mathcal{Q}_N$, the generalized velocity satisfies $\dot{q}_N\in T_{q_N}\mathcal{Q}_N$, and the conjugate momentum satisfies $p_N\in T^*_{q_N}\mathcal{Q}_N$. In the same paired-coordinate trivialization, $\dot{p}_N$ is stored as the momentum-rate component paired with p_N; comparisons across charts additionally require the associated connection or coordinate transformation. The full nonlinear material-reference formulation admits the configuration-dependent cotangent pullback

$$\boxed{\begin{aligned}&q_N=\Pi_{\mathrm{conf},N}(U),\qquad p_N=B_N(q_N)^{T}P_{\mathrm{full}},\\&\dot{q}_N=M_N(q_N)^{-1}p_N,\quad \dot{p}_N=D_qB_N(q_N)[\dot{q}_N]^{T}P_{\mathrm{full}}+B_N(q_N)^{T}\dot{P}_{\mathrm{full}}=:L_N^{\mathrm{can}}.\end{aligned}} \tag{3.13}$$

Here, $B_N(q_N)$ is the tangent lift of the translational–$SO(3)$ basis at the current configuration, P_{full} is the full-order momentum, and $M_N=B_N^{T}EB_N$. When B_N depends on configuration, the first contribution to L_N^{can} is the basis-transport (connection) term. The learning experiments instead use a fixed-basis finite-dimensional canonical approximation with constant M_N, for which $p_N=M_N\dot{q}_N$. The primary dataset therefore does not instantiate the full nonlinear cotangent-pullback map in Equation (3.13); this map remains the theoretical full nonlinear generalization.

3.2.4. *Construction of the Hamiltonian.* The Hamiltonian is defined as

$$H_{cons}\left(\mathbf{q},\mathbf{p}_0\right)=\mathbf{p}_0^T\dot{\mathbf{q}}-L_{cons}\left(\mathbf{q},\dot{\mathbf{q}}\right). \tag{3.14}$$

Substituting $\dot{\mathbf{q}}=\mathbf{M}^{-1}\mathbf{p}_0$ gives

$$\boxed{H_{cons}\left(\mathbf{q},\mathbf{p}_0\right)=\frac{1}{2}\mathbf{p}_0^T\mathbf{M}^{-1}\left(\mathbf{q}\right)\mathbf{p}_0+V_{cons}\left(\mathbf{q}\right).} \tag{3.15}$$

This Hamiltonian has the standard form $H=T+V$ and contains both translational and cross-sectional rotational degrees of freedom. Configuration dependence of the mass matrix does not invalidate the Legendre transformation; its contribution enters the geometric inertial terms through the derivative of the Hamiltonian with respect to $\mathbf{q}$.

3.2.5. *Nonlinear Potential and Restoring Force.* The discrete internal potential of the Cosserat rod is determined by the strain-energy functional in Section 2. Define

$$\boxed{\mathbf{f}_{int}\left(\mathbf{q}\right)=\nabla_{\mathbf{q}}V_{int,h}\left(\mathbf{q}\right),\qquad\mathbf{K}_{tan}\left(\mathbf{q}\right)=\frac{\partial\mathbf{f}_{int}}{\partial\mathbf{q}}=\nabla_{\mathbf{q}}^2V_{int,h}\left(\mathbf{q}\right).} \tag{3.16}$$

Relative to a reference configuration $\mathbf{q}_*$, the conservative potential may also be written as the path-independent line integral

$$V_{int,h}\left(\mathbf{q}\right)-V_{int,h}\left(\mathbf{q}_*\right)=\int_0^1\mathbf{f}_{int}\left(\mathbf{q}_*+s\left(\mathbf{q}-\mathbf{q}_*\right)\right)^T\left(\mathbf{q}-\mathbf{q}_*\right)\,ds. \tag{3.17}$$

Thus, the conservative potential can be represented by the line integral of the restoring force along a path in configuration space. In general, one cannot assume $\mathbf{f}_{int}=\mathbf{K}_{tan}\left(\mathbf{q}\right)\mathbf{q}$. A quadratic approximation is justified only in a linearized neighborhood or when the restoring force genuinely admits that factorization:

$$V_{int,h}\left(\mathbf{q}_*+\boldsymbol{\eta}\right)\approx V_{int,h}\left(\mathbf{q}_*\right)+\frac{1}{2}\boldsymbol{\eta}^T\mathbf{K}_{tan}\left(\mathbf{q}_*\right)\boldsymbol{\eta}. \tag{3.18}$$

A global quadratic form is generally invalid because both the potential and the restoring force are geometrically nonlinear.

3.3. Canonical Hamilton Equations and Geometric Inertia.

3.3.1. *Standard Symplectic Structure.* Define the phase-space variable and the standard symplectic matrix

$$\mathbf{z}_0=\begin{bmatrix}\mathbf{q}\\\mathbf{p}_0\end{bmatrix},\qquad\mathbf{J}=\begin{bmatrix}\mathbf{0}&\mathbf{I}\\-\mathbf{I}&\mathbf{0}\end{bmatrix},\qquad\mathbf{J}^T=-\mathbf{J}. \tag{3.19}$$

The autonomous conservative backbone satisfies

$$\boxed{\dot{\mathbf{z}}_0=\mathbf{J}\nabla_{\mathbf{z}_0}H_{cons},} \tag{3.20}$$

that is,

$$\boxed{\dot{\mathbf{q}}=\nabla_{\mathbf{p}_0}H_{cons},\qquad\dot{\mathbf{p}}_0=-\nabla_{\mathbf{q}}H_{cons}.} \tag{3.21}$$

The corresponding standard Poisson bracket is

$$\{F,G\}=\nabla_{\mathbf{q}}F^T\nabla_{\mathbf{p}_0}G-\nabla_{\mathbf{p}_0}F^T\nabla_{\mathbf{q}}G. \tag{3.22}$$

Because local orientation coordinates and their conjugate momenta are used, the standard canonical bracket applies. If global rotation matrices and material angular momenta were used instead, a Lie–Poisson or constrained symplectic structure would have to be constructed separately.

3.3.2. *Terms Generated by the Configuration-Dependent Mass Matrix.* For the i-th configuration component,

$$\frac{\partial H_{cons}}{\partial q_i} = \frac{1}{2}\mathbf{p}_0^T \frac{\partial \mathbf{M}^{-1}}{\partial q_i}\mathbf{p}_0 + \frac{\partial V_{cons}}{\partial q_i}. \tag{3.23}$$

Therefore, the momentum equation is

$$\boxed{\dot{p}_{0i} = -\frac{1}{2}\mathbf{p}_0^T \frac{\partial \mathbf{M}^{-1}}{\partial q_i}\mathbf{p}_0 - \frac{\partial V_{cons}}{\partial q_i}.} \tag{3.24}$$

Using the derivative identity for an inverse matrix,

$$\frac{\partial \mathbf{M}^{-1}}{\partial q_i} = -\mathbf{M}^{-1}\mathbf{M}_{,i}\mathbf{M}^{-1} \tag{3.25}$$

one obtains

$$-\frac{1}{2}\mathbf{p}_0^T \frac{\partial \mathbf{M}^{-1}}{\partial q_i}\mathbf{p}_0 = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}_{,i}\dot{\mathbf{q}}. \tag{3.26}$$

This term fully represents the contribution of the configuration-dependent mass matrix in the Hamiltonian momentum equation. Only when the momentum equation is converted back to a second-order acceleration equation must it be combined with $\dot{\mathbf{p}}_0$, specifically with the contribution $\dot{\mathbf{M}}\dot{\mathbf{q}}$ contained therein.

3.3.3. *Equivalence with the Euler–Lagrange Equations.* From

$$\dot{\mathbf{p}}_0 = \mathbf{M}\ddot{\mathbf{q}} + \dot{\mathbf{M}}\dot{\mathbf{q}} \tag{3.27}$$

and the preceding Hamiltonian momentum equation, one obtains

$$\boxed{\mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}_M(\mathbf{q}, \dot{\mathbf{q}}) + \nabla_{\mathbf{q}} V_{cons}(\mathbf{q}) = \mathbf{0},} \tag{3.28}$$

where

$$\boxed{\mathbf{g}_M = \dot{\mathbf{M}}\dot{\mathbf{q}} - \frac{1}{2}\nabla_{\mathbf{q}}\left(\dot{\mathbf{q}}^T \mathbf{M}\dot{\mathbf{q}}\right).} \tag{3.29}$$

This result agrees with the autonomous conservative Euler–Lagrange equation in Section 2, establishing the equivalence between the geometric-inertia representations in canonical momentum form and in second-order generalized-coordinate form.

3.3.4. *Energy Conservation.* Because $\mathbf{J}$ is skew-symmetric and H_{cons} has no explicit time dependence,

$$\frac{dH_{cons}}{dt} = \nabla H_{cons}^T \mathbf{J} \nabla H_{cons} = 0. \tag{3.30}$$

and hence

$$\boxed{H_{cons}(\mathbf{q}(t), \mathbf{p}_0(t)) = const.} \tag{3.31}$$

3.4. **Forced–Dissipative Hamiltonian Structure.**

3.4.1. *Unified Entry Point for Nonconservative Generalized Forces.* Nonconservative generalized forces enter uniformly through the right-hand side of the Hamiltonian momentum equation:

$$\boxed{\dot{\mathbf{q}} = \nabla_{\mathbf{p}_0} H_{cons}, \qquad \dot{\mathbf{p}}_0 = -\nabla_{\mathbf{q}} H_{cons} + \mathbf{Q}_{port}.} \tag{3.32}$$

Following the mutually exclusive power accounting established in Sections 1 and 2, write

$$\boxed{\mathbf{Q}_{port} = \mathbf{Q}_{diss} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_{boundary} + \mathbf{Q}_{contact} + \mathbf{r}_{\eta}.} \tag{3.33}$$

where:

- $\mathbf{Q}_{diss}$ contains only passive components whose power with respect to the structural mechanical energy is nonpositive;
- $\mathbf{Q}_{in}$ contains magnetic control, actuation, and other nonfluid inputs that have not been included in the potential;
- $\mathbf{Q}_{flow}$ contains background-flow drag input, Froude–Krylov input, and representation-dependent reactive-flow effects;
- $\mathbf{Q}_{boundary}$ separately collects terminal forces, terminal moments, and boundary actuation;
- $\mathbf{Q}_{contact}$ denotes contact reactions and the contact port;
- $\mathbf{r}_{\eta}$ denotes only residual model error after known model terms and ports have been removed.

The joint virtual-work projection of these forces and moments was derived in Section 2; the corresponding finite-dimensional generalized forces are used directly here.

3.4.2. *Equivalent Viscous-Damping Special Case.* If, for a particular operating condition, the nonconservative force can be represented equivalently as

$$\mathbf{Q}_{port} = \mathbf{F}_{ext}\left(\mathbf{q}, t\right) - \mathbf{C}\left(\mathbf{q}, \dot{\mathbf{q}}, t\right)\dot{\mathbf{q}}, \qquad \mathbf{C} = \mathbf{C}^{T} \succcurlyeq 0, \tag{3.34}$$

then $\dot{\mathbf{q}} = \nabla_{\mathbf{p}_0} H_{cons}$ gives

$$\boxed{\dot{\mathbf{p}}_0 = -\nabla_{\mathbf{q}} H_{cons} + \mathbf{F}_{ext} - \mathbf{C}\nabla_{\mathbf{p}_0} H_{cons}.} \tag{3.35}$$

These equations form a forced–damped Hamiltonian special case on the autonomous backbone. The operator $\mathbf{C}$ may be a state-dependent effective dissipation operator and need not be constant. For quadratic drag, only the passive dissipative component enters $\mathbf{C}\dot{\mathbf{q}}$; the background-flow input remains in $\mathbf{Q}_{flow}$. Coulomb friction, contact switching, and impact may be treated as separate generalized-force ports without requiring a single globally smooth dissipation matrix.

If the magnetic and elastic contact forces have not been included in the potential, one may set

$$\mathbf{F}_{ext} = \mathbf{F}_{mag} + \mathbf{F}_{cont} \tag{3.36}$$

If the same actions have already been generated by the gradients of $V_{mag,cons}$ or $V_{cont,el}$, they must not be counted again in $\mathbf{F}_{ext}$.

3.4.3. *Energy Rate.* For an autonomous Hamiltonian,[17, 20, 21]

$$\frac{dH_{cons}}{dt} = \nabla_{\mathbf{q}} H_{cons}^{T}\dot{\mathbf{q}} + \nabla_{\mathbf{p}_0} H_{cons}^{T}\dot{\mathbf{p}}_0. \tag{3.37}$$

Substitution of the forced canonical equations causes the symplectic terms to cancel, yielding

$$\boxed{\frac{dH_{cons}}{dt} = \dot{\mathbf{q}}^{T}\mathbf{Q}_{port}.} \tag{3.38}$$

Under the equivalent-damping condition of Section 3.4.2,

$$\boxed{\frac{dH_{cons}}{dt} = \nabla_{\mathbf{p}_0} H_{cons}^T \left(\mathbf{F}_{ext} - \mathbf{C}\nabla_{\mathbf{p}_0} H_{cons}\right).} \tag{3.39}$$

where

$$-\nabla_{\mathbf{p}_0} H_{cons}^T \mathbf{C} \nabla_{\mathbf{p}_0} H_{cons} \leq 0, \tag{3.40}$$

The sign of the external input power depends on the operating condition. This energy-rate expression clearly distinguishes external input from passive dissipation and is consistent with the port classification.

3.4.4. *Phase-Space Form of the Complete Forced System.* Define

$$\mathbf{B} = \begin{bmatrix} \mathbf{0} \\ \mathbf{I} \end{bmatrix}, \tag{3.41}$$

The complete forced–dissipative system on the autonomous backbone is then

$$\boxed{\dot{\mathbf{z}}_0 = \mathbf{J}\nabla_{\mathbf{z}_0} H_{cons} + \mathbf{B}\left(\mathbf{Q}_{diss} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_{boundary} + \mathbf{Q}_{contact} + \mathbf{r}_\eta\right).} \tag{3.42}$$

The system is therefore a forced–dissipative Hamiltonian system, or equivalently an open system with a canonical Hamiltonian backbone. It reduces to a classical closed conservative Hamiltonian system only when every port is closed and the Hamiltonian is autonomous. A closed Hamiltonian representation of the dissipative environment would require additional energy-storage or heat-bath degrees of freedom, which are not part of the finite-dimensional variable set constructed in Sections 1 and 2.

3.5. Canonical Momentum and Hamiltonian under a General Prescribed Flow.

3.5.1. *General Discrete Lagrangian.* The energy-consistent prescribed-flow formulation in Sections 1 and 2 generates a velocity-linear term. The general discrete Lagrangian is[17, 29]

$$\boxed{L_h\left(\mathbf{q}, \dot{\mathbf{q}}, t\right) = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}\left(\mathbf{q}\right)\dot{\mathbf{q}} - \dot{\mathbf{q}}^T \mathbf{b}_A\left(\mathbf{q}, t\right) + c_A\left(\mathbf{q}, t\right) - V_h\left(\mathbf{q}, t\right).} \tag{3.43}$$

Here, $\mathbf{M}$ still contains the velocity-quadratic added-mass contribution, while $\mathbf{b}_A$ and c_A represent the cross term and the pure-flow term induced by the prescribed flow. Added inertia already included in $\mathbf{M}_A, \mathbf{b}_A, c_A$ must not be reintroduced on the right-hand side as a complete Morison added-mass generalized force.

3.5.2. *General Canonical Momentum and Hamiltonian.* By definition,

$$\mathbf{p} = \frac{\partial L_h}{\partial \dot{\mathbf{q}}} \tag{3.44}$$

one obtains

$$\boxed{\mathbf{p} = \mathbf{M}\left(\mathbf{q}\right)\dot{\mathbf{q}} - \mathbf{b}_A\left(\mathbf{q}, t\right), \qquad \dot{\mathbf{q}} = \mathbf{M}^{-1}\left(\mathbf{q}\right)\left(\mathbf{p} + \mathbf{b}_A\right).} \tag{3.45}$$

When $\mathbf{b}_A = \mathbf{0}$, this relation reduces to $\mathbf{p}_0 = \mathbf{M}\dot{\mathbf{q}}$ for the autonomous backbone or the backbone–port representation. When the velocity-linear prescribed-flow term is retained explicitly, the general canonical momentum $\mathbf{p}$ must be used.

The Legendre transformation gives

$$\boxed{H\left(\mathbf{q}, \mathbf{p}, t\right) = \frac{1}{2}\left(\mathbf{p} + \mathbf{b}_A\right)^T \mathbf{M}^{-1}\left(\mathbf{q}\right)\left(\mathbf{p} + \mathbf{b}_A\right) + V_h\left(\mathbf{q}, t\right) - c_A\left(\mathbf{q}, t\right).} \tag{3.46}$$

where $-c_A$ follows directly from the Legendre transformation. If $\mathbf{M} = \mathbf{M}\left(\mathbf{q}\right)$ has no explicit time dependence, then

$$\boxed{\nabla_{\mathbf{p}} H = \mathbf{M}^{-1}\left(\mathbf{p} + \mathbf{b}_A\right) = \dot{\mathbf{q}},} \tag{3.47}$$

$$\boxed{\frac{\partial H}{\partial q_i} = -\frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}_{,i}\dot{\mathbf{q}} + \mathbf{b}_{A,i}^T \dot{\mathbf{q}} + \frac{\partial V_h}{\partial q_i} - \frac{\partial c_A}{\partial q_i},} \tag{3.48}$$

and, at fixed $(\mathbf{q}, \mathbf{p})$,

$$\boxed{\left.\frac{\partial H}{\partial t}\right|_{\mathbf{q},\mathbf{p}} = \dot{\mathbf{q}}^T \partial_t \mathbf{b}_A + \partial_t V_h - \partial_t c_A.} \tag{3.49}$$

3.5.3. *Equivalence with the General Euler–Lagrange Equations.* In the absence of external ports, the standard nonautonomous canonical equations still hold:

$$\dot{\mathbf{q}} = \nabla_{\mathbf{p}} H, \qquad \dot{\mathbf{p}} = -\nabla_{\mathbf{q}} H. \tag{3.50}$$

From

$$\dot{\mathbf{p}} = \mathbf{M}\ddot{\mathbf{q}} + \dot{\mathbf{M}}\dot{\mathbf{q}} - \dot{\mathbf{b}}_A \tag{3.51}$$

one recovers

$$\boxed{\mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}_M + \mathbf{g}_f + \nabla_{\mathbf{q}} V_h = \mathbf{0},} \tag{3.52}$$

where

$$\boxed{\mathbf{g}_f = -\dot{\mathbf{b}}_A + \left(\nabla_{\mathbf{q}} \mathbf{b}_A\right)^T \dot{\mathbf{q}} - \nabla_{\mathbf{q}} c_A.} \tag{3.53}$$

This result agrees with the prescribed-flow Euler–Lagrange equation in Section 2. Both $\mathbf{g}_M$ and $\mathbf{g}_f$ are generated automatically by the Legendre transformation and therefore must not be added again as separate generalized forces.

3.5.4. *Exact Representation and Backbone–Port Representation.* When $\mathbf{b}_A, c_A$ are retained, define[17, 29]

$$\mathbf{z} = \begin{bmatrix} \mathbf{q} \\ \mathbf{p} \end{bmatrix}, \tag{3.54}$$

The complete forced system is

$$\boxed{\dot{\mathbf{z}} = \mathbf{J}\nabla_{\mathbf{z}} H\left(\mathbf{z}, t\right) + \mathbf{B}\left(\mathbf{Q}_{diss} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_{boundary} + \mathbf{Q}_{contact} + \mathbf{r}_{\eta}\right).} \tag{3.55}$$

In the exact representation, $\mathbf{g}_f$ is already contained in the Hamiltonian, so $\mathbf{Q}_{flow}$ does not include an additional $-\mathbf{g}_f$.

The magnetic actuation is represented in one of two mutually exclusive ways. If the conservative magnetic potential is included in

$$H_{sk} = H_{cons} + V_{drv},$$

the same generalized force $-\nabla_q V_{drv}$ must not also enter $\mathbf{Q}_{in}$. If instead

$$H_{sk} = H_{cons}, \qquad \mathbf{Q}_{drv} = -\nabla_q V_{drv},$$

the force enters the input/magnetic port exactly once. When the magnetic action is represented by a potential, its generalized force must equal the gradient-induced force from that potential; when it is represented as an input, only the port force is included. These two descriptions are not used simultaneously, so the same conservative magnetic contribution is counted exactly once.

In the backbone–port representation, take

$$\mathbf{p}_0 = \mathbf{M}\dot{\mathbf{q}} \tag{3.56}$$

and define

$$H_{sk}\left(\mathbf{q},\mathbf{p}_0,t\right)=\frac{1}{2}\mathbf{p}_0^T\mathbf{M}^{-1}\mathbf{p}_0+V_h\left(\mathbf{q},t\right), \tag{3.57}$$

The prescribed-flow cross term is then transferred in its entirety to the port:

$$\boxed{\mathbf{Q}_{flow}^{sk}=\mathbf{Q}_{flow}-\mathbf{g}_f.} \tag{3.58}$$

The two representations generate the same configuration trajectories, but each must be used as a complete and internally consistent set:

$$\boxed{\left(\mathbf{p},H,\mathbf{Q}_{flow}\right)\neq\left(\mathbf{p}_0,H_{sk},\mathbf{Q}_{flow}^{sk}\right).} \tag{3.59}$$

The two variable sets and generating functions must remain paired. The shifted momentum $\mathbf{p}$ is used with the exact Hamiltonian H, whereas $\mathbf{p}_0$ is used with H_{sk}. The term $-\mathbf{g}_f$ must not be added again to the exact Hamiltonian representation.

3.5.5. *General Energy Rate.* The exact representation satisfies

$$\boxed{\frac{dH}{dt}=\frac{\partial H}{\partial t}+\dot{\mathbf{q}}^T\left(\mathbf{Q}_{diss}+\mathbf{Q}_{in}+\mathbf{Q}_{flow}+\mathbf{Q}_{boundary}+\mathbf{Q}_{contact}+\mathbf{r}_\eta\right).} \tag{3.60}$$

The backbone–port representation satisfies

$$\boxed{\frac{dH_{sk}}{dt}=\partial_t V_h+\dot{\mathbf{q}}^T\left(\mathbf{Q}_{diss}+\mathbf{Q}_{in}+\mathbf{Q}_{flow}^{sk}+\mathbf{Q}_{boundary}+\mathbf{Q}_{contact}+\mathbf{r}_\eta\right).} \tag{3.61}$$

These two expressions give the energy rates under nonautonomous potentials, prescribed flow, and multiple ports, and remain dynamically equivalent in their respective variable representations.

3.6. **Range of Applicability and the Strict Kirchhoff Constraint Branch.**

3.6.1. *Local Orientation Coordinates.* Exponential coordinates form a local chart on $SO(3)$. To use standard canonical variables within the current chart, require, for every $S\in[0,L]$,

$$\parallel\boldsymbol{\Psi}(S)\mathbf{b}\parallel<\pi. \tag{3.62}$$

This is a coordinate-chart restriction rather than a small-rotation mechanical approximation. When a trajectory crosses the chart boundary, the orientation coordinates must be switched and the conjugate momentum updated consistently through the corresponding cotangent transformation.

3.6.2. *Strict Kirchhoff Primary Branch and Finite-Shear Validation Branch.* The geometrically exact parent model in Section 1 is a finite-shear Cosserat rod. To remain consistent with the Kirchhoff slender-catheter assumptions and to provide Sections 4 and 5 with a nonredundant canonical state, the primary structured-learning and KAM branches preferentially use a locally reduced representation of the strict shear-free constraint. Write the constraint as[17, 18]

$$\mathbf{c}\left(\mathbf{q}\right)=\mathbf{0},\qquad\mathbf{G}\left(\mathbf{q}\right)=\frac{\partial\mathbf{c}}{\partial\mathbf{q}}, \tag{3.63}$$

If $\mathbf{G}$ has constant rank throughout the domain of interest, the constraint set is locally a smooth manifold. In the full coordinates, the equations are

$$\boxed{\dot{\mathbf{q}}=\nabla_{\mathbf{p}}H,\qquad\dot{\mathbf{p}}=-\nabla_{\mathbf{q}}H+\mathbf{Q}_{port}-\mathbf{G}^T\boldsymbol{\lambda}_c,\qquad\mathbf{c}\left(\mathbf{q}\right)=\mathbf{0}.} \tag{3.64}$$

We define the reaction generalized force as $\mathbf{Q}_c:=-\mathbf{G}^T\boldsymbol{\lambda}_c$. Equivalently, writing $\mathbf{Q}_c=+\mathbf{G}^T\boldsymbol{\lambda}_c$ moves the multiplier term to the opposite side; all weak-form and Hamiltonian labels below use the first convention.

The ideal constraint reaction satisfies $\dot{\mathbf{q}}^T\mathbf{G}^T\boldsymbol{\lambda}_c = 0$. To obtain a canonical system that can be used directly by the CHNN and standard symplectic analysis, introduce independent local coordinates $\mathbf{q} = \boldsymbol{\chi}(\mathbf{y})$ and define

$$\mathbf{C}(\mathbf{y}) = \frac{\partial \boldsymbol{\chi}}{\partial \mathbf{y}}, \qquad \mathbf{M}_r(\mathbf{y}) = \mathbf{C}^T\mathbf{M}(\boldsymbol{\chi}(\mathbf{y}))\,\mathbf{C}, \tag{3.65}$$

When $\mathbf{M}_r \succ 0$, construct the reduced Hamiltonian. The reduced momentum of the autonomous backbone is $\mathbf{p}_r = \mathbf{M}_r\dot{\mathbf{y}}$. If the reduced prescribed-flow model still contains a velocity-linear term, use instead $\mathbf{p}_r = \mathbf{M}_r\dot{\mathbf{y}} - \mathbf{b}_{A,r}$. Sections 4 and 5 use only this independent-coordinate canonical system; the differential–algebraic equations containing the multiplier $\boldsymbol{\lambda}_c$ cannot be treated directly as a standard Hamiltonian system on the full space.

The finite-shear-stiffness Cosserat ordinary differential equations are retained for COMSOL comparison, numerical regularization, and constraint-convergence analysis. Data from this regularized branch may replace locally reduced data in CHNN training only when the shear strains, trajectories, energies, and principal frequencies converge to the Kirchhoff results as the shear stiffness increases. This strategy preserves the advantages of the complete parent model while keeping the primary computational object consistent with the original Kirchhoff research objective.

3.7. **Interfaces to CHNN, KAM, and Structure-Preserving Discretization.** Section 3 does not anticipate the neural-network construction or KAM theorem proofs developed in Sections 4 and 5. Instead, based on the HNN, dissipative/port-Hamiltonian network, statistical-learning–KAM, symplectic-algorithm, and parameterization-method literature used there, it specifies the unique and verifiable mathematical objects that connect the preceding and subsequent parts.[19–28]

Second-order BDF with `tsteps` and `tstepsstore=1` is used only to store states at a common solver time step and to generate the vector-field quantities required by Equation (3.13). The three-level time-step comparison assesses temporal consistency and convergence of the canonical-variable mapping, not symplecticity of BDF. Any symplectic or KAM inference requires a separate symplectic map satisfying the assumptions in Section 5.

3.7.1. *Canonical Nondimensionalization, Complex Coordinates, and the Physical Hamiltonian Baseline.* A subsequent complex representation cannot add $\mathbf{q}$ and $\mathbf{p}$ directly when they have different physical dimensions and scales. Choose a stable reference state $(\mathbf{q}_*, \mathbf{p}_*)$, coordinate and momentum scale matrices $\mathbf{S}_q$ and $\mathbf{S}_p$, a time scale t_*, and an energy scale E_*, and define the action scale $\mathcal{A}_* = E_*t_*$. Require

$$\boxed{\mathbf{S}_q^T\mathbf{S}_p = \mathcal{A}_*\mathbf{I}.} \tag{3.66}$$

Define

$$\mathbf{q} = \mathbf{q}_* + \mathbf{S}_q\mathbf{Q}, \qquad \mathbf{p} = \mathbf{p}_* + \mathbf{S}_p\mathbf{P}, \qquad t = t_*\tau, \qquad H = E_*\widehat{H}. \tag{3.67}$$

The scale-compatibility condition then gives

$$\frac{d\mathbf{Q}}{d\tau} = \nabla_{\mathbf{P}}\widehat{H}, \qquad \frac{d\mathbf{P}}{d\tau} = -\nabla_{\mathbf{Q}}\widehat{H}, \tag{3.68}$$

so the nondimensional variables still satisfy the standard canonical Hamilton equations. Define the complex coordinates[17, 29]

$$\boxed{\boldsymbol{\xi} = \frac{\mathbf{Q} + i\mathbf{P}}{\sqrt{2}}, \qquad \overline{\boldsymbol{\xi}} = \frac{\mathbf{Q} - i\mathbf{P}}{\sqrt{2}}.} \tag{3.69}$$

The real and imaginary parts of $\boldsymbol{\xi}$ are canonically paired configuration and momentum variables with compatible scales. A nontrivial real-valued Hamiltonian is generally written as $\widehat{H}(\boldsymbol{\xi}, \overline{\boldsymbol{\xi}}) \in \mathbb{R}$, rather than as a real-valued holomorphic function of $\boldsymbol{\xi}$ alone.[23, 24]

Section 3 supplies Section 4 with the analytical physical baseline

$$\widehat{H}_{phys} = \widehat{H}_{cons}, \tag{3.70}$$

The real-valued readout in Section 4 should use the physical-residual form

$$\boxed{\widehat{H}_{\lambda} = \widehat{H}_{phys} + \Delta\widehat{H}_{\lambda}.} \tag{3.71}$$

The known mass matrix, Cosserat strain energy, and identified conservative magnetic and elastic-contact potentials are not relearned by the network. The term $\Delta\widehat{H}_{\lambda}$ represents only parameter corrections, conservative corrections induced by Galerkin truncation, and other unmodeled conservative effects. Because a Hamiltonian inferred from vector-field data is identifiable only up to an additive constant, a gauge must be fixed, for example $\widehat{H}_{\lambda}(\mathbf{z}_*) = 0$ or zero mean over the training domain.[19, 21]

3.7.2. *Backbone–Port Structure and Consistent Data Representations.* In the autonomous representation with time-independent actuation parameters $\boldsymbol{\lambda}(t) \equiv \overline{\boldsymbol{\lambda}}$,

$$V_{drv} = 0, \qquad H_{sk} = H_{cons}, \qquad \partial_t H_{sk} = 0, \tag{3.72}$$

and

$$\boxed{\dot{\mathbf{z}}_0 = \mathbf{J}\nabla_{\mathbf{z}_0} H_{cons} + \mathbf{B}\left(\mathbf{Q}_{diss} + \mathbf{Q}_{in} + \mathbf{Q}^{sk}_{flow} + \mathbf{Q}_{boundary} + \mathbf{Q}_{contact} + \mathbf{r}_{\eta}\right).} \tag{3.73}$$

Section 4 should parameterize or supervise the ports separately rather than allowing a single unstructured residual to absorb every physical contribution:

$$\mathbf{Q}_{port,\theta} = \mathbf{Q}_{diss,\mu} + \mathbf{Q}_{in,\nu} + \mathbf{Q}_{flow,\kappa} + \mathbf{Q}_{boundary} + \mathbf{Q}_{contact} + \mathbf{r}_{\eta}. \tag{3.74}$$

Here, $\mathbf{Q}_{diss,\mu}$ represents only the passive component with nonpositive power; magnetic control, background-flow input, and boundary actuation remain separate ports; and $\mathbf{r}_{\eta}$ represents only the residual remaining after the known model and known ports have been removed. This hierarchy is consistent with the structured modeling principles of dissipative SymODEN and port-Hamiltonian neural networks.[20, 21]

For time-varying or pulsatile operating conditions, $H_{sk} = H_{cons} + V_{drv}$. The following two representations are mutually exclusive:

1. If the complete H_{sk} is used as the generator, the same conservative actuation force must not also enter $\mathbf{Q}_{in}$;
2. If the autonomous H_{cons} is used as the generator, set $\mathbf{Q}_{drv} = -\nabla_{\mathbf{q}} V_{drv}$ and include it exactly once in $\mathbf{Q}_{in}$.

Under a general prescribed flow, use $\mathbf{p} = \mathbf{M}\dot{\mathbf{q}} - \mathbf{b}_A$ and the corresponding nonautonomous Hamiltonian; the autonomous backbone uses $\mathbf{p}_0 = \mathbf{M}\dot{\mathbf{q}}$. The two data types must be distinguished by a *representation* label. The backbone and nonautonomous data supplied to Section 4 are, respectively,

$$\mathcal{D}_{sk} = \left(\mathbf{Q}, \mathbf{P}, \boldsymbol{\xi}, \dot{\mathbf{z}}_0, \widehat{H}_{cons}, \nabla\widehat{H}_{cons}, \mathbf{Q}_{diss}, \mathbf{Q}_{in}, \mathbf{Q}^{sk}_{flow}, \mathbf{Q}_{boundary}, \mathbf{Q}_{contact}, \mathbf{r}_{\eta}\right), \tag{3.75}$$

$$\mathcal{D}_{na} = \left(\mathbf{z}, \dot{\mathbf{z}}, t, H, \nabla H, \partial_t H, \mathbf{Q}_{diss}, \mathbf{Q}_{in}, \mathbf{Q}^{H}_{flow}, \mathbf{Q}_{boundary}, \mathbf{Q}_{contact}, \mathbf{r}_{\eta}\right). \tag{3.76}$$

The statistical-learning–KAM interface preferentially uses derivative-supervised autonomous conservative samples and does not interpret the sum of trajectory loss, dissipation loss, and regularization terms directly as the KAM perturbation size.[22]

3.7.3. *Stable Equilibrium, Modal Canonicalization, and the Integrable Reference System.* The subsequent KAM analysis starts from the autonomous conservative backbone $H_{cons}(\mathbf{q},\mathbf{p}_0)$, without assuming that it is globally integrable. Choose an equilibrium $(\mathbf{q}_*,\mathbf{0})$ satisfying

$$\nabla_{\mathbf{q}} V_{cons}(\mathbf{q}_*)=\mathbf{0}, \qquad \mathbf{M}_*=\mathbf{M}(\mathbf{q}_*)\succ 0, \qquad \mathbf{K}_*=\nabla_{\mathbf{q}}^2 V_{cons}(\mathbf{q}_*)\succ 0. \tag{3.77}$$

Solve the generalized eigenvalue problem

$$\mathbf{K}_*\boldsymbol{\phi}_j=\omega_j^2\mathbf{M}_*\boldsymbol{\phi}_j, \tag{3.78}$$

and choose the modal matrix $\boldsymbol{\Phi}$ such that

$$\boldsymbol{\Phi}^T\mathbf{M}_*\boldsymbol{\Phi}=\mathbf{I}, \qquad \boldsymbol{\Phi}^T\mathbf{K}_*\boldsymbol{\Phi}=\boldsymbol{\Omega}^2. \tag{3.79}$$

The linear canonical transformation

$$\mathbf{q}-\mathbf{q}_*=\boldsymbol{\Phi}\mathbf{Q}_m, \qquad \mathbf{p}_0=\boldsymbol{\Phi}^{-T}\mathbf{P}_m \tag{3.80}$$

reduces the quadratic Hamiltonian to

$$H_2=\frac{1}{2}\left(\mathbf{P}_m^T\mathbf{P}_m+\mathbf{Q}_m^T\boldsymbol{\Omega}^2\mathbf{Q}_m\right)=\sum_{j=1}^{n}\omega_j I_j, \tag{3.81}$$

where

$$x_j=\sqrt{\omega_j}Q_{m,j}, \qquad y_j=\frac{P_{m,j}}{\sqrt{\omega_j}}, \qquad I_j=\frac{x_j^2+y_j^2}{2}. \tag{3.82}$$

With quadratic terms alone, the frequencies are independent of the actions, and hence

$$\frac{\partial^2 H_2}{\partial \mathbf{I}^2}=\mathbf{0}, \tag{3.83}$$

Thus the linear harmonic oscillator is isochronously degenerate. The condition $\mathbf{K}_*\succ 0$ cannot replace the Kolmogorov nondegeneracy condition; nondegeneracy must be supplied by geometric or material nonlinearities, conservative magnetic potentials, or other higher-order conservative terms.

If H_{cons} is analytic in the neighborhood of interest, or has the regularity required by the selected finitely differentiable KAM theorem, and the linear frequencies avoid low-order resonances to the required order, a finite-order Birkhoff normal form can be constructed:[18, 25, 29]

$$H_{cons}\circ\boldsymbol{\Psi}=H_*+\boldsymbol{\omega}\cdot\mathbf{I}+Z_4(\mathbf{I})+\cdots+Z_r(\mathbf{I})+R_{r+1}(\mathbf{I},\boldsymbol{\theta}). \tag{3.84}$$

Define

$$\boxed{H_0(\mathbf{I})=H_*+\boldsymbol{\omega}\cdot\mathbf{I}+Z_4(\mathbf{I})+\cdots+Z_r(\mathbf{I}), \qquad \varepsilon H_1=R_{r+1}.} \tag{3.85}$$

The lowest nontrivial case commonly has

$$Z_4(\mathbf{I})=\frac{1}{2}\mathbf{I}^T\mathbf{A}\mathbf{I}, \tag{3.86}$$

The Kolmogorov nondegeneracy condition over the action domain under consideration then requires

$$\boxed{\det\left(\frac{\partial^2 H_0}{\partial \mathbf{I}^2}\right)\neq 0,} \tag{3.87}$$

Near the origin, this reduces to $\det\mathbf{A}\neq 0$. If this matrix is singular, classical Kolmogorov KAM cannot be claimed; one must increase the normal-form order, choose another reference family, or invoke and verify an isoenergetic or degenerate KAM theorem.

On a fixed complex neighborhood or Sobolev domain $\mathcal{D}$, with a common function norm $\|\cdot\|_{\mathcal{X}(\mathcal{D})}$, write the nondimensional Hamiltonian as

$$\widehat{H}_{cons} = \widehat{H}_0 + \varepsilon P, \qquad \| P \|_{\mathcal{X}(\mathcal{D})} = 1, \tag{3.88}$$

and define

$$\boxed{\varepsilon = \| \widehat{H}_{cons} - \widehat{H}_0 \|_{\mathcal{X}(\mathcal{D})} .} \tag{3.89}$$

The frequency vector must also satisfy the corresponding Diophantine condition. If a low-order resonance is present, the analysis must use a resonant normal form and an appropriate theorem; the resonant term cannot simply be absorbed into an ordinary small perturbation.

3.7.4. *Learning Error, Discretization Error, and the KAM Smallness Budget.* To place the learning error from Section 4 and the symplectic-discretization error from Section 5 in one mathematical chain, all errors must be measured on the same domain and in compatible function norms. At the continuous Hamiltonian level, first define

$$\delta_{model} = \| H_{cons} - H_0 \|_{\mathcal{X}}, \qquad \delta_{learn} = \| H_\lambda - H_{cons} \|_{\mathcal{X}}, \tag{3.90}$$

so that

$$\boxed{\| H_\lambda - H_0 \|_{\mathcal{X}} \leq \delta_{model} + \delta_{learn}.} \tag{3.91}$$

At the discrete level, do not assume unconditionally that a globally exact "discrete Hamiltonian" exists. Let Φ_h^H denote the exact flow of Hamiltonian H over the time interval h, let $F_{\lambda,h}$ denote the conservative symplectic one-step map actually used in Section 5, and set the unperturbed reference map to $F_{0,h} = \Phi_h^{H_0}$. On a fixed compact domain, define in the C^r or analytic norm appropriate to the selected symplectic-map KAM theorem[18, 25]

$$\delta_{alg} = \| F_{\lambda,h} - \Phi_h^{H_\lambda} \|, \qquad \delta_{map} = \| F_{\lambda,h} - F_{0,h} \| . \tag{3.92}$$

The triangle inequality gives

$$\boxed{\delta_{map} \leq \delta_{alg} + \| \Phi_h^{H_\lambda} - \Phi_h^{H_{cons}} \| + \| \Phi_h^{H_{cons}} - \Phi_h^{H_0} \| .} \tag{3.93}$$

If the vector field depends locally Lipschitz-continuously on Hamiltonian perturbations and the flow remains in the same domain over the time interval considered, the last two terms can be bounded by $C_{learn}(h)\delta_{learn}$ and $C_{model}(h)\delta_{model}$, respectively. Hence,

$$\boxed{\delta_{map} \leq \delta_{alg} + C_{learn}(h)\delta_{learn} + C_{model}(h)\delta_{model}.} \tag{3.94}$$

If backward error analysis additionally constructs a modified Hamiltonian for a specified step size and finite time scale, an equivalent modified-Hamiltonian budget may be used. The domain of validity, truncation order, and remainder bound must accompany this object; global exact existence cannot be inferred solely from the statement that the integrator is symplectic.

Persistence of invariant tori can be concluded only when the continuous or discrete budget is below the concrete threshold of the selected KAM theorem, which depends on the analytic domain, Diophantine constants, nondegeneracy margin, dimension, and step size. The statistical-learning result of Chen–Matsubara–Yaguchi directly controls only the conservative HNN branch under its assumptions. Transferring a comparable probabilistic bound to the CHNN requires a real-valued readout and a corresponding complexity analysis of the function class.[22]

Quadratic drag, contact damping and friction, nonpotential magnetic actuation, background-flow input, boundary input, and contact activation or release are not included in the preceding classical autonomous-Hamiltonian perturbation. Their contributions to the complete safe operating domain must instead be treated through energy balance, port gains, structural defects, Lyapunov indicators, or a specialized theory of dissipative invariant tori.

3.7.5. *Symplectic Maps, Birkhoff Structure, and the Torus Equation.* For the autonomous conservative branch, a structure-preserving integrator produces the one-step map

$$F_h = \Phi_h^{cons}. \tag{3.95}$$

Section 5 first verifies whether F_h is an exact symplectic map and then applies the KAM conditions for symplectic maps and symplectic algorithms due to Shang together with the parameterization method of Haro et al.[25–28] The torus embedding $K : \mathbb{T}^d \to \mathcal{M}$ satisfies[28]

$$\boxed{F_h\left(K\left(\boldsymbol{\theta}\right)\right) = K\left(\boldsymbol{\theta} + \boldsymbol{\omega}\right).} \tag{3.96}$$

The invariance residual of a numerical approximation K_N is

$$E_N\left(\boldsymbol{\theta}\right) = F_h\left(K_N\left(\boldsymbol{\theta}\right)\right) - K_N\left(\boldsymbol{\theta} + \boldsymbol{\omega}\right). \tag{3.97}$$

For the complete data vector field, the known conservative backbone and known ports must first be removed:

$$\boxed{X_{res} = X_{data} - \mathbf{J}\nabla H_\lambda - \mathbf{B}\left(\mathbf{Q}_{diss,\mu} + \mathbf{Q}_{in,\nu} + \mathbf{Q}_{flow,\kappa} + \mathbf{Q}_{boundary} + \mathbf{Q}_{contact}\right).} \tag{3.98}$$

The "self-adjoint Birkhoff-system representation" used here and the "Hamiltonian Birkhoff normal form" used to construct H_0 in Section 3.7.3 are distinct concepts. The former concerns the variational self-adjointness and $K\left(\mathbf{z}, t\right)$-symplectic structure of a general first-order system, whereas the latter is the Hamiltonian normal form near an elliptic equilibrium. Section 5 tests only whether X_{res} admits a regular self-adjoint Birkhoff-system representation. If the closure, time-compatibility, or nondegeneracy conditions fail, X_{res} remains a general residual and the complete open system cannot be called a strict $K\left(\mathbf{z}, t\right)$-symplectic system. Explicit time dependence alone also does not necessarily imply a nonstandard symplectic structure: in appropriate canonical variables and extended phase space, the exact nonautonomous Hamiltonian branch should still be treated preferentially with standard symplectic theory.[18, 25]

3.7.6. *Structural Decomposition of the Difference between the Primary COMSOL Training Model and High-Fidelity Models.* Section 3 treats the Kirchhoff-consistent COMSOL branch as the primary training dynamics X_K, and the Weak Form, Timoshenko, or solid model as the high-fidelity reference X_{HF}. Define the paired-condition model discrepancy

$$\Delta X_{HF} = X_{HF} - X_K. \tag{3.99}$$

This discrepancy cannot be inserted wholesale into the Hamiltonian. First test its conservative closure in canonical variables. If a real-valued ΔH_{HF} exists such that

$$\Delta X_{HF}^{cons} = \mathbf{J}\nabla\Delta H_{HF}, \tag{3.100}$$

then that component may enter the conservative correction head in Section 4. If a momentum port satisfies

$$\nabla_{\mathbf{P}}\widehat{H}^T\Delta\mathbf{Q}_{HF}^{diss} \leq 0, \tag{3.101}$$

it may enter the dissipative correction head. Components that do not satisfy closure, power-sign, or smoothness conditions remain in the general model residual $\mathbf{r}_{HF}$. Thus,

$$\boxed{\Delta X_{HF} = \mathbf{J}\nabla\Delta H_{HF} + \mathbf{B}\Delta\mathbf{Q}_{HF}^{diss} + \mathbf{B}\Delta\mathbf{Q}_{HF}^{port} + \mathbf{r}_{HF}.} \tag{3.102}$$

This decomposition is a structural projection to be identified and validated, not an identity that holds automatically for an arbitrary high-fidelity discrepancy. The model error used in KAM analysis consists only of the conservatively closed component and its bound in the common function norm. Shear dissipation, contact switching, and local solid effects must not be included in H_1 without proof.

3.7.7. *Interfaces for Structured Learning and Stability Analysis.* The preceding derivations yield three interfaces required by the subsequent structured-learning and stability analyses:

$$\mathcal{I}_{CHNN} = \left\{ \begin{matrix} \mathbf{Q}, \mathbf{P}, \boldsymbol{\xi}, \widehat{H}_{phys}, \nabla \widehat{H}_{phys}, \mathbf{Q}_{diss}, \mathbf{Q}_{in}, \mathbf{Q}_{flow}, \\ \mathbf{Q}_{boundary}, \mathbf{Q}_{contact}, \mathbf{r}_{\eta}, model_level, representation \end{matrix} \right\}, \tag{3.103}$$

$$\mathcal{I}_{KAM} = \left(H_0, H_1, \varepsilon, \boldsymbol{\omega}\left(\mathbf{I}\right), \partial_{\mathbf{I}}\boldsymbol{\omega}, \gamma, \tau, \mathcal{D}, \mathcal{X}\right), \tag{3.104}$$

$$\mathcal{I}_{geom} = \left(\mathbf{J}, F_{0,h}, F_{\lambda,h}, X_{res}, E_N, \delta_{model}, \delta_{learn}, \delta_{alg}, \delta_{map}\right). \tag{3.105}$$

Each object is traceable to the physical model and Galerkin outputs in Sections 1 and 2 and to the canonicalization performed in this section. If an object has not yet been constructed or verified numerically, subsequent sections may list it only as a condition to be tested and must not present the conditional interface as an established theorem conclusion.

3.8. **Section Summary.** Using the complete joint coordinates for translation and cross-sectional orientation, the canonical momentum and Hamiltonian of the autonomous conservative backbone are

$$\boxed{\mathbf{p}_0 = \mathbf{M}\dot{\mathbf{q}}, \qquad H_{cons} = \frac{1}{2}\mathbf{p}_0^T\mathbf{M}^{-1}\mathbf{p}_0 + V_{cons},} \tag{3.106}$$

and the forced–dissipative canonical equations are

$$\boxed{\dot{\mathbf{z}}_0 = \mathbf{J}\nabla H_{cons} + \mathbf{B}\mathbf{Q}_{port}, \qquad \frac{dH_{cons}}{dt} = \dot{\mathbf{q}}^T\mathbf{Q}_{port}.} \tag{3.107}$$

The contribution of the configuration-dependent mass matrix in the Hamiltonian momentum equation is equivalent to the geometric-inertia vector $\mathbf{g}_M$ in Section 2. A nonlinear conservative potential can be represented as a path-independent line integral of the restoring force; under equivalent viscous damping, the energy rate reduces to the sum of external-input power and dissipative power.

Under a general prescribed flow, the canonical momentum and Hamiltonian become

$$\boxed{\mathbf{p} = \mathbf{M}\dot{\mathbf{q}} - \mathbf{b}_A, \qquad H = \frac{1}{2}\left(\mathbf{p} + \mathbf{b}_A\right)^T\mathbf{M}^{-1}\left(\mathbf{p} + \mathbf{b}_A\right) + V_h - c_A.} \tag{3.108}$$

This general representation retains cross-sectional orientation, independent torsion, rotary inertia, added mass, and external ports while unifying the autonomous backbone, prescribed-flow dynamics, and forced–dissipative dynamics. Canonical nondimensionalization, complex coordinates, a physical-residual Hamiltonian, Birkhoff normal form, a unified KAM error budget, and the three interfaces $\mathcal{I}_{CHNN}$, $\mathcal{I}_{KAM}$, and $\mathcal{I}_{geom}$ then enable Section 3 to provide the subsequent analyses with mutually consistent and traceable mathematical objects for structured CHNN learning, symplectic discretization, parameterized-torus validation, and safe-domain computation.

4. Structured Hamiltonian Neural Networks with Complex-Valued Representations

The reduced phase space makes it possible to compare representation and physical prior without changing the mechanical variables. We therefore build real and complex Hamiltonian networks around the same real-valued energy and keep conservative, dissipative, and driven contributions in separate channels.

4.1. **Learning Object and Upstream Interfaces.** The CHNN acts on the reduced canonical finite-dimensional system and connects to the subsequent KAM analysis. A single complex-valued scalar is not used to generate conservative, dissipative, and driven dynamics simultaneously. Section 3 provides the independent canonical variables, nondimensionalization, physical Hamiltonian baseline, and mutually exclusive port accounting used below.

Let the reduced nondimensional canonical state be

$$\mathbf{Z} = \begin{bmatrix} \mathbf{Q} \\ \mathbf{P} \end{bmatrix} \in \mathbb{R}^{2d}, \qquad \mathbf{J} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & \mathbf{0} \end{bmatrix}, \tag{4.1}$$

and define the complex coordinate

$$\boxed{\boldsymbol{\xi} = \frac{\mathbf{Q} + i\mathbf{P}}{\sqrt{2}}, \qquad \overline{\boldsymbol{\xi}} = \frac{\mathbf{Q} - i\mathbf{P}}{\sqrt{2}}.} \tag{4.2}$$

Here, $\mathbf{Q}$ and $\mathbf{P}$ have undergone a canonical scaling satisfying $\mathbf{S}_q^T \mathbf{S}_p = E_{ref} t_{ref} \mathbf{I}$, so they may be combined as the real and imaginary parts of a complex variable. The dimensional variables $\mathbf{q}$ and $\mathbf{p}$ are not added directly before canonical nondimensionalization.

The COMSOL data also carry

$$model_level \in \{K, WF, T, S\}, \qquad representation \in \{sk, na, red\}, \tag{4.3}$$

which identify, respectively, the Kirchhoff-consistent Beam model, strict weak-form model, Timoshenko model, and solid model, together with the autonomous-backbone, exact nonautonomous, and engineering-reduction representations. Data with different labels are not mixed without a documented compatibility analysis.

4.2. **Conservative Hamiltonian Head and Complex-Valued Internal Representation.**

4.2.1. *Real-Valued Physical Hamiltonian.* A standard HNN generates the conservative vector field from the gradient of a real-valued scalar Hamiltonian.[19] We use

$$\boxed{\widehat{H}_\lambda(\mathbf{Z}) = \widehat{H}_{phys}(\mathbf{Z}) + \Delta\widehat{H}_\lambda(\mathbf{Z}) \in \mathbb{R},} \tag{4.4}$$

where $\widehat{H}_{phys}$ is supplied by the mass matrix, Cosserat/Kirchhoff strain energy, identified conservative magnetic potential, and elastic contact potential for a fixed active set from Section 3. The network learns only material-parameter discrepancies, conservative corrections caused by Galerkin truncation, and other conservatively closed model discrepancies. To remove the additive gauge freedom, impose

$$\widehat{H}_\lambda(\mathbf{Z}_{ref}) = 0 \tag{4.5}$$

or require zero mean over the training domain.

4.2.2. *Complex-Valued Representation and Real-Valued Hamiltonian.* For the open-port formulation, complex-valued features are used only as internal representations, whereas the Hamiltonian supplied to the equations of motion is constrained to be real valued. Specifically,

$$\boxed{h_\theta = \mathcal{N}_\theta^{\mathbb{C}}(\boldsymbol{\xi}, \overline{\boldsymbol{\xi}}), \qquad \Delta H_\theta = \mathcal{R}_{H,\theta}(h_\theta) \in \mathbb{R}, \qquad H_\theta(\mathbf{Z}, \mu) = H_{\text{phys}}(\mathbf{Z}, \mu) + \Delta H_\theta(\mathbf{Z}, \mu) \in \mathbb{R}.} \tag{4.6}$$

The complex layer therefore has no independent interpretation as an energy channel, and its imaginary component is not assigned a physical Hamiltonian meaning. The Lean4 verification addresses the real-valued construction in Equation (4.6); alternative complex-to-scalar mappings lie outside the present analysis. Consequently, the formal result establishes a finite-dimensional representation property but does not address optimization behavior, the physical origin of nonlinear corrections, or the validity of the complete CHNN model.

Let

$$\mathbf{h}_\theta = \mathcal{N}_\theta^{\mathbb{C}}\left(\boldsymbol{\xi}, \overline{\boldsymbol{\xi}}\right), \qquad \Delta\widehat{H}_\lambda = \mathcal{R}_{H,\lambda}\left(\mathbf{h}_\theta\right) \in \mathbb{R}. \tag{4.7}$$

A complex linear layer represents amplitude scaling and phase rotation simultaneously through its real–imaginary block structure. For noncircular phase-space data, both $\boldsymbol{\xi}$ and $\overline{\boldsymbol{\xi}}$ are retained as inputs, or an equivalent augmented representation is used. A practical CVNN may be treated as a real-differentiable mapping $\mathbb{R}^{2d} \to \mathbb{R}$; global holomorphy of the network is not required.[23, 24] A nonconstant real-valued Hamiltonian generally cannot be a purely holomorphic function on a connected open set, so the physical readout may depend on both $\boldsymbol{\xi}$ and $\overline{\boldsymbol{\xi}}$.

For the theoretical analysis, the activation functions are chosen to be sufficiently smooth in their real and imaginary arguments, for example

$$\phi_{split}(z) = \tanh(Rez) + i\tanh(Imz). \tag{4.8}$$

modReLU, CReLU, and zReLU may be examined in numerical ablations, but their piecewise smoothness or boundary nondifferentiability prevents their direct use in statistical-learning–KAM arguments that require higher-order regularity. Complex weights use complex Glorot or semi-unitary initialization; the Hamiltonian backbone does not use BatchNorm based on batch statistics by default.[23, 24]

4.2.3. *Widely Linear Complex Layers and Real-Affine Completeness.* To retain an amplitude–phase inductive bias while preserving expressive compatibility with general real networks, the complex backbone uses a widely linear layer. Let

$$\mathbf{z}_{out} = \mathbf{W}\mathbf{z} + \mathbf{V}\overline{\mathbf{z}} + \mathbf{b} \tag{4.9}$$

where $\mathbf{z} = \mathbf{x} + i\mathbf{y}$, $\mathbf{W} = \mathbf{A} + i\mathbf{B}$, $\mathbf{V} = \mathbf{C} + i\mathbf{D}$, and $\mathbf{b} = \mathbf{b}_R + i\mathbf{b}_I$. Then

$$\mathbf{x}_{out} = (\mathbf{A} + \mathbf{C})\,\mathbf{x} + (-\mathbf{B} + \mathbf{D})\,\mathbf{y} + \mathbf{b}_R \tag{4.10}$$

$$\mathbf{y}_{out} = (\mathbf{B} + \mathbf{D})\,\mathbf{x} + (\mathbf{A} - \mathbf{C})\,\mathbf{y} + \mathbf{b}_I \tag{4.11}$$

Proposition 4.1 (Real-affine completeness)**.** *For any real affine mapping*

$$\mathbf{x}_{out} = \mathbf{M}_{11}\mathbf{x} + \mathbf{M}_{12}\mathbf{y} + \mathbf{c}_1 \tag{4.12}$$

$$\mathbf{y}_{out} = \mathbf{M}_{21}\mathbf{x} + \mathbf{M}_{22}\mathbf{y} + \mathbf{c}_2 \tag{4.13}$$

there exist unique $\mathbf{W}, \mathbf{V}, \mathbf{b}$ *for which the widely linear complex layer is exactly equivalent, with the explicit construction*

$$\mathbf{A} = \frac{\mathbf{M}_{11} + \mathbf{M}_{22}}{2}, \qquad \mathbf{C} = \frac{\mathbf{M}_{11} - \mathbf{M}_{22}}{2} \tag{4.14}$$

$$\mathbf{B} = \frac{\mathbf{M}_{21} - \mathbf{M}_{12}}{2}, \qquad \mathbf{D} = \frac{\mathbf{M}_{21} + \mathbf{M}_{12}}{2} \tag{4.15}$$

$$\mathbf{b} = \mathbf{c}_1 + i\mathbf{c}_2 \tag{4.16}$$

Proof. Substituting the stated construction into the real and imaginary equations recovers the four blocks $\mathbf{M}_{11}, \mathbf{M}_{12}, \mathbf{M}_{21}, \mathbf{M}_{22}$. Conversely, solving the four linear block equations gives uniqueness. □

When $\mathbf{V} = \mathbf{0}$, the real block matrix associated with a standard complex-linear layer must commute with the standard complex structure and is therefore a proper subclass of general real linear mappings. We use standard complex-linear layers as a stronger amplitude–phase structural bias and widely linear layers as the default theoretical model when general real-affine expressivity must be retained. This result establishes only a function-class inclusion and does not by itself imply fewer parameters or higher sample efficiency for the complex-valued model.

4.2.4. *Embedding of Real MLPs and Conditional Inheritance of Sobolev Approximation.*

Theorem 4.1 (Layerwise embedding of the real MLP class)**.** *Let a real-valued multilayer perceptron take* $(\mathbf{Q}, \mathbf{P})$ *as input and use a fixed real activation* σ *componentwise at every hidden layer, with* $\sigma(0) = 0$*. The complex network uses exactly* $\phi_\sigma(z) = \sigma(\operatorname{Re} z) + i\sigma(\operatorname{Im} z)$ *at the corresponding layers. Then there exists a network taking* $\boldsymbol{\xi} = (\mathbf{Q} + i\mathbf{P})/\sqrt{2}$ *and* $\overline{\boldsymbol{\xi}}$ *as inputs and using widely linear layers whose real-valued output is identical to that of the real MLP. This statement uses the widely linear class and does not assert the same coverage for standard complex-linear layers.*

Proof. The first layer treats $(Re\boldsymbol{\xi}, Im\boldsymbol{\xi})$ as the realified input and uses Proposition 4.1 to realize the target real affine mapping while setting the target imaginary preactivation to zero. The prescribed split activation then applies the same σ componentwise, and $\sigma(0) = 0$ keeps the imaginary channel zero. Starting from the second layer, the block-diagonal special case of Proposition 4.1 realizes each affine layer of the original real network. The final real-valued readout is therefore identical to that of the original MLP. □

Corollary 4.1 (Conditional inheritance of Sobolev approximation)**.** *Let* $\Omega \subset \mathbb{R}^{2d}$ *be a bounded Lipschitz domain,* $s \in \mathbb{N}$*, and* $1 \le r < \infty$*. If the class of real MLPs with the chosen activation is dense in* $W^{s,r}(\Omega)$*, then the class of real-valued readouts of the preceding widely linear complex networks is also dense in* $W^{s,r}(\Omega)$*.*

This corollary transfers an existing real-network density result and does not remove the original assumptions on the activation, domain, network width, or derivative approximation.[35] The standard complex-linear subclass is constrained by its block structure and does not automatically inherit the same Sobolev-density conclusion.

The theoretical advantages established by Zhang et al. for complex-reaction networks depend on their specific complex reaction structure, activation, and problem setting.[34] They cannot be generalized directly into the claim that an arbitrary complex-valued CHNN must outperform a real-valued HNN. Here, that work motivates only the possibility of a task-relevant inductive bias from complex structure; actual sample-efficiency and error advantages are evaluated through approximately parameter-matched ablation studies.

4.2.5. *Physical Limits of the Complex Representation.* Consider the complex scalar representation

$$\widetilde{\mathcal{H}} = H + iG \tag{4.17}$$

This is retained only as a special representation layer. A residual vector field may be represented by a real auxiliary potential G only if the corresponding one-form is closed and locally integrable. General quadratic drag, contact friction, magnetic control input, and background-flow effects do not satisfy this condition. The imaginary part is not interpreted directly as physical dissipation; physical energy is always given by the real-valued $\widehat{H}_\lambda$.

4.3. **Open-Port Dynamics and Structured Components.** The metriplectic framework of Coquinot–Morrison requires a Poisson bracket, a symmetric positive-semidefinite metric bracket, energy and entropy variables, and the corresponding degeneracy conditions and entropy-production relation.[30] The current catheter model does not establish an independent thermodynamic entropy field or these degeneracy structures. Therefore, the open-port model is not formulated metriplectically; that framework instead delineates the conditions required by future thermo-viscoelastic extensions and a rigorous entropy-production mechanism.

The complete structured model on the autonomous backbone is

$$\boxed{\dot{\mathbf{Z}} = \mathbf{J}\nabla H_\theta(\mathbf{Z}, \mu) + \mathbf{B}\left(\mathbf{Q}_{\text{diss}} + \mathbf{Q}_{\text{in}} + \mathbf{Q}_{\text{flow}} + \mathbf{Q}_{\text{boundary}} + \mathbf{Q}_{\text{contact}} + \mathbf{r}_\eta\right),} \tag{4.18}$$

where

$$\mathbf{B} = \begin{bmatrix} \mathbf{0} \\ \mathbf{I} \end{bmatrix}. \tag{4.19}$$

The additive terms in Equation (4.18) have distinct physical roles.

1. $\mathbf{Q}_{\text{diss}}$ represents only the passive component whose power with respect to the structural mechanical energy is nonpositive. For the equivalent-viscous subclass, one may take

$$\mathbf{Q}_{\text{diss}} = -\mathbf{D}_\theta \nabla_{\mathbf{P}} H_\theta, \qquad \mathbf{D}_\theta = \mathbf{L}_\theta \mathbf{L}_\theta^T \succcurlyeq 0, \tag{4.20}$$

- so that $P_{\text{diss}} = \nabla_{\mathbf{P}} H_\theta^T \mathbf{Q}_{\text{diss}} \leq 0$. This relation guarantees nonpositive instantaneous dissipative power for the adopted factorization. It does not imply monotone decay of the total open-system energy and is not, by itself, a dissipative-KAM hypothesis. Contact switching, Coulomb friction, and impact may remain separate ports rather than being forced into a single smooth damping matrix.[20]

2. $\mathbf{Q}_{\text{in}}$ represents magnetic control and other nonfluid inputs not included in the potential. Magnetic actions derivable from a potential at fixed magnetic parameters enter H_{phys}. A time-dependent conservative magnetic potential may either enter the nonautonomous Hamiltonian or enter the input port once as $-\nabla V_{drv}$; the two representations are mutually exclusive.
3. $\mathbf{Q}_{\text{flow}}$ represents the Froude–Krylov input, background-flow drag input, and reactive effects induced by the prescribed flow. The background-flow power is generally sign-indefinite and cannot be absorbed wholesale into a dissipation matrix.
4. $\mathbf{Q}_{\text{boundary}}$ is supplied by the known terminal force, terminal moment, and boundary actuation; $\mathbf{Q}_{\text{contact}}$ separately records contact and contact-reaction ports. Neither is relearned by the network in principle.
5. $\mathbf{r}_\eta$ represents only the residual remaining after the known physical backbone and all known ports have been removed.

This decomposition follows the separation of Hamiltonian, dissipation, and control in Dissipative SymODEN and the treatment of explicit time-dependent forcing and identifiability in port-Hamiltonian neural networks.[20, 21]

Numerical studies of pHNNs show that simultaneous identification of the Hamiltonian, time-dependent forcing, and damping from state trajectories alone may be non-identifiable, with one component compensating for another.[21] We therefore fix the physical Hamiltonian and known ports established in Sections 1 to 3; the network is allowed to learn only conservative corrections, sign-constrained dissipative corrections, and the residual after known terms have been removed. Staged training, port-power supervision, and sparsity or magnitude regularization are used to reduce mutual compensation among these components. These measures are structural constraints designed to mitigate leakage, but uniqueness and stability must still be tested through paired operating conditions and ablation experiments.

4.3.1. *Perturbation Types, Damping Classes, and Their KAM Interpretation.* To prevent distinct physical objects from being collapsed into one εH_1, we use the following type separation:

$$\boxed{\dot{\mathbf{z}} = X_{H_0} + \varepsilon X_{H_1} + \delta\, \mathbf{f}_{\text{diss}} + \mathbf{f}_{\text{ext}} + \mathbf{r}_\eta, \qquad X_{H_1} = \mathbf{J}\nabla H_1.} \tag{4.21}$$

Here X_{H_1} is a Hamiltonian perturbation, $\mathbf{f}_{\text{diss}}$ is a dissipative port, $\mathbf{f}_{\text{ext}}$ is an external/general port, and $\mathbf{r}_\eta$ is residual/model error. Only the Hamiltonian perturbation falls directly within classical Hamiltonian KAM theory. Dissipative terms, external inputs, and model residuals cannot be identified with H_1, nor can their KAM norm be inferred from the training loss alone. The Lean4 formalization represents these four classes separately so that this distinction is preserved in the finite-dimensional structural identities.

General learned damping may be written as $\mathbf{D}_\theta(\mathbf{Z}) \succeq 0$, but the present Lean result guarantees only nonpositive instantaneous dissipative power for the corresponding structure. It neither implies a conformally symplectic identity nor supplies the hypotheses required for dissipative KAM theory. The special choice $\mathbf{D} = \gamma\mathbf{I}$, $\gamma \geq 0$, may be examined as a weak-damping model in future dissipative or conformally symplectic analyses, but no such KAM result is claimed here.

4.3.2. *Lean4 Verification of Structural Identities.* Lean4 verifies finite-dimensional structural identities, not the complete physical model or the KAM analysis. The encoded relations include canonical skew structure, cancellation of internal Hamiltonian power, canonical port power, named-port decomposition, nonpositive passive dissipative power, exclusive magnetic accounting, and separation of open-port perturbation types. Results that depend on derivative existence, integrability, a nonautonomous chain rule, or specific physical provenance remain conditional. Appendix Section B states these boundaries in detail.

4.4. **COMSOL Data Hierarchy and Training Objects.**

4.4.1. *Kirchhoff-Consistent Primary Training Set.* The primary COMSOL training model uses the three-dimensional Euler–Bernoulli Beam interface with geometric nonlinearity, dynamic inertia, and torsional degrees of freedom. In COMSOL 6.3, the Beam interface applies the Euler–Bernoulli formulation to slender beams and neglects transverse shear while retaining three-dimensional displacement, cross-sectional rotation, and torque; geometric nonlinearity is represented in a corotational large-rotation, small-strain form.[4–6] The model is therefore consistent with the strict Kirchhoff branch in the shear-free assumption but is not unconditionally identical to the geometrically exact Cosserat equations.

The conservative training set is restricted to

$$\mathcal{D}_K^{cons} = \left\{ \left(\mathbf{Z}_i, \dot{\mathbf{Z}}_i, \widehat{H}_{phys,i} \right) : model_level = K,\ representation = sk,\ \mathbf{Q}_{port} = 0 \right\}. \tag{4.22}$$

The open-port training set is

$$\mathcal{D}_K^{port} = \left\{ \left(\mathbf{Z}_i, \dot{\mathbf{Z}}_i, \mathbf{Q}_{\mathrm{diss,i}}, \mathbf{Q}_{\mathrm{in,i}}, \mathbf{Q}_{\mathrm{flow,i}}, \mathbf{Q}_{\mathrm{boundary,i}}, \mathbf{Q}_{\mathrm{contact,i}}, \mathbf{r}_{\eta,i} \right) \right\}. \tag{4.23}$$

Only samples in $\mathcal{D}_K^{cons}$ are used when relating a learning-error bound to the small Hamiltonian perturbation required by KAM theory. The total loss of the open-port model has no such interpretation.

The primary experiment uses 96 complete trajectories in the fixed four-mode basis. A 64/16/16 whole-trajectory partition prevents samples from one physical trajectory from crossing the training, validation, and test subsets. The factorial comparison retains five random seeds for each of Real HNN, Gray Real HNN, Standard CHNN, and Gray CHNN; these optimization repetitions do not increase the number of independent physical samples. Residual MLP is evaluated under the same split as an external non-Hamiltonian baseline.

4.4.2. *Strict Weak-Form and High-Fidelity References.* When a term-by-term reproduction of the coupled weak form is required, Weak Form PDE or Weak Contribution is used to enter test functions, boundary contributions, and auxiliary variables.[7, 8] Timoshenko Beam and three-dimensional Solid Mechanics models are used to assess shear, local contact, and cross-sectional effects. For paired operating conditions, define

$$\Delta X_{HF} = X_{HF} - X_K. \tag{4.24}$$

Only the component that passes the closure test enters the conservative correction $\mathbf{J}\nabla\Delta H_{HF}$; a component with nonpositive power enters the dissipative correction; and all remaining components remain general residuals. The complete ΔX_{HF} must not be interpreted as a Hamiltonian perturbation or as H_1 in KAM theory.

4.5. **Loss Functions and Staged Training.**

4.5.1. *Conservative Derivative Supervision.* For $\mathcal{D}_K^{cons}$, define

$$\mathcal{L}_H = \frac{1}{N_c} \sum_{i=1}^{N_c} \left\| \dot{\mathbf{Z}}_i - \mathbf{J}\nabla \widehat{H}_\lambda \left(\mathbf{Z}_i \right) \right\|_2^2. \tag{4.25}$$

If the physical Hamiltonian is known, add supervision of the energy correction

$$\mathcal{L}_{\Delta H} = \frac{1}{N_c} \sum_{i=1}^{N_c} \left| \Delta \widehat{H}_\lambda \left(\mathbf{Z}_i \right) - \Delta \widehat{H}_{ref,i} \right|^2 . \tag{4.26}$$

4.5.2. *Port and Energy-Balance Supervision.* Define the port residual

$$\mathcal{L}_{port} = \frac{1}{N} \sum_{i=1}^{N} \| \dot{\mathbf{Z}}_i - \mathbf{J} \nabla H_\theta - \mathbf{B} \left(\mathbf{Q}_{\text{diss}} + \mathbf{Q}_{\text{in}} + \mathbf{Q}_{\text{flow}} + \mathbf{Q}_{\text{boundary}} + \mathbf{Q}_{\text{contact}} + \mathbf{r}_\eta \right) \|_2^2 . \tag{4.27}$$

The energy-balance loss is

$$\mathcal{L}_E = \frac{1}{N} \sum_{i=1}^{N} \left| \frac{dH_\theta}{d\tau} - \nabla_{\mathbf{P}} H_\theta^T \left(\mathbf{Q}_{\text{diss}} + \mathbf{Q}_{\text{in}} + \mathbf{Q}_{\text{flow}} + \mathbf{Q}_{\text{boundary}} + \mathbf{Q}_{\text{contact}} + \mathbf{r}_\eta \right) - \partial_\tau H_\theta \right|^2 . \tag{4.28}$$

The dissipation-sign constraint is imposed only on $\mathbf{Q}_{\text{diss}}$:

$$\mathcal{L}_{pass} = \frac{1}{N} \sum_{i=1}^{N} \left[\max \left\{ 0, \nabla_{\mathbf{P}} H_\theta^T \mathbf{Q}_{\text{diss}} \right\} \right]^2 . \tag{4.29}$$

4.5.3. *Trajectory, Gauge, and Symplectic-Structure Losses.* The trajectory loss is

$$\mathcal{L}_{traj} = \frac{1}{N} \sum_{i=1}^{N} \| \mathbf{Z}_{i,n+1} - Integrate \left(X_\theta, \mathbf{Z}_{i,n}, \Delta\tau \right) \|_2^2 . \tag{4.30}$$

The additive-gauge loss is

$$\mathcal{L}_{gauge} = \left| \widehat{H}_\lambda \left(\mathbf{Z}_{ref} \right) \right|^2 . \tag{4.31}$$

The symplectic-structure loss is applied only to the conservative submap $F_{h,\lambda}^{cons}$:

$$\mathcal{L}_{symp} = \frac{1}{N} \sum_{i=1}^{N} \| \left(DF_{h,\lambda}^{cons} \right)^T \mathbf{J} DF_{h,\lambda}^{cons} - \mathbf{J} \|_F^2 . \tag{4.32}$$

The complete dissipative map is generally not symplectic. The loss must therefore not be imposed on the complete open map and then interpreted as evidence of a strictly symplectic flow.

Training proceeds in three stages. The parameters of $\widehat{H}_\lambda$ are first estimated from the conservative data; the dissipative and external-input terms are then identified while the conservative component is held fixed or updated only weakly; finally, all components are optimized jointly. The connection to statistical-learning estimates and KAM theory uses only the conservative derivative error associated with $\mathcal{L}_H$ and its function-space generalization bound.

The formal E1 comparison isolates real versus complex representation and pure versus gray-box Hamiltonian prior under one data, optimization, and evaluation protocol; its independent-test results are reported in Section 6.3. E2 then treats the complete trajectory as the statistical unit. Joint learning under open-port conditions, higher-order function-space error certificates, and their implications for KAM perturbation estimates remain topics for future work.

4.5.4. *CHNN Implementation and Structural Validation.* To realize the preceding formulation numerically, we developed an independent CHNN implementation. The complex-valued operators were verified by unit tests and combined according to the canonical state representation, the gray-box Hamiltonian, the dissipative terms, and the external inputs defined above. The resulting dynamics are

$$H_\theta \left(\mathbf{x}, \boldsymbol{\mu} \right) = H_{phys} \left(\mathbf{x}, \boldsymbol{\mu} \right) + \Delta H_\theta \left(\mathbf{x}, \boldsymbol{\mu} \right) , \tag{4.33}$$

$$\dot{\mathbf{x}} = \mathbf{J}\nabla_{\mathbf{x}} H_\theta + \mathbf{B}\left(\mathbf{Q}_{diss} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_{\partial} + \mathbf{Q}_{contact} + \mathbf{r}_\eta\right), \qquad \mathbf{B} = \begin{bmatrix}\mathbf{0}\\ \mathbf{I}\end{bmatrix}. \tag{4.34}$$

The passive dissipative term is parameterized as

$$\mathbf{Q}_{diss} = -\mathbf{D}_\theta \nabla_{\mathbf{p}} H_\theta, \qquad \mathbf{D}_\theta = \mathbf{L}_\theta \mathbf{L}_\theta^T \succcurlyeq \mathbf{0}, \tag{4.35}$$

which gives

$$\left(\nabla_{\mathbf{p}} H_\theta\right)^T \mathbf{Q}_{diss} \leq 0. \tag{4.36}$$

The numerical implementation follows the preceding decomposition directly. The dimensional canonical variables $(\mathbf{q}, \mathbf{p})$ are first mapped to the nondimensional pair $(\mathbf{Q}, \mathbf{P})$, after which the complex coordinates are formed. The complex-valued network permits both standard complex-linear maps and widely linear maps of the form $\mathbf{W}\boldsymbol{\xi} + \mathbf{V}\overline{\boldsymbol{\xi}} + \mathbf{b}$. The Hamiltonian component always returns a real scalar and uses the gray-box form $H_{phys} + \Delta H_\theta$. Dissipation, magnetic actuation, flow, boundary effects, contact, and unresolved residual dynamics are represented as separate generalized-force terms that enter only the momentum equation through $\mathbf{B}$. Samples generated from different physical models or momentum definitions retain their model origin and canonical representation before training so that mechanically distinct quantities are not combined in the same loss.

The implementation was checked against the structural identities used in the analysis, including the equivalence between widely linear complex layers and real block-linear maps, round-trip consistency of canonical scaling, recovery of the prescribed Galerkin subspace, the additive decomposition of the Hamiltonian, energy orthogonality of the continuous conservative vector field, and nonpositive passive dissipative power. External inputs act only through the momentum equation and do not directly modify the $\dot{\mathbf{q}}$ channel. These checks establish consistency between the numerical implementation and the mathematical formulation; they do not constitute independent evidence for identification accuracy on the COMSOL catheter data or for KAM stability. In the general physical branch, the Hamiltonian $\widehat{H}_{phys}$, configuration-dependent mass–inertia matrix, nonautonomous terms, and COMSOL–Galerkin projection are determined from the physical model and numerical data considered below; the learning experiments instead use the fixed-basis constant-M_4 approximation stated in Section 6.

4.6. **From CHNN Learning Error to the KAM Smallness Condition.**

4.6.1. *The Integrable Reference System Is Not an Automatic Consequence of the Network Real Part.* Section 3 constructs, through a stable equilibrium, modal canonicalization, and a Hamiltonian Birkhoff normal form,

$$H_{cons} = H_0 + \varepsilon H_1. \tag{4.37}$$

The Hamiltonian H_0 must be verified to be integrable and nondegenerate. A real-valued H_λ is not integrable merely because it is real. The linear harmonic oscillator satisfies $\partial_{\mathbf{I}}^2 H_2 = 0$, so nondegeneracy must arise from higher-order conservative terms.

4.6.2. *Conservative Training Data, Function Class, and Additive Gauge.* The following statistical-learning statements apply only to the conservative training set $\mathcal{D}_K^{cons}$. Let $\Omega \subset \mathbb{R}^{2d}$ be a bounded connected Lipschitz domain, let the true conservative Hamiltonian be $H_\star$, and let the samples $\mathbf{Z}_1, \ldots, \mathbf{Z}_N$ be independent and identically distributed according to a probability measure $\mathbb{P}_Z$. The generalization theorem is a conditional function-class result. Temporally correlated states sampled along a single COMSOL trajectory are not interpreted as independent physical samples in the empirical evaluation. Assume noiseless derivative labels satisfy

$$\dot{\mathbf{Z}}_i = \mathbf{J}\nabla H_\star\left(\mathbf{Z}_i\right). \tag{4.38}$$

Represent the complex parameters of the Hamiltonian network by the real parameter vector $\boldsymbol{\vartheta} \in \Theta = [-B, B]^P$, and denote the learning error by

$$e_{\boldsymbol{\vartheta}} = \widehat{H}_{\boldsymbol{\vartheta}} - H_{\star}. \tag{4.39}$$

State derivatives alone do not identify the absolute constant of the Hamiltonian. Therefore, impose on the theoretical domain

$$\int_{\Omega} e_{\boldsymbol{\vartheta}}\left(\mathbf{Z}\right)\, d\mathbf{Z} = 0, \tag{4.40}$$

or use a known energy anchor to select the same representative of the additive equivalence class. Choose a statistical exponent $r > 2d$ and define the scalar gradient loss

$$g_{\boldsymbol{\vartheta}}\left(\mathbf{Z}\right) = \parallel \dot{\mathbf{Z}} - \mathbf{J}\nabla\widehat{H}_{\boldsymbol{\vartheta}}\left(\mathbf{Z}\right) \parallel_{\ell^r}^r = \parallel \nabla e_{\boldsymbol{\vartheta}}\left(\mathbf{Z}\right) \parallel_{\ell^r}^r . \tag{4.41}$$

To apply a uniform generalization bound for bounded losses, also assume $0 \leq g_{\boldsymbol{\vartheta}} \leq c$. This boundedness is an independent regularity assumption and does not follow automatically from a small empirical loss.

4.6.3. *Covering Numbers and Rademacher Complexity.* For a sample $S = (\mathbf{Z}_1, \ldots, \mathbf{Z}_N)$, define the normalized empirical metric

$$d_S(g, g\prime) = \left(\frac{1}{N} \sum_{i=1}^{N} \left| g\left(\mathbf{Z}_i\right) - g\prime\left(\mathbf{Z}_i\right) \right|^2 \right)^{1/2} . \tag{4.42}$$

Assumption 4.1 (Samplewise Lipschitz continuity from parameters to loss)**.** For a given sample S, there exists a finite constant L_S such that every $\boldsymbol{\vartheta}, \boldsymbol{\vartheta}\prime \in \Theta$ satisfies

$$d_S\left(g_{\boldsymbol{\vartheta}}, g_{\boldsymbol{\vartheta}\prime}\right) \leq L_S \parallel \boldsymbol{\vartheta} - \boldsymbol{\vartheta}\prime \parallel_2 . \tag{4.43}$$

When the input domain and parameter set are bounded, the network depth is finite, and the first and second derivatives of the activation are bounded on the relevant range, this assumption can hold on a compact parameter set. In general, L_S grows with depth, width, weight bounds, and derivative bounds.

Lemma 4.1 (Finite-parameter covering number)**.** *Under Assumption 4.1, let* $\mathcal{G} = \{g_{\boldsymbol{\vartheta}} : \boldsymbol{\vartheta} \in \Theta\}$*, and let* $\mathcal{G}_S$ *denote the restriction of this function class to the sample* S*. Then*

$$\mathcal{N}\left(\varepsilon, \mathcal{G}_S, d_S\right) \leq \left(1 + \frac{2B\sqrt{P}\, L_S}{\varepsilon} \right)^P \tag{4.44}$$

Proof. On the parameter cube $[-B, B]^P$, place a grid of Euclidean radius ε / L_S, and use Assumption 4.1 to push the parameter covering forward to a covering of the loss functions. □

Because the parameter set is compact and the loss is continuous in the parameters, the supremum is attained. Define the empirical Rademacher complexity by

$$\widehat{\mathfrak{R}}_S\left(\mathcal{G}\right) = \mathbb{E}_{\boldsymbol{\sigma}} \left[\max_{g \in \mathcal{G}} \frac{1}{N} \sum_{i=1}^{N} \sigma_i g\left(\mathbf{Z}_i\right) \right] \tag{4.45}$$

Let

$$R_S = \max_{g \in \mathcal{G}} d_S(g, 0) \tag{4.46}$$

Dudley's entropy integral gives[36]

$$\widehat{\mathfrak{R}}_S\left(\mathcal{G}\right) \leq \inf_{0 < \alpha < R_S} \left[4\alpha + \frac{12}{\sqrt{N}} \int_{\alpha}^{R_S} \sqrt{\log \mathcal{N}\left(\varepsilon, \mathcal{G}_S, d_S\right)}\, d\varepsilon \right] \tag{4.47}$$

A purely complex-linear network is a structurally constrained subclass of the corresponding real network. A widely linear network must instead be counted using its actual number P of

real parameters and its actual Lipschitz constant L_S; "one complex parameter" must not be incorrectly counted as one real degree of freedom.

4.6.4. *From Gradient Risk to Uniform Hamiltonian Error.* Assume that the loss class is fixed before sampling. A standard uniform Rademacher generalization bound then implies that, with probability at least $1-\delta$, simultaneously for every $\boldsymbol{\vartheta} \in \Theta$,

$$\mathbb{E}_{\mathbb{P}_Z} g_{\boldsymbol{\vartheta}} \leq \widehat{\mathcal{L}}_r(\boldsymbol{\vartheta}) + 2\widehat{\mathfrak{R}}_S(\mathcal{G}) + 3c\sqrt{\frac{\log(2/\delta)}{2N}} =: B_{N,\delta}(\boldsymbol{\vartheta}). \tag{4.48}$$

Assume further that $\mathbb{P}_Z$ has a density f_Z with respect to Lebesgue measure and that throughout Ω,

$$f_Z(\mathbf{Z}) \geq m_Z > 0. \tag{4.49}$$

Then, on the preceding uniform event,

$$\| \nabla e_{\boldsymbol{\vartheta}} \|_{L^r(\Omega;\ell^r)} \leq \left(\frac{B_{N,\delta}(\boldsymbol{\vartheta})}{m_Z} \right)^{1/r}. \tag{4.50}$$

Using the additive gauge, the Poincaré inequality, and the Morrey–Sobolev embedding for $r > 2d$,[37] one obtains:

Theorem 4.2 (Uniform generalization bound for the Hamiltonian)**.** *Under the preceding assumptions, with probability at least $1-\delta$, simultaneously for every $\boldsymbol{\vartheta} \in \Theta$,*

$$\boxed{\| \widehat{H}_{\boldsymbol{\vartheta}} - H_\star \|_{L^\infty(\Omega)} \leq C_{\Omega,r}(1 + C_P) \left(\frac{B_{N,\delta}(\boldsymbol{\vartheta})}{m_Z} \right)^{1/r}.} \tag{4.51}$$

where C_P is the Poincaré constant. The overall $1/r$ power arises from converting the r-th-power risk back to a gradient norm and cannot be absorbed into a constant independent of the training error and sample size.

4.6.5. *Higher-Order Sobolev or Analytic Error Certificate.* Theorem 4.2 provides only a C^0-level uniform error. A high-frequency, small-amplitude function may converge to zero in C^0 while its higher derivatives diverge. Therefore, neither a first-derivative risk nor proximity of Hamiltonian values alone is sufficient to claim the KAM smallness condition.

Assume that Section 3 has constructed an integrable reference and coordinate transformation on the local action–angle domain

$$\mathcal{A} = \mathbb{T}^M \times U \tag{4.52}$$

If the original training data are represented in physical canonical coordinates, the learning error must be pulled back to $\mathcal{A}$, and bounds on the higher derivatives of the coordinate transformation and its inverse must be included in the constants. Choose integers s, ℓ and a statistical exponent r satisfying

$$s > \ell + \frac{2M}{r}. \tag{4.53}$$

Let the training algorithm output $\widehat{\boldsymbol{\vartheta}}$. In addition to the first-order statistical certificate from Theorem 4.2, require a higher-order certificate for this specific output: on an event of probability at least $1-\eta$,

$$\sum_{2 \leq |\alpha| \leq s} \| D^\alpha \left(\widehat{H}_{\widehat{\boldsymbol{\vartheta}}} - H_\star \right) \|^r_{L^r(\mathcal{A})} \leq C_{s,\eta} \left(\widehat{\boldsymbol{\vartheta}} \right)^r. \tag{4.54}$$

The certificate must control higher derivatives of the true learning error, not merely the magnitude of the network's own derivatives. Periodic Sobolev embedding then gives, on the intersection of the two events,

$$\| \widehat{H}_{\widehat{\boldsymbol{\vartheta}}} - H_\star \|_{C^\ell(\mathcal{A})} \leq \mathfrak{C}_{N,\delta,\eta}\left(\widehat{\boldsymbol{\vartheta}}\right), \tag{4.55}$$

where one may take

$$\mathfrak{C}_{N,\delta,\eta} = C_{s,r,\mathcal{A}} \left[(C_P^r + 1) \frac{B_{N,\delta}\left(\widehat{\boldsymbol{\vartheta}}\right)}{m_Z} + C_{s,\eta}\left(\widehat{\boldsymbol{\vartheta}}\right)^r \right]^{1/r}. \tag{4.56}$$

If the higher-order certificate is obtained by deterministic interval validation or interpolation with a rigorous remainder, set $\eta = 0$. If it is itself probabilistic, independence from the generalization event is not assumed, and the union bound yields only a joint probability of at least $1 - \delta - \eta$.

If an analytic KAM theorem is selected, the preceding finitely smooth certificate is still insufficient; the analytic supremum norm of the learning error on a specified complex neighborhood must be controlled separately. Explicit dependence of a widely linear network on $\boldsymbol{\xi}$ and $\overline{\boldsymbol{\xi}}$ does not prevent the real-variable Hamiltonian from admitting a real-analytic extension, but smallness in a real-domain C^ℓ norm does not automatically imply smallness in a complex-neighborhood analytic norm.

4.6.6. *Accounting for the Integrable Reference, Pre-Existing Physical Perturbation, and Learning Error.* Fix the integrable reference Hamiltonian $h(\mathbf{I})$ constructed and verified in Section 3. Write the true conservative system and the learned system as

$$H_\star = h + P_0, \qquad \widehat{H}_{\widehat{\boldsymbol{\vartheta}}} = H_\star + e_{\widehat{\boldsymbol{\vartheta}}} = h + P_{\widehat{\boldsymbol{\vartheta}}}, \tag{4.57}$$

where

$$P_{\widehat{\boldsymbol{\vartheta}}} = P_0 + e_{\widehat{\boldsymbol{\vartheta}}}. \tag{4.58}$$

Therefore, in the common $C^\ell(\mathcal{A})$ norm,

$$\| P_{\widehat{\boldsymbol{\vartheta}}} \|_{C^\ell} \leq \| P_0 \|_{C^\ell} + \mathfrak{C}_{N,\delta,\eta}\left(\widehat{\boldsymbol{\vartheta}}\right). \tag{4.59}$$

If the true system is already only nearly integrable, P_0 consumes part of the KAM perturbation budget; the entire threshold cannot be assigned to the learning error. Coordinate-transformation, Galerkin-truncation, parameter-identification, and model-closure errors may be added through the triangle inequality only after they have been controlled in the same function-space norm used by the KAM theorem. General dissipation, friction, background-flow input, and nonpotential magnetic control are not Hamiltonian perturbations here and remain in the open-port branches of Section 4.

4.6.7. *Probabilistic Implication for the KAM Small-Perturbation Condition.* Select an external KAM theorem in advance. Let $\mathsf{A}_{KAM}(h, P)$ denote all of its structural assumptions other than perturbation smallness, including the action–angle domain, regularity order, type of nondegeneracy, Diophantine condition, distance to the boundary, and admissible perturbation class. Let ε_{KAM} be its smallness threshold and let $\mathcal{K}(h + P)$ denote the torus-persistence conclusion supplied by that theorem.

Proposition 4.2 (Probabilistic implication)**.** *Under the preceding assumptions, with probability at least* $1 - \delta - \eta$*,*

$$\boxed{\mathsf{A}_{KAM}\left(h, P_{\widehat{\boldsymbol{\vartheta}}}\right) \ \wedge \ \left(\| P_0 \|_{C^\ell(\mathcal{A})} + \mathfrak{C}_{N,\delta,\eta}\left(\widehat{\boldsymbol{\vartheta}}\right) < \varepsilon_{KAM} \right) \ \Rightarrow \ \mathcal{K}\left(\widehat{H}_{\widehat{\boldsymbol{\vartheta}}}\right).} \tag{4.60}$$

Proof. On the intersection of the uniform-generalization event and the higher-order-certificate event, Section 4.6.5 gives a C^ℓ bound for the learning error. The triangle inequality in Section 4.6.6 then makes the total Hamiltonian perturbation smaller than ε_{KAM}. If all remaining conditions in A_{KAM} also hold, the selected KAM theorem applies. By the union bound, the probability of the joint event is at least $1 - \delta - \eta$. □

This proposition converts the stated learning-error bound only into the small-perturbation hypothesis of an external KAM theorem. The regularity class, measure estimate, frequency properties, and geometry of the persistent tori remain those of the chosen KAM theorem and cannot be inferred from the training loss alone.

4.6.8. *Continuous and Discrete Error Budgets.* On a common analytic domain $\mathcal{D}_\rho$, define

$$\varepsilon_{model} = \| X_{cons} - X_0 \|_\rho, \qquad \varepsilon_{learn} = \| X_\lambda - X_{cons} \|_\rho . \tag{4.61}$$

Let Φ_h^0 be the exact map of X_0 over a time step h, and let $\Psi_{h,\lambda}$ be an order-s_a symplectic algorithm applied to the learned Hamiltonian. If the vector fields and their first derivatives are uniformly bounded and the trajectories remain in a smaller analytic domain $\mathcal{D}_{\rho\prime}$, then there exists a constant $C_{\rho,\rho\prime}$ such that

$$\boxed{\| \Psi_{h,\lambda} - \Phi_h^0 \|_{\rho\prime} \leq C_{\rho,\rho\prime} h \left(\varepsilon_{model} + \varepsilon_{learn} + h^{s_a}\right) =: \varepsilon_{map}.} \tag{4.62}$$

Here, h^{s_a} arises from the local algorithmic error. This budget does not assume unconditionally that a globally exact "discrete Hamiltonian" exists. Only when

$$\varepsilon_{map} < \delta_{KAM} \left(\rho\prime, \gamma, \tau, \text{nondegeneracy margin}, h\right) \tag{4.63}$$

and the analyticity, Diophantine nonresonance, and step-size nonresonance conditions are satisfied may a specific symplectic-map KAM theorem be invoked.[25–27, 33]

4.7. **Three Discrete-Dynamics Branches.**

1. **Autonomous conservative case.** Apply a generating-function method, implicit midpoint rule, or another algorithm proved to be symplectic to $\dot{\mathbf{Z}} = \mathbf{J}\nabla\widehat{H}_\lambda$, yielding F_h^{cons}. This map is used for Shang's KAM theory for symplectic algorithms and for validation by the Haro parameterization method.
2. **Exact nonautonomous Hamiltonian case.** Apply the standard nonautonomous canonical equations to $H(\mathbf{Z}, \tau)$, or introduce the extended variables (τ, p_τ) and $\mathcal{H} = H + p_\tau$ and use a symplectic algorithm in extended phase space. Explicit time dependence alone does not require a nonstandard $K(\mathbf{z}, t)$-symplectic structure.
3. **Open-port case.** Integrate the complete forced–dissipative system consistently while monitoring its energy balance. Its map is generally neither a standard symplectic map nor automatically a Birkhoff system. A $K(\mathbf{z}, t)$-symplectic formulation of the type discussed in Section 5 is applicable only if the residual vector field, after subtraction of the known Hamiltonian and port contributions, satisfies the required self-adjoint representation condition.

4.8. **Output Objects for Invariant-Torus Analysis.** The objects required by the invariant-torus analysis in Section 5 are therefore

$$\mathcal{O}_{4\to5} = \left(F_h^{cons}, \varepsilon_{map}, X_{res}, \mathcal{D}, \gamma, \tau, \text{nondegeneracy margin}, model_level, representation\right), \tag{4.64}$$

where

$$\boxed{X_{res} = X_{data} - \mathbf{J}\nabla H_\theta - \mathbf{B}\left(\mathbf{Q}_{\text{diss}} + \mathbf{Q}_{\text{in}} + \mathbf{Q}_{\text{flow}} + \mathbf{Q}_{\text{boundary}} + \mathbf{Q}_{\text{contact}}\right).} \tag{4.65}$$

If any quantity has not yet been constructed or verified numerically, subsequent sections list it only as a condition to be tested rather than as an established conclusion.

5. Symplectic Discretization, Birkhoff Structures, and Invariant-Torus Validation

The learned conservative dynamics become relevant to long-time analysis only after the dynamical object and preserved structure have been specified. This section separates canonical Hamiltonian flows, nonautonomous extensions, and Birkhoff systems before developing the local quartic KAM result.

5.1. **Analysis branches.** The KAM analysis follows a fixed-dimensional Galerkin truncation and therefore uses finite-dimensional Hamiltonian-flow or symplectic-map results. Kuksin's infinite-dimensional KAM theory[32] is relevant only to possible infinite-mode or PDE limits. Moser's invariant-curve theorem for area-preserving annulus maps[31] concerns two-dimensional twist maps and cannot replace the Pöschel/Shang conditions for multi-degree-of-freedom Hamiltonian systems or high-dimensional symplectic maps.[26, 27, 33]

The systems considered here do not all share the same symplectic structure. We therefore distinguish among:

- standard $\mathbf{J}$-symplectic discretization of an autonomous conservative Hamiltonian system;
- standard nonautonomous or extended-phase-space symplectic discretization of an exact nonautonomous Hamiltonian system;
- time-dependent $\mathbf{K}(\mathbf{z}, t)$-symplectic discretization of a Birkhoff system that passes the variational self-adjointness test.

The Hamiltonian Birkhoff normal form is used in Sections 3 and 4 to construct $H_0 + \varepsilon H_1$. The self-adjoint Birkhoff representation considered here is used to test the structure of a general first-order system. The names are similar, but the mathematical objects are different.

5.2. **Standard symplectic formulation and generating functions.** For the autonomous conservative CHNN

$$\dot{\mathbf{Z}} = \mathbf{J}\nabla \widehat{H}_{\lambda}(\mathbf{Z}), \tag{5.1}$$

the exact flow satisfies

$$(D\Phi_t)^T \mathbf{J}(D\Phi_t) = \mathbf{J}. \tag{5.2}$$

If the one-step map F_h^{cons} is an exact symplectic map sufficiently close to the identity and the relevant Jacobian block of the chosen projection is nondegenerate, a local generating function exists. Retaining the leading term of a center-type generating function gives the implicit midpoint rule

$$\frac{\mathbf{Z}_{n+1} - \mathbf{Z}_n}{h} = \mathbf{J}\nabla \widehat{H}_{\lambda}\left(\frac{\mathbf{Z}_{n+1} + \mathbf{Z}_n}{2}\right), \tag{5.3}$$

which is a second-order symplectic method. Higher-order methods may be constructed through generating-function expansions, composition methods, or other proven symplectic algorithms[25].

For a nonautonomous Hamiltonian $H(\mathbf{Z}, t)$, introduce the extended phase space

$$\mathcal{H}(\mathbf{Z}, t, p_t) = H(\mathbf{Z}, t) + p_t, \tag{5.4}$$

and integrate with respect to the extended canonical symplectic form. A transition to the formulation of the next subsection is needed only when the variables are noncanonical or when the system possesses a special Birkhoff representation.

The second-order BDF scheme used to generate canonical data is not treated as a symplectic integrator. It is used only to obtain time-consistent COMSOL state samples and vector-field labels. The three-level convergence study assesses the temporal consistency of this data mapping; neither the observed energy drift nor the second-order convergence of BDF is used to infer symplecticity. Subsequent numerical KAM calculations must employ a separately justified symplectic map and quantify its discretization error.

5.3. **Self-adjoint representation of a Birkhoff system.** Let $\mathbf{z} \in \mathbb{R}^{2n}$, let $F_i(\mathbf{z}, t)$ be Birkhoff functions, let $B(\mathbf{z}, t)$ be the Birkhoffian, and define

$$K_{ij}(\mathbf{z}, t) = \frac{\partial F_j}{\partial z_i} - \frac{\partial F_i}{\partial z_j}. \tag{5.5}$$

The Birkhoff equations are

$$\boxed{K_{ij}\left(\mathbf{z},t\right)\dot{z}_j-\frac{\partial B}{\partial z_i}-\frac{\partial F_i}{\partial t}=0.} \tag{5.6}$$

Consider a general system written as

$$K_{ij}\dot{z}_j+D_i=0. \tag{5.7}$$

On a local star-shaped domain, if $\mathbf{K}$ is nondegenerate and satisfies skew symmetry, the spatial closure condition, and the temporal compatibility condition, then the system admits a regular self-adjoint Birkhoff representation. Equivalently, functions F_i and B can be found such that the preceding equation holds[38]. Explicitly, the required conditions are $K_{ij}+K_{ji}=0$, $\partial_{z_k}K_{ij}+\partial_{z_i}K_{jk}+\partial_{z_j}K_{ki}=0$, and $\partial_t K_{ij}=\partial_{z_j}D_i-\partial_{z_i}D_j$, together with nondegeneracy and a local star-shaped (hence locally contractible) domain. Define the time-dependent two-form

$$\Omega_K=\frac{1}{2}K_{ij}\left(\mathbf{z},t\right)\,dz_i\wedge dz_j. \tag{5.8}$$

Its phase flow satisfies the time-dependent structure-preservation relation

$$\left(D\Phi_{t,t_0}\right)^T\mathbf{K}\left(\Phi_{t,t_0},t\right)D\Phi_{t,t_0}=\mathbf{K}\left(\mathbf{z}_0,t_0\right). \tag{5.9}$$

When $\mathbf{K}=\mathbf{J}^{-1}$ is constant, this condition reduces to the canonical symplectic condition.

5.4. **Identification of a Birkhoff structure for the residual vector field.** The analysis is applied only to X_{res}, as defined in Section 4. Parameterize the one-form directly as

$$\mathcal{F}_\rho=\sum_{i=1}^{2n}F_{\rho,i}\left(\mathbf{z},t\right)\,dz_i,\qquad K_{\rho,ij}=\partial_{z_i}F_{\rho,j}-\partial_{z_j}F_{\rho,i}, \tag{5.10}$$

introduce B_β, and define the candidate vector field

$$\widetilde{X}_{\rho,\beta}=\mathbf{K}_\rho^{-1}\left(\nabla B_\beta+\partial_t\mathbf{F}_\rho\right). \tag{5.11}$$

The training loss is

$$\mathcal{L}_{Birk}=\frac{1}{N}\sum_{\ell=1}^{N}\;\|\;\mathbf{K}_\rho X_{res}-\nabla B_\beta-\partial_t\mathbf{F}_\rho\;\|_2^2+\mu_{nd}\frac{1}{N}\sum_{\ell=1}^{N}\left[\max\{0,\kappa-\sigma_{min}\left(\mathbf{K}_\rho\right)\}\right]^2. \tag{5.12}$$

On the domain $\mathcal{D}$, define the defect of the self-adjoint Birkhoff representation by

$$\boxed{\delta_{Birk}=\inf_{F,B,\;\inf_{\mathcal{D}}\sigma_{min}(K_F)\geq\kappa}\;\|\;K_F X_{res}-\nabla B-\partial_t F\;\|_{\mathcal{X}(\mathcal{D})}\,.} \tag{5.13}$$

If $\delta_{Birk}=0$, an exact representation is obtained. If $\delta_{Birk}>0$, only the projected vector field $\widetilde{X}$ has an exact K-symplectic structure, and the remaining error must be controlled separately. The complete open system containing a nonstructured residual cannot therefore be described as a strictly $K\left(\mathbf{z},t\right)$-symplectic system.

5.5. **A $K\left(\mathbf{z},t\right)$-symplectic discretization for Birkhoff systems.** Under regularity, self-adjointness, analyticity, and nondegeneracy conditions, the Birkhoff phase flow admits a gradient map and a scalar generating function under an appropriate local projection. If the generating function is analytic in the step size, it can be expanded as

$$S\left(\mathbf{Z},h\right)=\sum_{r=1}^{\infty}S_r\left(\mathbf{Z}\right)h^r. \tag{5.14}$$

An exact generating-function implementation gives the exact one-step relation

$$\left(D\Psi_h\right)^T\mathbf{K}\left(\Psi_h\left(\mathbf{z}\right),t+h\right)D\Psi_h=\mathbf{K}\left(\mathbf{z},t\right). \tag{5.15}$$

Let Φ_h denote the exact flow. An order-m generating-function truncation defines a theoretical map Ψ_h with local approximation error

$$\Psi_h(\mathbf{z}) - \Phi_h(\mathbf{z}) = O(h^{m+1}). \tag{5.16}$$

Provided that the truncated generating function still defines a regular K-symplectic map, Ψ_h continues to satisfy Equation (5.15) exactly: truncation order controls flow accuracy, not structural preservation. For an implemented map $\widetilde{\Psi}_h$, the structural defect is measured separately as

$$R_{K,h} := \left(D\widetilde{\Psi}_h\right)^T \mathbf{K}\left(\widetilde{\Psi}_h(\mathbf{z}), t+h\right) D\widetilde{\Psi}_h - \mathbf{K}(\mathbf{z}, t). \tag{5.17}$$

Only implementation approximations that break the structural identity contribute to this defect. A corresponding decomposition and norm bound may be written as

$$R_{K,h} = R_{\text{solve}} + R_{\text{diff}} + R_{\text{proj}} + R_{\text{impl}}, \qquad \|R_{K,h}\| \le C_{\text{solve}}\tau_{\text{solve}} + C_{\text{diff}}\tau_{\text{diff}} + C_{\text{proj}}\tau_{\text{proj}} + C_{\text{impl}}\tau_{\text{impl}}. \tag{5.18}$$

An $O(h^{m+1})$ term is not inserted into this structural-defect bound without an independent derivation showing that a particular implementation approximation breaks Equation (5.15). The present data package contains no independent numerical certificate for $R_{K,h}$; therefore exact K-symplecticity, numerical order, and implementation residual are not conflated. None of these statements by itself implies persistence of classical Hamiltonian KAM tori.

5.6. **Local quartic Birkhoff normal form and qualitative KAM existence.** For the selected material-scalar conservative reference, the equilibrium is first recentered in the common modal coordinates. The four linear frequencies of the reconstructed local Hamiltonian are

$$\frac{\boldsymbol{\omega}}{2\pi} = (5.56299685,\ 5.60626502,\ 34.66176454,\ 35.09209478)\ \text{Hz}. \tag{5.19}$$

The cubic and quartic Hamiltonian terms are then reconstructed and a fourth-order nonresonant Birkhoff normalization is performed. After the local canonical Lie transformation, the fourth-order jet of the selected exact quartic conservative approximation satisfies

$$J_4K = \boldsymbol{\omega}\cdot\mathbf{I} + Z_4(\mathbf{I}), \qquad R_{\text{exact}} = K - J_4K = O(\|W\|^5). \tag{5.20}$$

The polynomial-algebra check gives a quartic homological residual of 2.58368×10^{-80}. For the primary J_{ABC} reconstruction, the twist matrix

$$T = \nabla^2_{\mathbf{I}} Z_4(\mathbf{I}) \tag{5.21}$$

has full rank, $\operatorname{rank} T = 4$, with dimensionless minimum singular value 6.35822×10^{-11}. The three auxiliary matrices J_{AB}, J_{AC}, and J_{BC}, each omitting one modal pack, also have rank four. Because a rigorous bound on model-identification error is not yet available, these values establish local nondegeneracy only for the selected quartic approximation and are not interpreted as a uniform robust twist bound for the material-reference system.

The 50-, 80-, 120-, and 160-digit recomputations are summarized in Figure 5.1. Stability of the rank, determinant sign, singular spectrum, and low-order divisor supports the arithmetic reproducibility of the stored decimal coefficients; it does not bound uncertainty in the upstream physical coefficients.

With $W = \varepsilon w$, the Hamiltonian normalized by the quartic homogeneous scale is

$$K_\varepsilon(w) = \varepsilon^{-4}K(\varepsilon w) = \varepsilon^{-2}H_2(w) + Z_4(\mathbf{J}) + \varepsilon R_{5,\varepsilon}(w). \tag{5.22}$$

Because the selected exact quartic conservative approximation and the generators χ_3 and χ_4 are polynomial, the local canonical transformations are analytic. On a fixed small complex neighborhood there is a finite constant C such that

$$\|P_\varepsilon\| \le C\varepsilon \longrightarrow 0. \tag{5.23}$$

The normal-form frequency map is

$$\boldsymbol{\Omega}_\varepsilon(\mathbf{J}) = \varepsilon^{-2}\boldsymbol{\omega} + T\mathbf{J}, \qquad D\boldsymbol{\Omega}_\varepsilon = T, \tag{5.24}$$

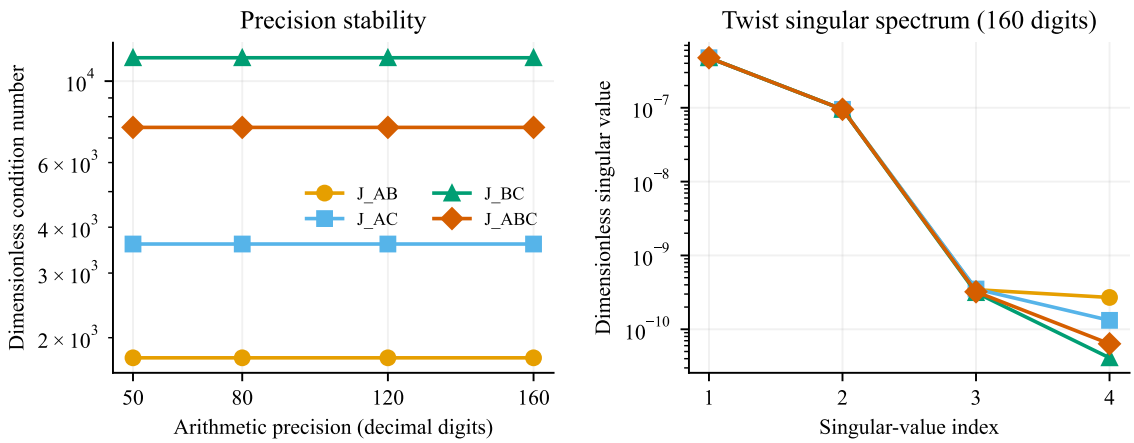


Figure 5.1. Precision comparison of the dimensionless quartic twist matrices. Left: condition numbers at 50–160 decimal digits. Right: singular spectra at 160 digits. The curves assess arithmetic reproducibility of the stored decimal coefficients, not physical-model uncertainty or an explicit invariant torus.

so $\det T \neq 0$ yields a locally open frequency image.

Proposition 5.1 (Local qualitative KAM existence for the exact quartic conservative approximation)**.** *For the exact quartic conservative approximation defined above, there exists an unspecified $\varepsilon_* > 0$ such that, for sufficiently small nonzero amplitudes $0 < \varepsilon < \varepsilon_*$, a positive-measure Cantor family of Lagrangian invariant tori exists near and approaches the elliptic equilibrium.*

The proposition follows the classical Kolmogorov KAM route: analyticity follows from the finite polynomial Hamiltonians and local canonical transformations, Equation (5.23) provides the small-perturbation condition, and $\operatorname{rank} T = 4$ gives Kolmogorov nondegeneracy. Removing resonant frequencies from the open frequency image leaves a Diophantine Cantor subset[33, 39]. This local statement does not require the entire prescribed action region to be resonance free. In fact, the finite-order K32 scan detects a resonance crossing within that region, so no numerical KAM conclusion is drawn for the full region. The proposition provides no explicit ε_*, Diophantine constant, torus measure fraction, or specific torus embedding.

The result is restricted to the selected J_{ABC} quartic conservative approximation. On the sampled local8 set, its maximum relative error with respect to the full material-reference Hamiltonian is 4.43237×10^{-8}; this number is a finite-sample error estimate rather than a supremum bound on a neighborhood. Because no rigorous $O(5+)$ neighborhood bound is available, Proposition 5.1 cannot be extended to a KAM theorem for the full material-reference Hamiltonian.

5.7. **Invariant-torus computation by the parameterization method.** For the exact symplectic one-step map F_h^{cons} of the autonomous conservative system, seek an embedding

$$K : \mathbb{T}^d \to \mathcal{M} \tag{5.25}$$

satisfying

$$\boxed{F_h^{cons}\left(K\left(\boldsymbol{\theta}\right)\right) = K\left(\boldsymbol{\theta} + \boldsymbol{\omega}\right).} \tag{5.26}$$

Use the Fourier truncation

$$K_N\left(\boldsymbol{\theta}\right) = \sum_{|k| \leq N} \widehat{K}_k e^{2\pi i k \cdot \boldsymbol{\theta}}, \qquad \widehat{K}_{-k} = \overline{\widehat{K}_k}, \tag{5.27}$$

and define the invariance residual

$$E_N\left(\boldsymbol{\theta}\right) = F_h^{cons}\left(K_N\left(\boldsymbol{\theta}\right)\right) - K_N\left(\boldsymbol{\theta} + \boldsymbol{\omega}\right). \tag{5.28}$$

FFT analysis, Poincaré sections, and long trajectories are used only to initialize K_N and $\boldsymbol{\omega}$; a Newton-like method is then used for correction. The a posteriori KAM theorems of Haro et al. require the map to be analytic and exact symplectic, the frequency to satisfy a Diophantine condition, the dynamics to be nondegenerate, and $\| E_N \|$ to satisfy an explicit smallness condition. Existence of a true invariant torus can be certified only after all constants have been verified. A small residual alone supports only the statement that a numerical approximate torus has been obtained[28].

5.8. **Parameter continuation and safe operating domain.** Continuation is performed with respect to magnetic-field strength, insertion speed, flow velocity, contact clearance, and material parameters. At each parameter point, record:

- ε_{map} and the Hamiltonian learning error;
- the symplectic or K-symplectic structural defect;
- the invariance residual $\| E_N \|$;
- the minimum singular value of the twist matrix/frequency map;
- the finite-order Diophantine margin and decay of the Fourier coefficients;
- the energy balance, port power, peak contact response, and trajectory error;
- the discrepancy between the Kirchhoff model and the high-fidelity model.

Define the KAM-theoretic guarantee domain as

$$\mathcal{S}_{KAM} = \{\mu : \text{the symplecticity, nondegeneracy, nonresonance, and a posteriori smallness conditions all hold}\}. \tag{5.29}$$

The complete engineering safe domain is defined as

$$\boxed{\mathcal{S}_{safe} = \mathcal{S}_{KAM} \cap \mathcal{S}_{energy} \cap \mathcal{S}_{contact} \cap \mathcal{S}_{model} \cap \mathcal{S}_{control}.} \tag{5.30}$$

Here, KAM theory provides only a conditional long-time stability guarantee for the conservative backbone. Open ports, dissipation, contact switching, and high-fidelity model discrepancy are addressed by additional indicators. The safe domain is neither a single "chaos threshold" nor evidence of clinical efficacy.

At present, Proposition 5.1 establishes local qualitative KAM existence for the selected quartic conservative approximation at sufficiently small amplitude, but $\mathcal{S}_{KAM}$, $\mathcal{S}_{safe}$, a specific invariant-torus embedding K_N, and its a posteriori constants have not been instantiated numerically. The result therefore does not constitute finite-domain numerical KAM certification, a rigorous KAM result for the material-reference system, or an engineering safe domain.

6. Numerical Experiments and Results

6.1. **Dataset, canonical variables, and evaluation design.** The numerical study follows the sequence in Figure 6.1. The continuum and Galerkin models provide the mechanical construction, while the primary trajectory dataset uses a fixed-basis canonical representation

$$q \in \mathbb{R}^4, \qquad p = M_4 \dot{q}, \tag{6.1}$$

with a constant symmetric positive-definite M_4. This relation holds across all 96 trajectories, with a maximum canonical residual of 1.058791×10^{-22}. The configuration-dependent full nonlinear cotangent pullback remains the theoretical branch described in Section 3; it is not numerically instantiated by this dataset.

The 96 complete trajectories contain 801 states over 0.2 s at $\Delta t = 2.5 \times 10^{-4}$ s. The whole-trajectory split is 64/16/16 for training, validation, and independent testing. Test trajectories are excluded from tuning, checkpoint selection, seed selection, and architecture selection. Table 6.1 summarizes the training and evaluation design.

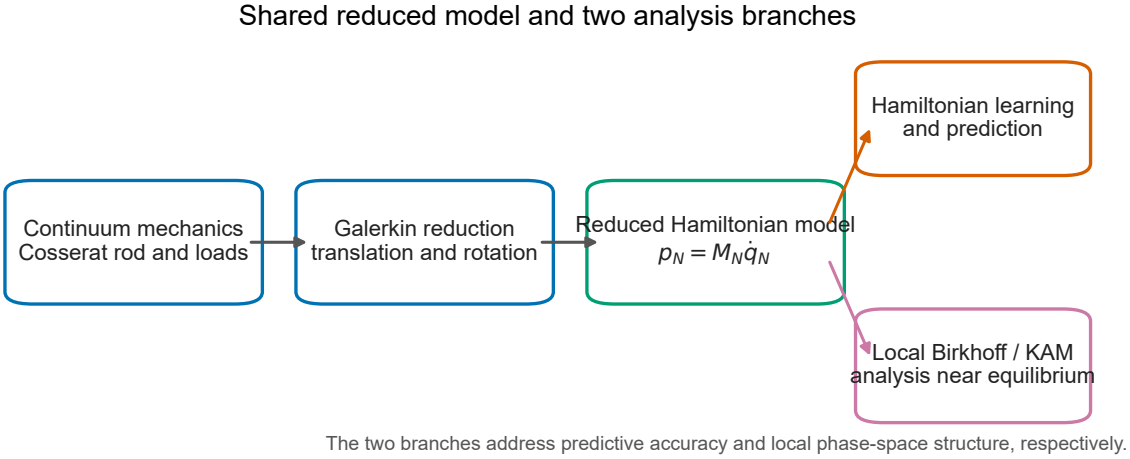


Figure 6.1. Scientific workflow. The primary trajectory experiments follow the fixed-basis reduced canonical branch; the full nonlinear cotangent pullback is displayed separately as a theoretical generalization.

Table 6.1. Data and evaluation protocol. All five seeds are retained as repeated runs; validation and test performance are not used to select a seed.

Item	Setting or result
COMSOL trajectories	96 complete trajectories; each spans 0.2 s with $\Delta t = 2.5 \times 10^{-4}$ s, 801 sampled states, and 800 transitions.
Whole-trajectory partition	64 training, 16 validation, and 16 independent-test trajectories; assignments are fixed and mutually exclusive.
Canonical representation	Fixed-basis reduced canonical coordinates with constant M_4 and $p = M_4\dot{q}$.
Trainable models	Residual MLP, Real HNN, Gray Real HNN, Standard CHNN, and Gray CHNN.
Training repetitions	Seeds 20260701–20260705, 300 epochs per run, for 25 completed runs.
Optimization	Adam optimizer, batch size 256, learning rate 2×10^{-3}, weight decay 10^{-6}, gradient-clipping threshold 10, and no early stopping.
Checkpoint	Epoch 300 is used for every run; early stopping and seed selection are not used.
Independent test	All five seeds of every architecture are evaluated on the same 16 trajectories. Per-trajectory summaries use the median across seeds, and uncertainty resamples complete trajectories.
Parameter accounting	Real trainable degrees of freedom are 9416, 8961, 8961, 8822, and 8822 in the model order above; complex parameters are counted as two real scalars.

6.2. **Factorial architecture and parameter accounting.** E1 separates real versus complex representation from the presence or absence of a physical Hamiltonian prior. The four factorial cells are Real HNN, Gray Real HNN, Standard CHNN, and Gray CHNN; Residual MLP is retained as a non-Hamiltonian external baseline. Both gray models use the same physical term,

$$H_{\text{gray}} = H_{\text{phys}} + \Delta H_\theta. \tag{6.2}$$

The real models have 8961 trainable real degrees of freedom and the complex models have 8822, giving a real/complex ratio of 1.016. The two representations use the same inputs, loss, optimizer family, epochs, seeds, normalization, data identities, rollout integrator, and metrics; complex parameters are counted as two real scalars.

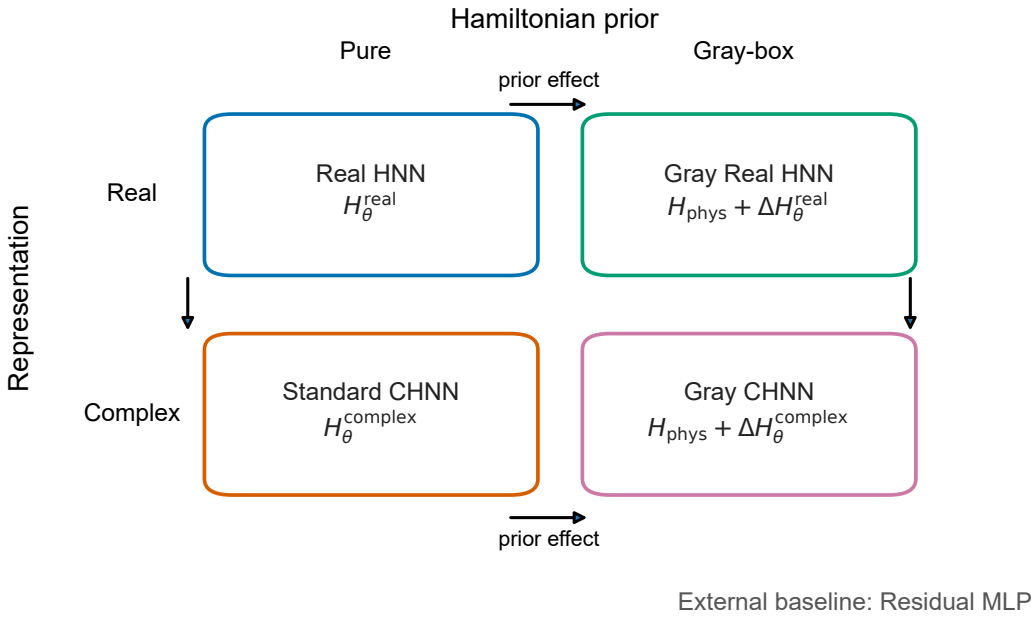


Figure 6.2. Approximately parameter-matched factorial architecture. The prior factor adds the same H_{phys} to the real and complex residual Hamiltonians.

Every architecture is trained for 300 epochs at seeds 20260701–20260705. For each test trajectory and model, E1/E2 first takes the median metric across the five seeds and then treats the 16 trajectories as the independent empirical units. Median training times are 493.04, 935.66, 2224.10, and 2610.27 seconds for Real HNN, Gray Real HNN, Standard CHNN, and Gray CHNN, respectively; the corresponding complete-test inference times are 14.60, 44.93, 83.20, and 112.92 seconds. Parameter matching therefore does not imply equal computational cost.

6.3. **Independent-test performance.** Table 6.2 reports all five models. The mean rollout errors are 1665.33 for Real HNN and 1665.28 for Standard CHNN, compared with 12.6587 for Gray Real HNN and 12.6640 for Gray CHNN. The same pattern appears in one-step MSE and endpoint error. The dominant contrast is therefore the addition of the physical Hamiltonian, rather than the choice of real or complex representation.

Table 6.2. Independent-test results after taking the median across five seeds for each trajectory and then the mean across 16 trajectories. The interval is a trajectory-level 95% bootstrap confidence interval for mean rollout error.

Model	Real DOF	One-step MSE	Rollout mean	95% CI	Endpoint	Train (s)	Test (s)
Residual MLP	9416	3.93437×10^{6}	2.20953×10^{9}	$[2.20948, 2.20958] \times 10^{9}$	1.03378×10^{7}	367.90	6.94
Real HNN	8961	9.01606×10^{-2}	1665.33	[1048.33,2324.91]	490.607	493.04	14.60
Gray Real HNN	8961	1.62604×10^{-3}	12.6587	[6.99943,19.4900]	4.57008	935.66	44.93
Standard CHNN	8822	9.01605×10^{-2}	1665.28	[1051.88,2341.64]	490.503	2224.10	83.20
Gray CHNN	8822	1.62689×10^{-3}	12.6640	[7.09015,19.4131]	4.57009	2610.27	112.92

The gray-box rollout improvement is 1652.68 in the real comparison and 1652.62 in the complex comparison. Their trajectory-bootstrap 95% confidence intervals, $[1040.79, 2312.99]$ and $[1037.38, 2310.08]$, exclude zero. By contrast, the pure complex-representation effect is 0.05316 with interval $[-0.08830, 0.17817]$, and the gray complex-representation effect is -0.00532 with interval $[-0.14161, 0.12748]$. The interaction contrast is -0.05848 with interval $[-0.17323, 0.06401]$. The physical prior is therefore supported, whereas an additional benefit of complex representation is not statistically resolved under this design.

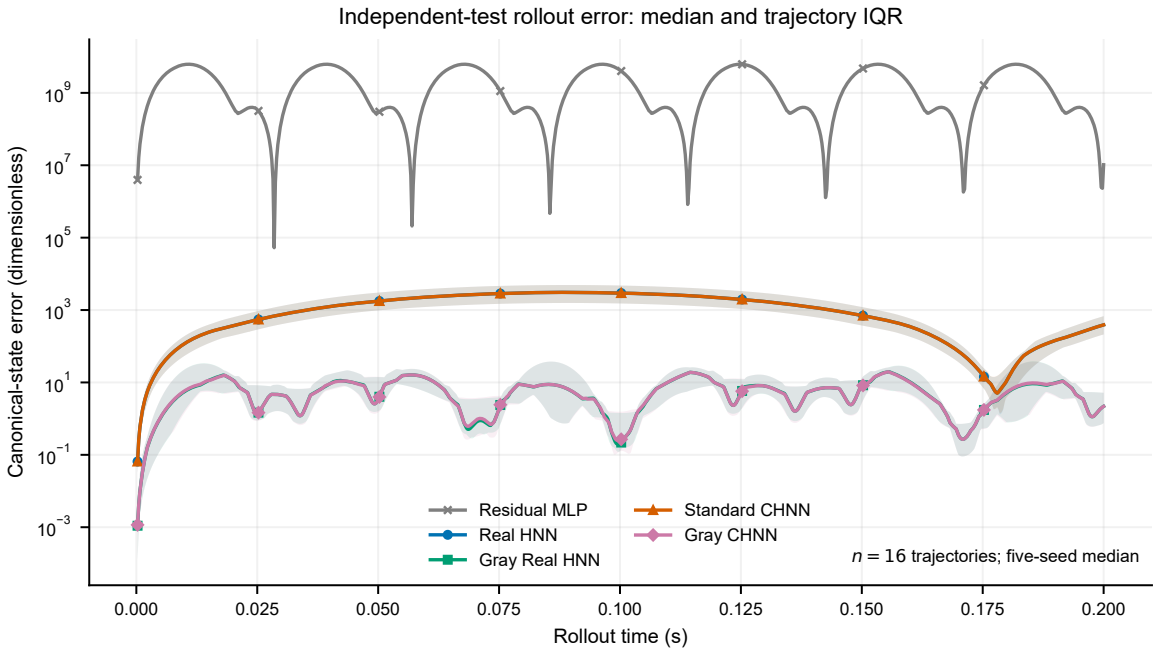


Figure 6.3. Rollout error versus time for the five models. Curves summarize complete test trajectories after five-seed aggregation; the gray-box models remain close to one another and below the pure Hamiltonian models.

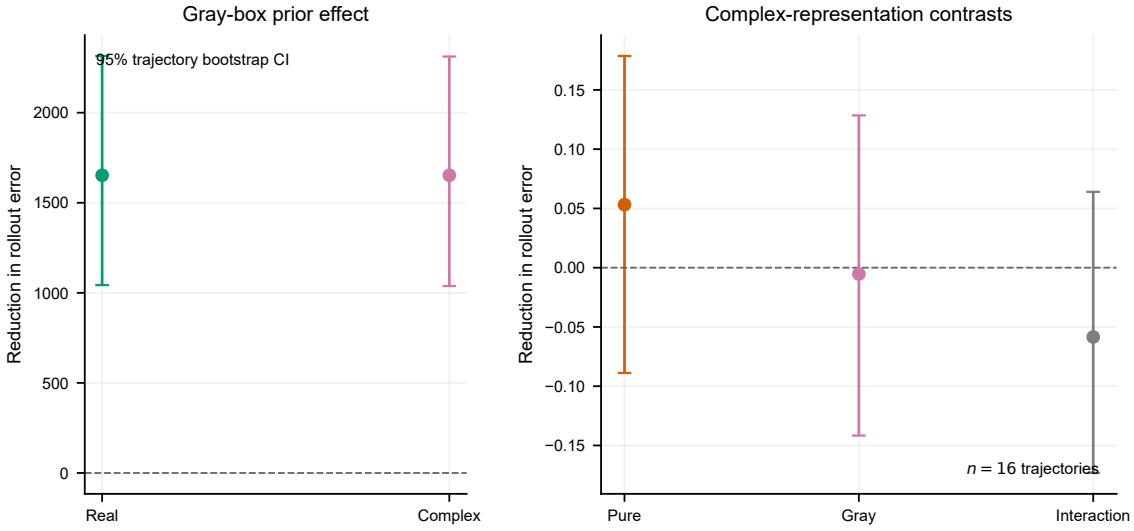


Figure 6.4. Factorial contrasts for prior, representation, and their interaction. Confidence intervals resample the 16 complete test trajectories.

The maximum trajectory-time prediction distance between the Real HNN and Standard CHNN checkpoints is approximately 1.94×10^{-10} in the reported L2 comparison. Their checkpoint hashes and parameter tensors are distinct. The near equality is therefore an empirical observation under this design, rather than evidence of checkpoint reuse.

6.4. **Trajectory-level uncertainty and paired comparisons.** The complete trajectory is the statistical unit. Each of the 16 rows per model contains one-step MSE, rollout mean and endpoint errors, energy error, and maximum error after the within-trajectory five-seed median. Means, standard deviations, medians, interquartile ranges, and 95% confidence intervals are

computed across trajectories; bootstrap resampling never treats the 801 time points from one trajectory as independent samples.

Table 6.3. Paired comparisons of rollout mean error over the same 16 test trajectories. Differences are model A minus model B; lower error is better. Confidence intervals resample complete paired trajectories.

Model A	Model B	Mean difference	Median difference	95% CI	A better	B better
Real HNN	Standard CHNN	0.0531609	0.102769	[-0.0882979,0.178173]	4	12
Gray Real HNN	Gray CHNN	-0.00532128	-0.00654212	[-0.141609,0.127481]	9	7
Real HNN	Gray Real HNN	1652.68	1359.97	[1040.79,2312.99]	0	16
Standard CHNN	Gray CHNN	1652.62	1359.88	[1037.38,2310.08]	0	16

For Gray Real HNN minus Gray CHNN, the mean paired rollout difference is -0.005321, the median is -0.006542, and the 95% interval is $[-0.141609, 0.127481]$; Gray Real HNN is lower on 9 trajectories and Gray CHNN on 7. Figure 6.5 displays all 16 paired points and their differences. The paired evidence does not resolve a directional performance difference at the evaluated scale; it is not a formal equivalence test.

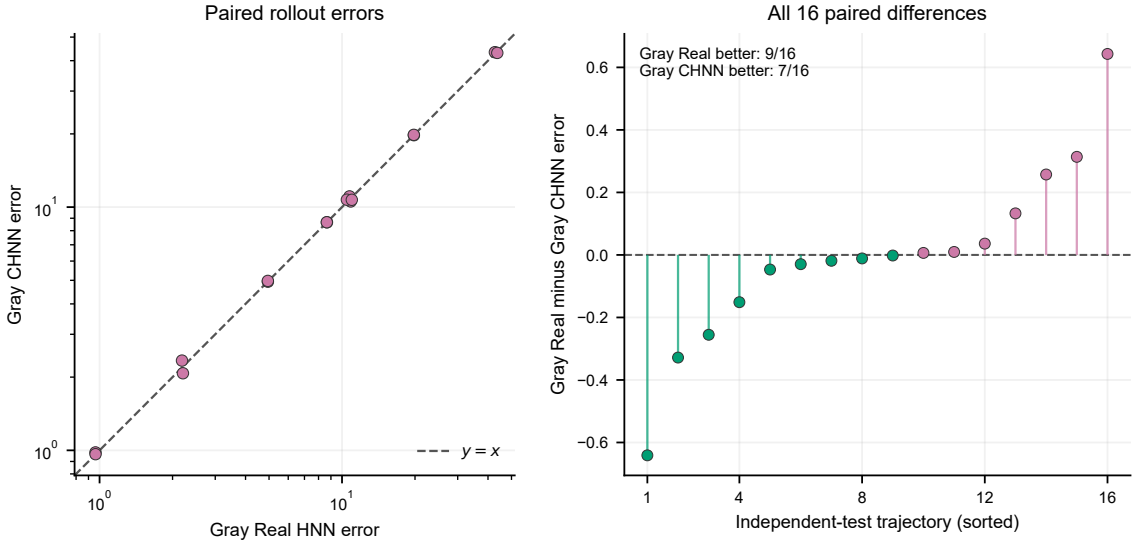


Figure 6.5. Paired Gray Real HNN and Gray CHNN rollout errors. All 16 trajectories are shown together with trajectory-wise differences; the diagonal denotes equal error.

6.5. **Auxiliary modal-order sensitivity.** The auxiliary analysis uses three complete trajectories (mode1, mode2, and mixed12), each containing 997 samples from 0.002 to 0.998 s, in nested $N = 2, 4, 6$ coordinates. These trajectories are separate from the primary learning split. With $N = 6$ as the nested reference, e_{recon} is the root-mean-square of trajectory Frobenius relative errors in zero-padded q and p; e_T and e_{H_2} are mean absolute relative errors in quadratic kinetic energy and quadratic Hamiltonian; and e_Q is the trajectory Frobenius relative error of the zero-padded stored $\dot{p}$ representation. Every metric is dimensionless and is calculated per complete trajectory before averaging across the three trajectories.

At $N = 2$, the respective means are 0.011228, 0.0002436, 0.0004259, and 0.201533. At $N = 4$, they decrease to 0.003644, 2.6915×10^{-5}, 4.9927×10^{-5}, and 0.090368. The generalized-force representation therefore remains above the predeclared representation threshold at both lower orders. These representation-level results do not support the sufficiency of the $N = 2$ truncation for the full predictive task. The stored conservative masks are all zero. A common $N = 2, 4, 6$ dataset with the same training, validation, and test identities is unavailable. Consequently, no

predictive model is trained on these auxiliary trajectories and the predictive panel in Figure 6.6 is marked as not evaluated.

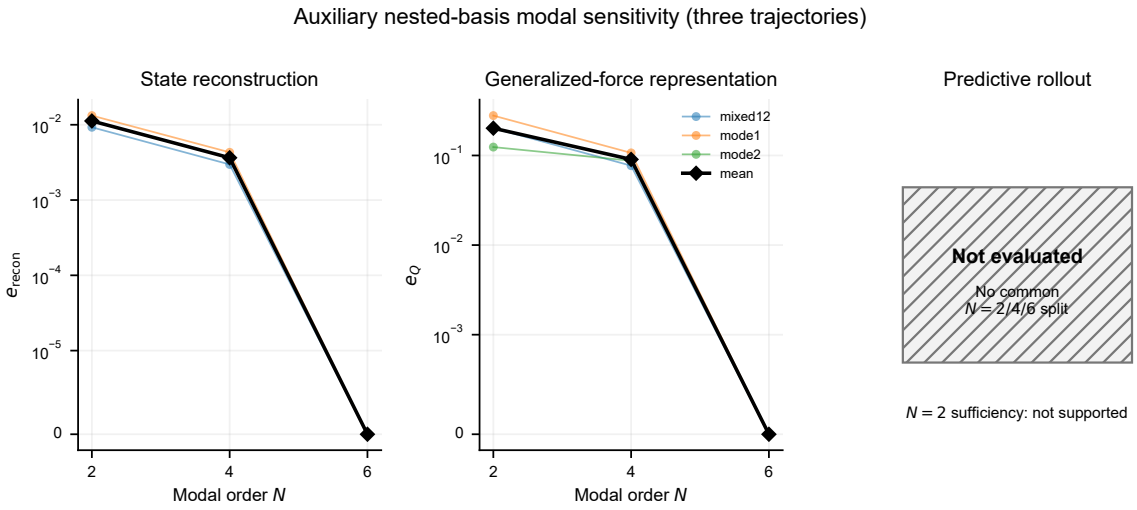


Figure 6.6. Auxiliary nested-basis sensitivity. Representation errors are measured relative to $N = 6$; predictive rollout sensitivity is not evaluated because a common split is unavailable.

6.6. **High-precision local KAM robustness.** The quartic J_{ABC} twist matrix is recomputed at 50, 80, 120, and 160 decimal digits from the stored decimal coefficients. Across all precisions, its smallest singular value is $6.3582193813 \times 10^{-11}$, its condition number is 7480.800334, and its rank is four under the recorded 10^{-40} threshold. The minimum scanned divisor for $\|k\|_1 \leq 4$ is 0.2718619503. The auxiliary J_{AB}, J_{AC}, and J_{BC} matrices are also full rank, although their conditioning differs. These stable high-precision quantities support the local nondegeneracy calculation for the selected quartic conservative approximation.

The high-precision twist-conditioning result is summarized once in Figure 5.1 and is not repeated here.

The result remains local and conditional. It neither supplies a uniform higher-order neighborhood bound for the full material-reference Hamiltonian nor constructs and certifies a specific invariant torus. Accordingly, the KAM computation complements the learning experiments but does not convert their trajectory errors into a global or engineering stability guarantee.

7. Discussion and scope

7.1. **Two canonical levels.** The canonical construction has two related but distinct levels. The full nonlinear material-reference formulation admits a configuration-dependent tangent lift and cotangent pullback, including the associated transport term. The primary trajectories instead use a fixed four-mode basis with constant M_4 and define $p = M_4\dot{q}$. This representation is sufficient for the reported E1/E2 experiments. The nonlinear pullback remains a theoretical generalization because the data do not contain the full-order momentum and basis-transport fields needed to evaluate it. This distinction defines the scope of the experiment without weakening the fixed-basis result.

The same separation applies to the learning theorem. Its iid assumption defines a conditional function-class statement. The empirical study does not count the 801 temporally correlated samples along one trajectory as independent physical samples; model uncertainty and paired comparisons use 16 complete test trajectories. The trajectory-bootstrap intervals therefore quantify variation over the test cases rather than the much smaller pointwise variation that would result from pseudoreplication.

7.2. **What the factorial experiment resolves.** The factorial design clarifies the learning results. Gray Real HNN and Gray CHNN reduce mean rollout error by approximately 1653 relative to their pure-Hamiltonian counterparts, and both paired prior-effect intervals exclude zero. The real and complex models within each prior level are almost indistinguishable at the current trajectory resolution. In particular, the gray representation contrast has a confidence interval spanning both signs and a 9/7 trajectory count. The shared physical prior is therefore associated with the dominant improvement, whereas the experiment does not resolve an additional benefit from complex representation.

Parameter matching narrows one confound but does not equalize computation: the complex architectures take substantially longer to train and evaluate on the CPU implementation. The real-affine completeness result for widely linear layers also does not predict the outcome of the Standard CHNN experiment, which uses the stated standard complex architecture. The theorem gives a conditional expressive relationship; the factorial experiment assesses this implementation on this dataset.

The near equality of Real HNN and Standard CHNN is reproduced with distinct checkpoints. It is therefore an empirical feature of the present optimization and task, rather than evidence of checkpoint reuse. The observation should not be generalized to other widths, optimizers, canonical scalings, or excitation distributions without a new prospective comparison.

7.3. **Modal and KAM boundaries.** E3 resolves only representation-level modal sensitivity. The nested auxiliary data show meaningful reductions from $N = 2$ to $N = 4$, especially in generalized-force representation, and the stated criterion does not support $N = 2$ sufficiency. Because these three trajectories are not a common $N = 2, 4, 6$ split, training predictive models on them would change the comparison. Predictive ranking stability across modal orders therefore remains unknown.

The KAM result is similarly scoped. High-precision stability of twist singular values supports the arithmetic robustness of the local quartic nondegeneracy calculation. Classical KAM conclusions still require the analytic local transformation, nonresonance, perturbation smallness, and an appropriate neighborhood. The present evidence does not provide a uniform higher-order remainder bound from the full material-reference Hamiltonian to the quartic approximation, nor an a posteriori invariance certificate for a specific torus. The KAM analysis therefore establishes a local conditional result for the quartic conservative model and does not define an engineering safe domain.

Future work should construct a common $N = 2, 4, 6$ dataset before predictive modal-order comparison, examine representation effects under additional approximately parameter-matched architectures and operating conditions, and obtain rigorous or validated-numerical higher-order bounds for specific invariant-torus certification. Open-port learning likewise requires independently validated state, inertia, and generalized-force mappings before new labels or predictive claims are introduced.

8. Conclusions

This work connects geometrically exact rod mechanics, a fixed-basis canonical learning experiment, and a local quartic KAM analysis. In the primary trajectories, a constant four-mode mass matrix defines $p = M_4\dot{q}$ for all 96 cases. The configuration-dependent full nonlinear cotangent pullback remains a theoretical generalization and is not reported as numerically instantiated by these data.

The five-model comparison uses one 64/16/16 trajectory split, five seeds per architecture, epoch-300 checkpoints, and trajectory-level uncertainty. Gray Real HNN and Gray CHNN reduce mean rollout error from approximately 1665 to approximately 12.66. Their paired difference is near zero, with a 95% interval spanning both signs. The physical gray-box prior is therefore supported, whereas an additional advantage of complex representation is not statistically resolved under this design.

The auxiliary modal analysis does not support $N = 2$ sufficiency: the generalized-force representation error remains 0.2015 at $N = 2$ and 0.09037 at $N = 4$ relative to $N = 6$. Predictive

modal-order sensitivity is left unevaluated because a common $N = 2, 4, 6$ train/validation/test dataset is unavailable. No predictive curve or ranking is inferred from the auxiliary trajectories.

High-precision recomputation gives a full-rank quartic twist matrix with stable singular values from 50 to 160 digits. This supports the local nondegeneracy calculation for the selected quartic conservative approximation, subject to the stated analytic and perturbative assumptions. It does not certify the full material-reference Hamiltonian or a specific invariant torus. Within these boundaries, the fixed-basis experiment, factorial learning evidence, modal sensitivity assessment, and local KAM computation form a coherent submission-level result.

Appendix A. Reduced-Model and Numerical Details

This appendix provides the reduced-model and numerical details needed to reproduce the canonical simulations used in the learning study.

A.1. **Reduced coordinates and canonical mapping.** The fixed-basis reduced model uses configuration coordinates obtained by projection,

$$q = \Pi_{\mathrm{conf}}(U), \qquad p = B^T P_{\mathrm{full}}, \tag{1.1}$$

$$\dot{q} = M_N^{-1} p, \qquad \dot{p} = B^T L. \tag{1.2}$$

Here q is a configuration coordinate and p is its associated cotangent coordinate. In the learning experiments, $N = 4$, the basis is fixed, and the reduced mass matrix M_4 is constant, so the canonical momentum is $p = M_4 \dot{q}$. The general geometrically exact formulation may instead use a configuration-dependent cotangent pullback; that construction is not used for the fixed-basis learning experiments.

A.2. **Numerical discretization and consistency measures.** Exported states and vector-field labels are evaluated at common solver times. When both are required from the same solver step, order-2 BDF with `tsteps` is used for storage and label generation. This sampling choice does not imply that BDF is symplectic and does not identify the stored discrete trajectory with a symplectic map.

Consistency residuals compare the exported $\dot{q}$ with $M_N^{-1} p$ and the exported $\dot{p}$ with $B^T L$. Time-discretization comparisons additionally examine q, p, and total energy on the same physical trajectory and at the same sampling times. For errors e_h and $e_{h/2}$ on adjacent refinement levels, the observed order is defined by

$$r_{\mathrm{obs}} = \log_2 \left(\frac{e_h}{e_{h/2}} \right).$$

These quantities define the numerical checks without turning solver diagnostics or completion records into additional scientific claims.

Appendix B. Learning and Statistical Evaluation Details

This appendix summarizes the training and statistical procedures underlying the factorial model comparison.

B.1. **Data partition and model evaluation.** The primary trajectory dataset contains 96 prospective COMSOL trajectories, each spanning 0.2 s with $\Delta t = 2.5 \times 10^{-4}$ s and 801 sampled states. The fixed whole-trajectory partition is 64 training, 16 validation, and 16 independent-test trajectories. The five random seeds are retained as repeated runs, and the same test trajectories are used for every architecture. The complete training and evaluation settings are specified in the numerical-results section.

The four trainable architecture families are trained for 300 epochs with five seeds per family. Candidate models are selected using the validation primary metric before the independent test trajectories are loaded. Final testing consists only of forward and rollout evaluation followed by metric aggregation; it does not update parameters, modify checkpoints, reselect a model, or tune hyperparameters. Uncertainty is estimated by resampling complete trajectories rather than individual time points.

B.2. **Structural checks of the learning equations.** The implementation is checked against the finite-dimensional identities used in the analysis: canonical skew structure, cancellation of internal Hamiltonian power, canonical port power, nonpositive passive dissipative power, exclusive magnetic accounting, and separation of open-port perturbation types. These checks establish consistency between the implementation and the mathematical formulation; they are not evidence for identification accuracy or for KAM stability. Identities that require derivative existence, integrability, a nonautonomous chain rule, or specific physical provenance remain conditional on those assumptions.

The real and complex representations are compared under the same data, optimization, and evaluation protocol. The paired confidence interval for their performance difference contains zero, so the present experiment does not resolve an additional advantage from complex representation. The decrease in representation error from $N = 2$ to $N = 4$ indicates that the two-mode truncation cannot be regarded as converged for the complete predictive task. Predictive modal-order sensitivity is not evaluated because no common $N = 2, 4, 6$ train/validation/test dataset is available.

Appendix C. Local Birkhoff and KAM Computation

This appendix collects the numerical ingredients used in the local Birkhoff and KAM calculation near the selected equilibrium.

C.1. **Quartic conservative approximation.** The calculation uses a recentered four-degree-of-freedom conservative reference and a quartic Birkhoff normal form. The twist matrix is evaluated in the action variables of this approximation using high-precision arithmetic. Table 3.1 reports the frequencies, residuals, twist rank, and the domain of validity.

Table 3.1. Birkhoff normal form and qualitative KAM analysis for the local quartic conservative approximation. The reported quantities support only the local conclusion for this approximation.

Item	Value or conclusion	Domain of validity
Low-frequency pair	5.562997, 5.606265 Hz	Recentered four-degree-of-freedom conservative reference
High-frequency pair	34.661765, 35.092095 Hz	Same conservative reference
Maximum full-Hamiltonian relative error on the local sample set	4.43237×10^{-8}	Finite-sample error estimate; not a uniform bound on a neighborhood
Quartic BNF homological residual	2.58368×10^{-80}	Numerical residual of the quartic homological equation
Primary J_{ABC} twist rank	4	Kolmogorov twist nondegeneracy
Primary dimensionless twist $\sigma_{\min}$	6.35822×10^{-11}	Magnitude of the weakest singular direction of the twist matrix
Resonance crossing within the prescribed action region	Present	No numerical KAM conclusion is drawn for the entire region
Local qualitative KAM existence	Established	Selected quartic conservative approximation only
Extension to the full material-reference Hamiltonian	Not established	Missing a uniform neighborhood bound for the $O(5+)$ remainder

The computed twist matrix is nondegenerate for the selected quartic conservative approximation, supporting the stated local KAM conclusion. The resonance crossing within the prescribed action region prevents a uniform conclusion over that entire region. Extension to the full material-reference Hamiltonian would require a uniform neighborhood bound for the higher-order remainder, together with the corresponding nonresonance and invariance estimates. The present result is therefore a local conditional statement for the selected conservative approximation, not a claim about the complete dissipative open system.

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